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Nonparametric Goodness-of-fit Testing under Covariate Shift

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Abstract

This paper develops procedures for nonparametric goodness-of-fit testing under covariate shift, where labelled data are drawn from a *source* population but goodness-of-fit is evaluated for a *target* population. The distribution mismatch is quantified by either a bounded moment condition or a sub-exponential tail condition on the target-to-source density ratio. Our method combines truncated importance-weighting kernel ridge regression with a multiplier bootstrap to construct confidence sets for the regression function. The truncation stabilizes the importance-weighting kernel ridge regression as well as the bootstrap calibration, making our approach applicable even when the density ratio has heavy tails. We prove nonasymptotic validity and sharpness of the resulting confidence sets under suitable operator compatibility conditions, and establish explicit error rates for coverage probability under specific conditions on the target-to-source density ratio and on the spectral decay of the kernel integral operator. Numerical experiments corroborate our theoretical findings.

Keywords: Covariate shift, kernel ridge regression, multiplier bootstrap, goodness-of-fit, importance weighting.

1 Introduction

Covariate shift arises when labelled data are collected from a source population but scientific conclusions are sought for a target population with a different covariate distribution. This problem has become increasingly prominent as machine learning methods trained on labelled source data are deployed in specialized target domains [Guan and Liu, 2022], where target-domain labels are often scarce or expensive [Wang et al., 2024]. For instance, when general-purpose language models are deployed in specialized biomedical and legal domains, the prompts, documents, and questions can differ drastically from the broad corpora used for pretraining or instruction tuning [Gururangan et al., 2020]. These applications raise important statistical questions, for example, *which statistical methods perform well under target distributions and are robust to target-to-source mismatch?* and *is it possible to evaluate the performance of these methods under the target distribution even when they are trained under the source distribution?* While the first question has been extensively studied in the literature, the second is far less understood. This paper provides an affirmative answer through a

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general inference framework based on truncated importance-weighted kernel ridge regression (KRR) and the multiplier bootstrap [Bickel and Freedman, 1981, Efron and Tibshirani, 1994, Hall, 2013].

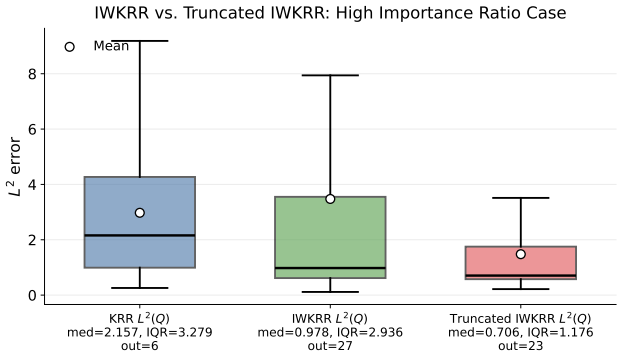
Let $\mathbb{P}$ and $\mathbb{Q}$ denote the *source* and *target* distributions, respectively, on the joint space $\mathfrak{X} \times \mathbb{R}$. For simplicity, we assume that both $\mathbb{P}_X$ and $\mathbb{Q}_X$ admit *known* densities with respect to a common dominating measure, and that $\mathbb{Q}_X$ is absolutely continuous with respect to $\mathbb{P}_X$. Our methods and main theorems hold as long as the density ratio $\rho(x) = (d\mathbb{Q}_X/d\mathbb{P}_X)(x)$ is known. Moreover, $\rho(x)$ can be estimated from unlabelled source and target covariates by direct importance-estimation or distribution-matching methods, such as those in Sugiyama et al. [2008], Huang et al. [2007]. See further discussion in Section 7. We assume that the marginal distributions of the covariates may differ, $\mathbb{P}_X \neq \mathbb{Q}_X$, whereas the conditional distributions of the response remain the same, $\mathbb{P}_{Y|X} = \mathbb{Q}_{Y|X}$. With a slight abuse of notation, we shall write $\mathbb{P}$ and $\mathbb{Q}$ for the covariate marginal distributions $\mathbb{P}_X$ and $\mathbb{Q}_X$, respectively, whenever no ambiguity arises. Suppose that an i.i.d. sample $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n \subset \mathfrak{X} \times \mathbb{R}$ drawn from the source distribution $\mathbb{P}$ is available, where the response and covariates satisfy the regression model $f_*(x) = \mathbb{E}[Y|X = x]$, where $f_*(\cdot) : \mathfrak{X} \mapsto \mathbb{R}$ is an *unknown* regression function. Our goal is to evaluate f_* under the target covariate distribution. More specifically, given a candidate function f_0, we aim to test

$$H_0 : \|f_0 - f_*\|_{\mathsf{L}^2(\mathbb{Q})} = 0 \quad v.s. \quad H_1 : \|f_0 - f_*\|_{\mathsf{L}^2(\mathbb{Q})} > 0, \tag{1.1}$$

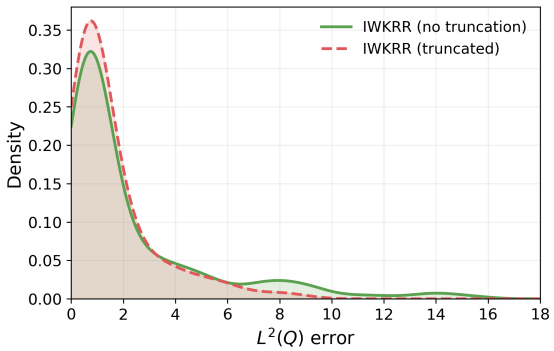
where the $\mathsf{L}^2(\mathbb{Q})$-norm is defined by $\|g\|^2_{\mathsf{L}^2(\mathbb{Q})} := \int g(x)^2 d\mathbb{Q}(x)$. Equivalently, the null hypothesis states that $\mathbb{Q}_X\big(\{x : f_*(x) = f_0(x)\}\big) = 1$. For ease of exposition, we focus on the simple test, but our framework can be easily generalized to composite tests and two-sample tests. See numerical experiments in Section 6.3 and the Appendix.

The seminal work Shimodaira [2000] proposed importance weighting (IW) for estimating the target function $f_*(x)$ under covariate shift in parametric likelihood models, showing that the asymptotically optimal weighting function is the target-to-source density ratio $\rho(x)$. Non-asymptotic analyses of importance weighting have also attracted considerable attention across diverse fields and applications, including the generalization bounds for statistical learning with function classes of finite pseudo-dimension [Cortes et al., 2010], local convergence rates for learning Lipschitz functions under local doubling conditions [Schmidt-Hieber and Zamolodtchikov, 2024], minimax-optimal rates for learning Hölder-continuous functions under covariate shift [Pathak et al., 2022], and kernel-based regression and classification [Ma et al., 2023, Wang, 2026].

Importance weighting is generally necessary for unbiased estimation of the target risk under covariate shift, but it may be unnecessary for identifying the Bayes-optimal predictor when the conditional distribution is invariant and the model is correctly specified. See, for example, Gogolashvili et al. [2023] and the references therein. However, importance weighting can degrade finite-sample performance when the density ratio is unbounded or highly variable. A common remedy is to truncate the density ratio, replacing $\rho(x)$ by $w_n(x) := \min\{\rho(x), \tau_n\}$ where $\tau_n > 0$ is a pre-specified truncation level. Ma et al. [2023] showed that the importance weighted kernel ridge regression (IWKRR) estimator achieves the minimax optimal convergence rate under covariate shift using a truncated importance weight for variance reduction. Figure 1 illustrates the stabilizing effect of truncation when the density ratio is large in target-relevant regions. Indeed, a small number of source observations may receive extremely large importance weights, causing the IWKRR estimator to be highly *variable*. Capping the weights can therefore substantially stabilize the $\mathsf{L}^2(\mathbb{Q})$ error.



(a) $\mathsf{L}^2(\mathbb{Q})$-norm error rates: KRR, IWKRR, and IWKRR with truncation.



(b) IWKRR $\mathsf{L}^2(\mathbb{Q})$-norm error distributions, without and with truncation.

Figure 1: truncating large density-ratio weights $\rho(x) := (d\mathbb{Q}/d\mathbb{P})(x)$ stabilizes the finite-sample performance of the importance-weighting kernel ridge regression (IWKRR). Source and target distributions are *truncated normal* $\mathrm{TN}(-1, 0.25; [-2, 2])$ and $\mathrm{TN}(1, 0.25; [-2, 2])$, respectively. Panel (a) summarizes the validation-tuned errors of KRR, IWKRR, and truncated IWKRR. The true regression function is $f_*(x) = |x|$. Panel (b) shows the Monte Carlo densities of the $\mathsf{L}^2(\mathbb{Q})$ errors over 200 repetitions, using raw importance weights which differ only in truncation.

The aforementioned works primarily focus on efficient and robust estimation under covariate shift. While they provide practical guidance and theoretical justification for statistical methods that improve predictive performance under target distributions, it remains largely unclear how to evaluate the performance of these methods under the target distribution and how to quantify the uncertainty of the resulting estimators. For instance, in studies of conditional average treatment effects, one may wish not only to estimate the treatment-effect function but also to test whether treatment effects are constant over the target population [Crump et al., 2008, Ding et al., 2016]. These inferential conclusions depend critically on the target and source distribution mismatch: heterogeneity that is negligible under the source distribution may be substantial under the target distribution if the regions in which the effect varies have substantially different probability densities under the two distributions.

In this paper, we develop statistical procedures to quantify the uncertainty of the truncated IWKRR estimator and to evaluate its performance under the target covariate distribution. More specifically, we construct confidence sets for f_* with respect to the $\mathsf{L}^2(\mathbb{Q})$-norm distance based on the truncated IWKRR estimator, establish non-asymptotic bounds on the absolute coverage error of these confidence sets, and study the goodness-of-fit testing problem in (1.1). The main challenge is that the IWKRR estimator, the inferential loss, and the key technical tools naturally live in different geometries. The IWKRR estimator is governed by the truncated weighted-source risk, whereas coverage of the confidence sets is evaluated in the target $\mathsf{L}^2(\mathbb{Q})$ metric. On the technical side, the covariance operator of the stochastic fluctuation must be analyzed in the target $\mathsf{L}^2(\mathbb{Q})$ geometry, whereas its algebraic structure is most transparent through RKHS operators. We address this mismatch using unitary isomorphisms generated by the square roots of the target and weighted-source integral operators. These maps realize the relevant L^2 geometries inside the RKHS while preserving Hilbert-space norms and inner products, thereby allowing the Gaussian and bootstrap

approximations to be compared within a common RKHS operator calculus. This operator-theoretic treatment captures two effects of covariate shift: the bias of the truncated IWKRR estimator and its covariance structure relevant for coverage guarantees of the proposed confidence sets.

Our contributions are as follows. First, based on the truncated IWKRR estimator, trained on finite-sample data from source distribution, and multiplier bootstrap, we propose an inference procedure to construct confidence sets of the unknown regression function in the target distribution. The procedure is computationally efficient where computing each bootstrap statistics mainly involves matrix and vector multiplications. Second, we provide a general theoretical framework establishing the coverage guarantee of the bootstrap confidence sets, where the error derives from the bias, Gaussian approximation, and bootstrap approximation. The target-to-source distribution mismatch, kernel integrals, and truncation level jointly determine the validity of multiplier bootstrap method. We identify the *regularized transfer coverage* condition ensuring that the source risk is transferable to the target risk and the *non-degenerate covariance* condition that prevents the Gaussian covariance from degenerating in the target geometry. Third, we establish explicit error rates for the coverage probability under distinct density ratio and kernel spectral conditions, covering the cases of sub-exponential type density ratio and polynomially decaying kernel eigenvalues. Finally, our inference methods yield interesting findings in real data experiments.

1.1 Related work

Covariate shift. The seminal work Shimodaira [2000] initiated this field by analyzing weighted likelihood methods for parametric predictive inference under covariate shift and identifying importance weighting as the asymptotically optimal weighting scheme. Finite-sample generalization bounds for importance weighting were developed by Cortes et al. [2008], covering kernel-based algorithms and function classes with finite pseudo-dimension. The unknown density ratio, as well as its variants, can be estimated using unlabelled samples from source and target populations such as the Kullback–Leibler importance estimation [Sugiyama et al., 2008] and kernel mean matching [Huang et al., 2007]. The γ-transfer exponent was introduced by Kpotufe and Martinet [2018] as a measure of source and target similarity. A systematic review of machine learning methods under covariate shift is provided by Sugiyama and Kawanabe [2012]. More recently, Zamolodtchikov [2026] established the minimax rates for nonparametric regression with both source and target samples, characterized through the notion of a transfer function. Model selection and cross validation under covariate shift have also gained considerable attention. An information criterion was derived in Shimodaira [2000] for model selection under covariate shift. Sugiyama et al. [2007] showed that the ordinary cross-validation can be biased under covariate shift and proposed importance-weighted cross-validation. The instability of importance-weighted cross-validation and its robust variants were studied by Kouw et al. [2019].

Nonparametric inference. Chernozhukov et al. [2014b, 2017, 2014a] established nonasymptotic Gaussian and bootstrap approximations for suprema of empirical processes, with applications to confidence bands. Zaitsev [1987] and Buzun et al. [2022] proved Gaussian approximation results for sums of independent random vectors under bounded and sub-Gaussian conditions, respectively. Yang et al. [2017] studied posterior coverage and sup-norm credible sets in Gaussian process regression. Singh and Vijaykumar [2023] developed inference procedures for KRR without covariate shift. Similarly, Shang et al. [2025] studied data integration based on KRR and constructed adaptive confidence intervals for $f_*(x)$ via local and global Bootstrap procedures. Constructing uniformly

valid pointwise confidence intervals is generally more challenging than testing a global null such as $H_0 : \|f_* - f_0\|_{\mathsf{L}^2(\mathbb{P})} = 0$. The former requires uniform control of the estimation error, typically in sup norm, and is subject to well-known impossibility and lower-bound phenomena in nonparametric models. See, e.g., Low [1997], Genovese and Wasserman [2008], Giné and Nickl [2010], Cai and Low [2006], Juditsky and Lambert-Lacroix [2003], Li [1989], Robins and van der Vaart [2006], Spokoiny [1996] on the methods and challenges of constructing honest and adaptive non-parametric confidence sets.

Predictive intervals. A related but distinct line of work studies predictive intervals and sets under covariate shift. Tibshirani et al. [2019] developed weighted conformal prediction when the likelihood ratio between the target and source covariate distributions is known or can be accurately estimated, and Barber et al. [2023] further studied conformal prediction beyond exchangeability. These methods provide prediction sets for a future response and are largely distribution-free. By contrast, our goal is function-level inference: we construct a confidence ball for the regression function f_*, with a radius calibrated in $\mathsf{L}^2(\mathbb{Q})$-norm, and therefore must control the bias, covariance structure, and bootstrap approximation of the IWKRR estimator with truncated weights.

1.2 Notations

Throughout, H denotes the RKHS associated with a kernel function $k(\cdot,\cdot)$, $\mathsf{L}^2(\mathbb{Q})$ denotes the L^2 space with respect to the target covariate distribution $\mathbb{Q}_X$, and we write $k_x = k(x,\cdot)$. For a Hilbert space G, let $\langle\cdot,\cdot\rangle_{\mathsf{G}}$ and $\|\cdot\|_{\mathsf{G}}$ denote its inner product and norm, respectively; for $a, b \in \mathsf{G}$, $a \otimes_{\mathsf{G}} b$ denotes the rank-one operator $(a \otimes_{\mathsf{G}} b)c = \langle b, c\rangle_{\mathsf{G}}\, a$ for $c \in \mathsf{G}$. For a bounded linear operator $\mathcal{A} : \mathsf{G} \to \mathsf{S}$, $\|\mathcal{A}\|_{\mathsf{G}\to\mathsf{S}}$ denotes its operator norm and $\|\mathcal{A}\|_{\mathrm{HS},\mathsf{G}\to\mathsf{S}}$ its Hilbert-Schmidt norm, when finite; if $\mathcal{A} : \mathsf{G} \to \mathsf{G}$ is trace class, $\mathrm{tr}_{\mathsf{G}}(\mathcal{A})$ denotes its trace; for self-adjoint operators $\mathcal{A}, \mathcal{B}$ on G, we write $\mathcal{A} \preceq_{\mathsf{G}} \mathcal{B}$ if $\mathcal{B}-\mathcal{A}$ is positive semidefinite. When the subscript is omitted, inner products, norms, tensor products, operator norms, Hilbert-Schmidt norms, traces, and Loewner order are understood in H. For a compact positive self-adjoint operator $\mathcal{A}$ on G, $(\mu_k(\mathcal{A},\mathsf{G}), \phi_k(\mathcal{A},\mathsf{G}))$ denotes its k-th eigenpair, with eigenvalues arranged in nonincreasing order and counted with multiplicity; we use the shorthand $(\mu_k, \phi_k) := (\mu_k(\mathcal{L}_Q, \mathsf{L}^2(\mathbb{Q})), \phi_k(\mathcal{L}_Q, \mathsf{L}^2(\mathbb{Q})))$, define the spectral tail $\sigma^2_{\mathsf{G}}(m,\mathcal{A}) := \sum_{j=m+1}^{\infty} \mu_j(\mathcal{A},\mathsf{G})$ for $m \geq 0$, and also write $\sigma^2(m,\mathcal{A},\mathsf{G})$ to specify the ambient space. For a positive trace-class operator $\mathcal{C}$ on G, $\mathrm{N}_{\mathsf{G}}(0,\mathcal{C})$ denotes the centered Gaussian element in G with covariance operator $\mathcal{C}$. Finally, we write $a \lesssim b$ if $a \leq Cb$ for a universal constant $C > 0$, and $a \asymp b$ if both $a \lesssim b$ and $b \lesssim a$ hold; the empirical average is denoted by $\mathbb{E}_n[g] = \mathbb{E}_{\mathbb{P}_n}[g] = n^{-1}\sum_{i=1}^n g_i$.

2 Problem Formulation and Preliminaries

2.1 Confidence balls in target distribution

A pair of random variables $(X, Y) \in \mathfrak{X} \times \mathbb{R}$ satisfies

$$Y = f_*(X) + \varepsilon, \tag{2.1}$$

where X and Y denote the covariate and response, respectively, and the regression function f_* is unknown. The random error ε carries no systematic information in the sense that $\mathbb{E}[\varepsilon|X] = 0$.

Let $(X, Y), \{(X_i, Y_i)\}_{i=1}^n \in \mathfrak{X} \times \mathbb{R}$ be i.i.d. observations, where the covariate X is drawn from a source distribution $\mathbb{P}$ on $\mathfrak{X}$. We aim to construct a confidence set for f_* in the $\mathsf{L}^2(\mathbb{Q})$ geometry. Given an $\mathsf{L}^2(\mathbb{Q})$-norm consistent estimator $\widehat{f}$, a natural confidence set is the $\mathsf{L}^2(\mathbb{Q})$-norm ball centered at $\widehat{f}$, with a radius $\hat{c}_{\alpha,n}$ calibrated using the quantiles of $\|\widehat{f} - f_*\|_{\mathsf{L}^2(\mathbb{Q})}$. More precisely, the inferential goal is to construct the $100(1-\alpha)\%$ confidence set

$$\mathsf{CS}_\alpha(\widehat{f}) = \Big\{ f : \|\widehat{f} - f\|_{\mathsf{L}^2(\mathbb{Q})} \le \hat{c}_{\alpha,n} \Big\}, \tag{2.2}$$

with asymptotically valid coverage probability; that is, $\mathbb{P}\big(f_* \in \mathsf{CS}_\alpha(\widehat{f})\big) \to 1-\alpha$ as $n \to \infty$. We suppress the dependence of $\widehat{f}$ and $\mathsf{CS}_\alpha(\widehat{f})$ on the sample size n for simplicity. Once a valid confidence set is available, testing the hypothesis in (1.1) is straightforward. For example, one can reject the null hypothesis at significance level α if $f_0 \notin \mathsf{CS}_\alpha(\widehat{f})$.

We emphasize that the choice of metric is part of the inferential goal, rather than merely a technical convention. A confidence set contains precisely those candidate functions that are close to the point estimator in the chosen geometry. Consequently, different metrics may lead to different confidence sets. There are three natural metrics in KRR: the reproducing kernel Hilbert space (RKHS) norm, the $\mathsf{L}^2(\mathbb{P})$-norm induced by the source distribution, and the $\mathsf{L}^2(\mathbb{Q})$-norm induced by the target distribution. The RKHS norm measures discrepancy in the smoothness geometry imposed by the kernel, whereas the $\mathsf{L}^2(\mathbb{P})$- and $\mathsf{L}^2(\mathbb{Q})$-norms weight pointwise errors according to the source and target distributions, respectively. Under covariate shift, regions that are important under $\mathbb{Q}$ may have little mass under $\mathbb{P}$. Hence, a function can be close to f_* in $\mathsf{L}^2(\mathbb{P})$ but not in $\mathsf{L}^2(\mathbb{Q})$.

2.2 Reproducing kernel Hilbert space and integral operators

Let H be a reproducing kernel Hilbert space (RKHS, Schölkopf and Smola [2002]) over $\mathfrak{X}$ with positive definite kernel $k(\cdot,\cdot) : \mathfrak{X} \times \mathfrak{X} \mapsto \mathbb{R}$. Throughout, we assume that $\mathfrak{X}$ is compact. More precisely, H is the Hilbert space of functions on $\mathfrak{X}$ obtained as the completion of the linear span of $\{k_x := k(x,\cdot)\}_{x\in\mathfrak{X}}$. For any $\{x_i\}_{i=1}^p, \{\tilde{x}_j\}_{j=1}^q \subset \mathfrak{X}$, consider functions of the form $f(\cdot) = \sum_{i=1}^p a_i k(x_i,\cdot)$ and $g(\cdot) = \sum_{j=1}^q b_j k(\tilde{x}_j,\cdot)$. Their inner product in H is given by $\langle f, g\rangle_{\mathsf{H}} = \sum_{i=1}^p \sum_{j=1}^q a_i b_j k(x_i, \tilde{x}_j)$.

For any $f \in \mathsf{H}$, the reproducing property holds: $f(x) = \langle f, k(x,\cdot)\rangle_{\mathsf{H}}$, for $\forall x \in \mathfrak{X}$. Throughout the paper, we assume that the kernel $k(\cdot,\cdot)$ is *bounded*, namely, $\sup_{x\in\mathfrak{X}} k(x,x) \le \kappa^2$ for some constant $\kappa > 1$. This immediately implies that point evaluation is bounded by the RKHS norm. Indeed, for any $f \in \mathsf{H}$, we have $\sup_x |f(x)| = \sup_x \big|\langle f, k(x,\cdot)\rangle_{\mathsf{H}}\big| \le \|f\|_{\mathsf{H}} \cdot \sup_x \|k(x,\cdot)\|_{\mathsf{H}} \le \kappa \|f\|_{\mathsf{H}}$.

The *kernel integral operator* is central to the analysis of kernel methods, as its spectral structure encodes both the geometry of the data distribution and the effective complexity of the induced feature space. Given the probability measure $\mathbb{Q}$, define the integral operator $\mathcal{L}_Q : \mathsf{L}^2(\mathbb{Q}) \to \mathsf{L}^2(\mathbb{Q})$ by

$$\mathcal{L}_Q f = \int_{\mathfrak{X}} k(x,\cdot) f(x) d\mathbb{Q}(x), \quad \forall f \in \mathsf{L}^2(\mathbb{Q}).$$

Note that the integral operator $\mathcal{L}_Q$ is closely related to the RKHS *covariance operator* $\Sigma_{\mathbb{Q}} = \mathbb{E}_{\mathbb{Q}}[k_X \otimes_{\mathsf{H}} k_X] : \mathsf{H} \to \mathsf{H}$, where, for $g, h \in \mathsf{H}$, the rank-one operator $g \otimes_{\mathsf{H}} h : \mathsf{H} \to \mathsf{H}$ is defined by $(g \otimes_{\mathsf{H}} h) f = \langle h, f\rangle_{\mathsf{H}} g$, $\forall f \in \mathsf{H}$.

By the reproducing property, we have $\Sigma_{\mathbb{Q}} f = \int_{\mathfrak{X}} k(x,\cdot) f(x) d\mathbb{Q}(x)$, $\forall f \in \mathsf{H}$. Under the bounded-kernel assumption, both $\mathcal{L}_Q$ and $\Sigma_{\mathbb{Q}}$ are bounded, positive, self-adjoint, and trace-class operators.

Since these two operators are given by the same integral expression but act on different spaces, we abuse notation and write $\mathcal{L}_Q$ for both whenever the underlying domain is clear from context.

The spectral decomposition of $\mathcal{L}_Q$ is often used to characterize the kernel $k(\cdot,\cdot)$, the distribution $\mathbb{Q}$, and the associated RKHS H.

Lemma 2.1. *[Mercer [1909]] Suppose that $\mathfrak{X}$ is compact, $k(\cdot,\cdot)$ is continuous, symmetric, and positive definite, $\mathbb{Q}$ is a Borel probability measure on $\mathfrak{X}$, and $\operatorname{supp}(\mathbb{Q}) = \mathfrak{X}$. Then there exist nonnegative eigenvalues $\{\mu_j\}_{j\geq 1}$ and corresponding eigenfunctions $\{\phi_j\}_{j\geq 1} \subset \mathsf{L}^2(\mathbb{Q})$, with $\langle \phi_i, \phi_j\rangle_{\mathsf{L}^2(\mathbb{Q})} = \delta_{ij}$ and $k(x,y) = \sum_{j=1}^{\infty} \mu_j \phi_j(x)\phi_j(y)$, $x, y \in \mathfrak{X}$. The convergence is absolute and uniform on $\mathfrak{X}\times\mathfrak{X}$.*

Mercer's theorem immediately implies several useful identities. For $\mu_i, \mu_j > 0$, $\langle \phi_i, \phi_j\rangle_{\mathsf{H}} = \mu_i^{-1}\delta_{ij}$. Thus, $\{\mu_j^{1/2}\phi_j : \mu_j > 0\}_{j=1}^{\infty}$ is an orthonormal basis of H. In particular, if $f = \sum_{j=1}^{\infty} a_j\phi_j$, then $\|f\|_{\mathsf{H}}^2 = \sum_{j:\mu_j>0} a_j^2\mu_j^{-1}$.

The fractional powers of $\mathcal{L}_Q$ are defined via the spectral theorem. For $r > 0$, define the operator $\mathcal{L}_Q^r : \mathsf{L}^2(\mathbb{Q}) \to \mathsf{L}^2(\mathbb{Q})$ by

$$\mathcal{L}_Q^r f = \sum_{j\geq 1} \mu_j^r \langle f, \phi_j\rangle_{\mathsf{L}^2(\mathbb{Q})}\phi_j, \quad f \in \mathsf{L}^2(\mathbb{Q}).$$

Although $\mathcal{L}_Q^{1/2}$ is defined as an operator on $\mathsf{L}^2(\mathbb{Q})$, its range admits a natural RKHS interpretation. Define $\operatorname{Ran}(\mathcal{L}_Q) = \big\{\sum_{j:\mu_j>0} a_j\phi_j : \sum_{j:\mu_j>0} a_j^2\mu_j^{-2} < \infty\big\}$. Indeed, if $f \in \overline{\operatorname{Ran}(\mathcal{L}_Q)}$, then $\big\|\mathcal{L}_Q^{1/2} f\big\|_{\mathsf{H}}^2 = \sum_{j:\mu_j>0}\langle f, \phi_j\rangle_{\mathsf{L}^2(\mathbb{Q})}^2 = \|f\|_{\mathsf{L}^2(\mathbb{Q})}^2$. Hence $\mathcal{L}_Q^{1/2}$ can be viewed as an isometry from $\overline{\operatorname{Ran}(\mathcal{L}_Q)}$ to H.

Lemma 2.2. *The operator $\mathcal{L}_Q^{1/2} : \overline{\operatorname{Ran}(\mathcal{L}_Q)} \to \mathsf{H}$ is a unitary isomorphism onto H. In particular, it preserves inner products: $\langle f, g\rangle_{\mathsf{L}^2(\mathbb{Q})} = \langle \mathcal{L}_Q^{1/2} f, \mathcal{L}_Q^{1/2} g\rangle_{\mathsf{H}}$, for $f, g \in \overline{\operatorname{Ran}(\mathcal{L}_Q)}$.*

We can similarly define the integral operator and covariance operator by the probability measure $\mathbb{P}$ on $\mathfrak{X}$, denoted both by $\mathcal{L}_P$ for notational simplicity. These notations can be generalized to the sampling distribution $\mathbb{P}_n$, defined from the i.i.d. sample $X_1, \cdots, X_n \sim \mathbb{P}$. For instance, we write the integral operator

$$\mathcal{L}_{P_n} f = \frac{1}{n}\sum_{i=1}^{n} f(X_i) k_{X_i}, \quad f \in \mathsf{L}^2(\mathbb{P}),$$

which may also represent the *sample covariance operator* $f \mapsto n^{-1}\sum_{i=1}^n (k_{X_i} \otimes_{\mathsf{H}} k_{X_i}) f, \forall f \in \mathsf{H}$.

Let $w : \mathfrak{X} \to (0,\infty)$ be a weight function. Define the weighted measure $\mathbb{P}^w$ on $\mathfrak{X}$ by $\mathbb{P}^w(\mathcal{A}) = \int_{\mathcal{A}} w(x)\, d\mathbb{P}(x), \mathcal{A} \subseteq \mathfrak{X}$. For instance, if w is the target-to-source density ratio $\rho(x) = (d\mathbb{Q}/d\mathbb{P})(x)$, then $\mathbb{P}^w = \mathbb{Q}$, assuming that $\mathbb{Q} \ll \mathbb{P}$. The measure $\mathbb{P}^w$ and its empirical counterpart induce the following integral operators:

$$\mathcal{L}_{P^w} f := \int_{\mathfrak{X}} k_x f(x) w(x)\, d\mathbb{P}(x), \quad \text{and} \quad \mathcal{L}_{P_n^w} f := \frac{1}{n}\sum_{i=1}^{n} w(X_i) k_{X_i} f(X_i).$$

2.3 Importance weighted kernel ridge regression

Given the i.i.d. sample $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n \subset \mathfrak{X}\times\mathbb{R}$, the kernel ridge regression (KRR) seeks a function $f \in \mathsf{H}$ that minimizes the empirical risk $n^{-1}\sum_{i=1}^n \big(f(X_i) - Y_i\big)^2$ with an RKHS-norm

penalization $\lambda_n \|f\|_{\mathsf{H}}^2$, where $\lambda_n > 0$ is the regularization parameter controlling the trade-off between fitting the data and model complexity. The regularization prevents KRR from overfitting, especially when H is dense in the space of continuous functions. The error rates and optimality of KRR are well understood. See Wahba [1990], Caponnetto and De Vito [2007], Vapnik [2013] and references therein.

Importance weighting (IW, Sugiyama et al. [2007]) is a technique for unbiased estimation of target risk under covariate shift. In particular, the importance-weighted kernel ridge regression (IWKRR) estimator is defined by

$$\widehat{f} := \arg\min_{f \in \mathsf{H}} \ \frac{1}{n} \sum_{i=1}^{n} w_n(X_i)\{Y_i - f(X_i)\}^2 + \lambda_n \|f\|_{\mathsf{H}}^2, \tag{2.3}$$

where $w_n : \mathfrak{X} \to (0, \infty)$ is a weight function. If equipped with a proper weight function, IW transports the source risk to the target risk. For instance, using the density ratio weight $w_n(x) = \rho(x)$, we get $\mathbb{E}_{\mathbb{P}}[w_n(X)(Y - f(X))^2] = \mathbb{E}_{\mathbb{Q}}[(Y - f(X))^2]$. This density-ratio weighting thus corrects the population mismatch at the level of the squared-error risk. For notational simplicity, we may write λ and $w(\cdot)$ and suppress their dependence on n.

However, the IWKRR estimator can have large finite-sample variances if the target-to-source density ratio is unbounded or has heavy tails, in which case a few observations carry excessively large weights and destabilize the estimator. A common remedy is to use the truncated density ratio $w_n(x) := \min\{\rho(x), \tau_n\}$, where $\tau_n > 0$ is the truncation level. The truncation introduces a controlled bias but reduces variance. Under suitable density ratio and spectral conditions, with appropriate choices of regularization parameter λ_n and truncation parameter τ_n, the truncated IWKRR attains the minimax optimal rate for the target $\mathsf{L}^2(\mathbb{Q})$-norm risk. See Ma et al. [2023] and Section 4 for detailed discussions.

In this paper, we aim to construct the confidence ball of f_* with respect to the $\mathsf{L}^2(\mathbb{Q})$-norm distance based on the truncated IWKRR estimator (2.3). We begin with presenting the explicit matrix form of the IWKRR estimator. Denote the $n \times n$ matrix $\mathbf{K}$ with its (i,j)-th entry $k(X_i, X_j)$, the weight matrix $\mathbf{W} = \mathrm{diag}\big(w_n(X_1), \cdots, w_n(X_n)\big)$, and $\mathbf{y} = (Y_1, \cdots, Y_n)^\top$. Then, the IWKRR estimator can be compactly written as

$$\widehat{f} = \mathbf{k}_{\mathbf{x}}^\top (\mathbf{W}\mathbf{K} + n\lambda_n \mathbf{I})^{-1} \mathbf{W}\mathbf{y} = \mathbf{k}_{\mathbf{x}}^\top (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1} \mathbf{y}, \tag{2.4}$$

where the vector function $\mathbf{k}_{\mathbf{x}} = \big(k(X_1, \cdot), \cdots, k(X_n, \cdot)\big)^\top$ and the second equation holds provided that $\mathbf{W}$ is nonsingular.

The IWKRR estimator is biased from two sources: the RKHS-norm regularization and the truncated density-ratio weight. The population IWKRR estimator is

$$f_{P^w} = \arg\min_{f \in \mathsf{H}} \ \mathbb{E}_{\mathbb{P}}\left[w_n(X)(f(X) - Y)^2\right] + \lambda_n \|f\|_{\mathsf{H}}^2 = (\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1} \mathcal{L}_{P^w} f_*. \tag{2.5}$$

If the weight $w_n(x) = (d\mathbb{Q}/d\mathbb{P})(x)$ without truncation, the population IWKRR estimator reduces to

$$f_Q = \arg\min_{f \in \mathsf{H}} \mathbb{E}_{\mathbb{Q}}\left[(f(X) - Y)^2\right] + \lambda_n \|f\|_{\mathsf{H}}^2 = (\mathcal{L}_Q + \lambda_n \mathcal{I})^{-1} \mathcal{L}_Q f_*, \tag{2.6}$$

which minimizes the regularized risk in target distribution. The explicit formulas in (2.5) and (2.6) follow from the first-order optimality conditions for the two population objectives. Finally, we remark that the IWKRR estimator (2.4) can be explicitly written as $\widehat{f} = (\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1} \left(n^{-1} \mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\mathbf{y}\right)$.

3 Inference Procedure

3.1 Bias-variance decomposition and Bahadur representation

We begin with the bias-variance decomposition of $\widehat{f}$ and derive its Bahadur representation. Notice that $\sqrt{n}(\widehat{f} - f_*) = \sqrt{n}(f_{P^w} - f_*) + \sqrt{n}(\widehat{f} - f_{P^w})$, where the first and second terms capture the bias and variance, respectively. Moreover, the bias term can be explicitly written as

$$\sqrt{n}(f_{P^w} - f_*) = -\sqrt{n}\lambda_n(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1} f_*,$$

implying that the bias term can be estimated by $-\sqrt{n}\lambda_n(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1}\widehat{f}$. However, we will show in Section 4 that the bias term is asymptotically negligible if λ_n is small, i.e., under-smoothing IWKRR. Our bootstrap inference procedure therefore focuses on the variance term $\sqrt{n}(\widehat{f} - f_{P^w})$. Nevertheless, we also provide a bias-corrected inference procedure and numerical experiments in Appendix B.

The second term, $\sqrt{n}(\widehat{f} - f_{P^w})$, captures the stochastic fluctuation of the IWKRR estimator around its population counterpart. We show in Section 4 that its leading term, in $\mathsf{L}^2(\mathbb{Q})$-norm, admits the Bahadur representation:

$$\sqrt{n}(\widehat{f} - f_{P^w}) \approx_{\mathsf{L}^2(\mathbb{Q})} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} U_i, \tag{3.1}$$

where $U_i := (\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1}\big(w_n(X_i)\varepsilon_i k_{X_i} + \big(w_n(X_i)(k_{X_i} \otimes_{\mathsf{H}} k_{X_i}) - \mathcal{L}_{P^w}\big)(f_* - f_{P^w})\big)$. It suggests that $\sqrt{n}(\widehat{f} - f_{P^w})$ can be approximated by a centered Gaussian process in $\mathsf{L}^2(\mathbb{Q})$ with the covariance operator $\mathcal{V} = \mathbb{E}[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i]$, and Gaussian multiplier bootstrap can be used to calibrate the radius of $\mathsf{L}^2(\mathbb{Q})$-norm confidence balls for f_*.

3.2 Multiplier bootstrap

The random element U_i is not directly applicable because it relies on the unknown f_* and the population IWKRR estimator f_{P^w}. Nevertheless, the Bahadur representation (3.1) motivates us to consider the *oracle* bootstrap function, defined by the anti-symmetric multiplier process:

$$B_{\mathsf{ora}} := \frac{1}{n} \sum_{1 \le i \ne j \le n} h_{ij} \cdot \frac{U_i - U_j}{\sqrt{2}} = \frac{1}{n} \sum_{1 \le i \ne j \le n} h_{ij} \cdot \frac{V_i - V_j}{\sqrt{2}},$$

where $h_{ij} \overset{\text{i.i.d.}}{\sim} N(0,1)$ are independent of the data $\mathcal{D}$ and

$$V_i := U_i + (\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1}\mathcal{L}_{P^w}(f_* - f_{P^w}) = w_n(X_i)\big(Y_i - f_{P^w}(X_i)\big)(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1} k_{X_i}.$$

Conditioned on $\mathcal{D}$, the oracle bootstrap function B_{ora} is a centered Gaussian process in $\mathsf{L}^2(\mathbb{Q})$ with covariance operator $\widehat{\mathcal{V}}_{\mathsf{ora}} = \mathbb{E}_{\mathbb{P}_n}[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i] - \mathbb{E}_{\mathbb{P}_n}[U_i] \otimes_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}_{\mathbb{P}_n}[U_i]$, which is the sample version of the covariance operator $\mathcal{V} = \mathbb{E}[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i]$ defined after (3.1). The conditional covariance operator of B_{ora} can be equivalently written as $\widehat{\mathcal{V}}_{\mathsf{ora}} = \mathbb{E}_{\mathbb{P}_n}[V_i \otimes_{\mathsf{L}^2(\mathbb{Q})} V_i] - \mathbb{E}_{\mathbb{P}_n}[V_i] \otimes_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}_{\mathbb{P}_n}[V_i]$.

The oracle bootstrap function B_{ora} remains unavailable in practice, due to the unknown function f_{P^w}. By plugging in the empirical counterparts, we define

$$\widehat{V}_i := w_n(X_i)\big(Y_i - \widehat{f}(X_i)\big)(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1} k_{X_i} = w_n(X_i)\widehat{\varepsilon}_i(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1} k_{X_i},$$

which leads to the computable multiplier bootstrap function:

$$B := \frac{1}{n} \sum_{1 \le i \ne j \le n} h_{ij} \cdot \frac{\widehat{V}_i - \widehat{V}_j}{\sqrt{2}} = \sqrt{n}\, \mathbf{k}_{\mathbf{x}}^\top (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1} \mathrm{diag}(\widehat{\boldsymbol{\varepsilon}}) \mathbf{z}, \tag{3.2}$$

with $\mathbf{h} := (h_{ij})_{n\times n}$ and $\mathbf{z} := (2n)^{-1/2}(\mathbf{h} - \mathbf{h}^\top)\mathbf{1}_n \sim N(0, \mathbf{I}_n - \mathbf{1}\mathbf{1}^\top/n)$. Similarly, conditioned on data, B is a centered Gaussian process in $\mathsf{L}^2(\mathbb{Q})$ with covariance operator $\widehat{\mathcal{V}} := \mathbb{E}_{\mathbb{P}_n}[\widehat{V}_i \otimes_{\mathsf{L}^2(\mathbb{Q})} \widehat{V}_i] - \mathbb{E}_{\mathbb{P}_n}[\widehat{V}_i] \otimes_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}_{\mathbb{P}_n}[\widehat{V}_i]$.

Fast computation of $\mathsf{L}^2(\mathbb{Q})$-norm. We use bootstrap percentiles of $n^{-1/2}\|B\|_{\mathsf{L}^2(\mathbb{Q})}$ to calibrate the radius of confidence balls. It is computationally expensive to evaluate these norms by definition. Fortunately, B is a linear combination of functions k_{X_i}'s, which permits fast computation of $\|B\|^2_{\mathsf{L}^2(\mathbb{Q})}$. Indeed, given the sampled vector $\mathbf{z}$, we have $B = \sqrt{n}\,\mathbf{k}_{\mathbf{x}}^\top \mathbf{a}$ with $\mathbf{a} = (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\mathrm{diag}(\widehat{\boldsymbol{\varepsilon}})\mathbf{z}$. Then, $\|B\|^2_{\mathsf{L}^2(\mathbb{Q})} = n\|\mathbf{k}_{\mathbf{x}}^\top \mathbf{a}\|^2_{\mathsf{L}^2(\mathbb{Q})} = n\mathbf{a}^\top \mathbf{K}_{\mathbb{Q}} \mathbf{a}$, where $\mathbf{K}_{\mathbb{Q}}$ is an $n \times n$ matrix with its (i,j)-th entry being $(\mathbf{K}_{\mathbb{Q}})_{ij} = \langle k_{X_i}, k_{X_j} \rangle_{\mathsf{L}^2(\mathbb{Q})}$. Since the matrix $\mathbf{K}_{\mathbb{Q}}$ is fixed throughout bootstrap repetitions, computing $\|B\|^2_{\mathsf{L}^2(\mathbb{Q})}$ amounts to evaluating a quadratic form.

Algorithm 1 Multiplier bootstrap for $\mathsf{L}^2(\mathbb{Q})$-norm confidence balls

Require: Data $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$, kernel $k(\cdot,\cdot)$, density ratio $\rho(x) = q(x)/p(x)$, weight $w_n(x) = \min\{\rho(x), \tau_n\}$, confidence level $1-\alpha$, bootstrap repetitions M, regularization parameter λ_n;

1: Form the matrix $\mathbf{K} = \big(k(X_i, X_j)\big)_{n\times n}$, the weight matrix $\mathbf{W} = \mathrm{diag}(w_n(X_1), \ldots, w_n(X_n))$, the vector $\mathbf{y} = (Y_1, \ldots, Y_n)^\top$, and the vector function $\mathbf{k}_{\mathbf{x}} = \big(k(X_1,\cdot), \cdots, k(X_n,\cdot)\big)^\top$;
2: Compute the IWKRR estimator $\widehat{f} = \mathbf{k}_{\mathbf{x}}^\top (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\mathbf{y}$;
3: Compute the residual vector $\widehat{\boldsymbol{\varepsilon}} = \big(Y_1 - \widehat{f}(X_1), \ldots, Y_n - \widehat{f}(X_n)\big)^\top$;
4: Form the $n \times n$ matrix $\mathbf{K}_{\mathbb{Q}}$ with its (i,j)-th entry being $(\mathbf{K}_{\mathbb{Q}})_{ij} = \langle k_{X_i}, k_{X_j} \rangle_{\mathsf{L}^2(\mathbb{Q})}$;
5: **for** $b = 1, \ldots, M$ **do**
6: Sample $\mathbf{z}^{(b)} \sim N(0, \mathbf{I}_n - \mathbf{1}\mathbf{1}^\top/n)$.
7: Set the coefficient vector $\mathbf{a}^{(b)} = (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\mathrm{diag}(\widehat{\boldsymbol{\varepsilon}})\mathbf{z}^{(b)}$.
8: Compute the bootstrap statistic $T^{(b)} = \|\mathbf{k}_{\mathbf{x}}^\top \mathbf{a}^{(b)}\|_{\mathsf{L}^2(\mathbb{Q})} = \big((\mathbf{a}^{(b)})^\top \mathbf{K}_{\mathbb{Q}} \mathbf{a}^{(b)}\big)^{1/2}$.
9: **end for**
10: Let $\hat{c}_{\alpha,n}$ be the empirical $(1-\alpha)$%-percentile of $\{T^{(b)}\}_{b=1}^M$.
11: Define the confidence set $\mathsf{CS}_\alpha(\widehat{f}) = \big\{f \in \mathsf{H} : \|\widehat{f} - f\|_{\mathsf{L}^2(\mathbb{Q})} \le \hat{c}_{\alpha,n}\big\}$.

Ensure: the IWKRR $\widehat{f}$ and the radius $\hat{c}_{\alpha,n}$.

The multiplier bootstrap procedure is detailed in Algorithm 1. Let $\widehat{f}$ and $\hat{c}_{\alpha,n}$ be the IWKRR estimator and radius output by Algorithm 1. The resulting $100(1-\alpha)\%$ confidence ball is

$$\mathsf{CS}_\alpha(\widehat{f}) = \left\{ f \in \mathsf{H} : \|\widehat{f} - f\|_{\mathsf{L}^2(\mathbb{Q})} \le \hat{c}_{\alpha,n} \right\}. \tag{3.3}$$

The computational cost of (IW)KRR grows fast when the sample size n increases. Fortunately, the multiplier bootstrap Algorithm 1 requires solving IWKRR only once, and the subsequent bootstrap repetitions involve just matrix-vector multiplications. We remark that the residual bootstrap [Hall, 2013] is also applicable for constructing the $\mathsf{L}^2(\mathbb{Q})$-norm confidence balls. See Appendix C for details.

4 Theoretical Guarantees

This section establishes the asymptotic validity of the bootstrap confidence balls (3.3). The proof mainly involves four components. First, we introduce key covariate-shift *compatibility conditions* connecting the target $\mathsf{L}^2(\mathbb{Q})$ geometry with the regularized truncated weighted-source geometry $\mathsf{L}^2(\mathbb{P}^w)$. Second, we provide a Bahadur representation of the IWKRR estimator $\widehat{f}$ and use Gaussian coupling to approximate the stochastic fluctuation of $\widehat{f}$ by a Gaussian element in $\mathsf{L}^2(\mathbb{Q})$. Third, a multiplier bootstrap function is developed to couple with the same Gaussian element. Finally, we establish the coverage-error bounds for the bootstrap confidence balls using Gaussian approximation and anti-concentration.

4.1 Assumptions and compatibility conditions

4.1.1 Statistical and spectral assumptions

We first collect the statistical and spectral assumptions needed for Bahadur representation and Gaussian approximation under covariate shift and truncated importance weighting.

Assumption 4.1 (uniformly bounded noise)**.** *There exist constants* $0 < \underline{\sigma} \le \overline{\sigma} < \infty$ *such that*

$$|\varepsilon| \le \overline{\sigma} \quad \text{and} \quad \mathbb{E}[\varepsilon^2 \mid X] \ge \underline{\sigma}^2 \quad \textit{almost surely.}$$

For technical convenience, we assume the random error is uniformly bounded. It is possible to generalize the main theorems to sub-Gaussian noise, but we leave it for future work. The major challenge is to control the bias and variance induced by truncated importance weighting.

Assumption 4.2 (source condition)**.** *There exist* $s \in [0,1]$, $\tau_0 > 0$, *and* $g_* \in \mathsf{H}$ *such that*

$$f_* = \mathcal{L}^s_{P^{w_0}}(g_*), \text{ where } \mathcal{L}_{P^{w_0}}(g) = \mathbb{E}_{\mathbb{P}}\big[w_0(X)g(X)k_X\big] \text{ with } w_0(X) = \min\big\{\rho(X), \tau_0\big\}.$$

The source condition is typical for KRR that characterizes the smoothness of f_* relative to the kernel integral operator. Assumption 4.2 differs slightly from the standard ones in KRR literature due to the truncated importance weighting. See, e.g., Caponnetto and De Vito [2007], Ma et al. [2023]. The larger smoothness parameter s ensures sharper control on the $\mathsf{L}^2(\mathbb{Q})$-norm bias and the RKHS-norm quantities appearing in the bootstrap approximation. Note that we assume the truncation level used in Algorithm 1 satisfies $\tau_n \ge \tau_0$ so that $\mathcal{L}_{P^{w_0}} \preceq_{\mathsf{H}} \mathcal{L}_{P^w}$.

Assumption 4.3 (effective dimension)**.** *There exists* $d \in [0,1]$ *such that*

$$E_d := \max\Big\{1, \sup_{\lambda\in(0,1]} \big(\mathcal{N}_\lambda\, \lambda^d\big)^{1/2}\Big\} < \infty, \text{ where } \mathcal{N}_\lambda := \mathrm{tr}_{\mathsf{H}}\big((\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q\big)\,.$$

Assumption 4.3 always holds with $d = 1$ under the bounded-kernel condition, since $\mathrm{tr}_{\mathsf{H}}(\mathcal{L}_Q) = \sum_i \mu_i \le \kappa^2$ by Lemma 2.1. If $d \in (0,1)$, Assumption 4.3 holds if the eigenvalues of $\mathcal{L}_Q$ decrease fast enough, for instance, $\mu_i = O(i^{-1/d})$; see Caponnetto and De Vito [2007] for more discussions.

4.1.2 Compatibility conditions under covariate shift

The covariate shift and truncated importance weighting require compatibility conditions on the integral operators defined on the measure $\mathbb{Q}$ and $\mathbb{P}^w$. The two compatibility conditions play distinct roles. The first compatibility condition ensures the *regularized transfer coverage* so that the source-side risk and RKHS-norm regularization control the target-side risk. The second compatibility condition guarantees a *non-degenerate covariance* in the target distribution, which essentially prevents the Gaussian covariance from degenerating in target geometry and is generally necessary for anti-concentration properties.

Assumption 4.4 (regularized transfer coverage)**.** *There exists a constant $\mathfrak{C}_1 \geq 1$, independent of n, such that $\left\|\mathcal{L}_Q^{1/2}(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1/2}\right\|_{\mathsf{H}\to\mathsf{H}} \leq \mathfrak{C}_1$, for the chosen τ_n and λ_n.*

Equivalently, Assumption 4.4 requires the *regularized operator domination*: $\mathcal{L}_Q \preceq_{\mathsf{H}} \mathfrak{C}_1^2(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})$, or, in quadratic-form notation, $\|f\|^2_{\mathsf{L}^2(\mathbb{Q})} \leq \mathfrak{C}_1^2\big(\mathbb{E}_{\mathbb{P}}\left[w_n(X) f(X)^2\right] + \lambda_n \|f\|^2_{\mathsf{H}}\big)$ for every $f \in \mathsf{H}$. Moreover, by Lemma 2.2, the operator norm in Assumption 4.4 admits the equivalent variational representation

$$\left\|\mathcal{L}_Q^{1/2}(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1/2}\right\|^2_{\mathsf{H}\to\mathsf{H}} = \sup_{f\in\mathsf{H},\, f\neq 0} \frac{\|f\|^2_{\mathsf{L}^2(\mathbb{Q})}}{\mathbb{E}_{\mathbb{P}}[w_n(X) f(X)^2] + \lambda_n \|f\|^2_{\mathsf{H}}}.$$

Therefore, Assumption 4.4 essentially states that the weighted source distribution $\mathbb{P}^w$ covers the target distribution $\mathbb{Q}$ in all kernel directions, up to the regularization level λ_n. If the constant $\mathfrak{C}_1$ is large, there exist directions that have small regularized size in the weighted source distribution but large size in the target distribution, which may lead to unstable transfer from source distribution to the target geometry. The constant $\mathfrak{C}_1$ measures the residual mismatch after importance weighting, truncation, and regularization; it is therefore determined jointly by the distribution shift, the kernel geometry, and the choices of τ_n and λ_n, rather than by the density ratio alone.

Remark 1 (sufficient conditions of Assumption 4.4)**.** *The following three typical density-ratio conditions are sufficient to ensure Assumption 4.4. Recall the truncated weight function $w_n(x) = \min\{\rho(x), \tau_n\}$.*

(i) Bounded density ratio. Suppose that $\rho(X) \leq \bar{\rho}$ $\mathbb{P}$-almost surely. If $\tau_n \leq \bar{\rho}$, then $\rho(x) \leq (\bar{\rho}/\tau_n) w_n(x)$. Therefore, $\mathcal{L}_Q \preceq_{\mathsf{H}} (\bar{\rho}/\tau_n)\mathcal{L}_{P^w} \preceq_{\mathsf{H}} (\bar{\rho}/\tau_n)(\mathcal{L}_{P^w} + \lambda\mathcal{I})$, and Assumption 4.4 holds with $\mathfrak{C}_1 \leq (\bar{\rho}/\tau_n)^{1/2}$. If $\tau_n > \bar{\rho}$, then $w_n(x) = \rho(x)$. Then, $\mathcal{L}_{P^w} = \mathcal{L}_Q$, and Assumption 4.4 holds with $\mathfrak{C}_1 = 1$. Combining the two regimes gives $\mathfrak{C}_1 \leq \max\left\{1, (\bar{\rho}/\tau_n)^{1/2}\right\}$.

(ii) Sub-exponential density ratio. The bounded density ratio condition is stringent. Assumption 4.4 still holds if the density ratio has light tails.

Definition 4.1 (Bernstein-type moment condition)**.** *There exist $\gamma \geq 1, W > 0$, and $\sigma > 0$ such that, for every integer $m \geq 2$, $\left(\mathbb{E}_{\mathbb{Q}}\left[\rho(X)^{\gamma(m-1)}\right]\right)^{1/\gamma} \leq (m!/2) W^{m-2}\sigma^2$.*

The density ratio may be unbounded, but has light tails under Definition 4.1. Nevertheless, Lemma F.29 in the Appendix verifies Assumption 4.4 with $\mathfrak{C}_1 \leq \sqrt{2}$, if the truncation level

$$\tau_n \geq \left(2^{2\gamma-1}\kappa^{2\gamma} E_d^{2\gamma} m_0! W^{m_0-2}\sigma^2\right)^{\frac{1}{m_0-1}} \lambda^{-\frac{(1+d)\gamma}{m_0-1}},$$

where $m_0 \geq 2$ is some integer and E_d is the effective dimension defined in Assumption 4.3.

(iii) Heavy-tailed density ratio. Even if the density ratio has heavy tails, Assumption 4.4 still holds if the truncation level is properly chosen.

Definition 4.2 (Bounded θ-th moment)**.** *For a $\theta \geq 2$, there exists some $\Omega < \infty$ such that $\mathbb{E}_{\mathbb{P}}\left[\rho(X)^{\theta}\right] \leq \Omega$. If $\theta = 2$, then $\chi^2(\mathbb{Q} \,\|\, \mathbb{P}) \leq \Omega - 1$.*

Lemma F.31 in the Appendix verifies Assumption 4.4 with $\mathfrak{C}_1 \leq \sqrt{2}$ for $\theta > 2$, if $\tau_n^{(\theta-1)} \geq 4\kappa^2 \Omega E_d^2 \lambda^{-(1+d)}$, where E_d is the effective dimension defined in Assumption 4.3.

Note that similar comparison inequality between integral operators already appeared in the literature to quantify the information transferred from the source distribution to the target task. In particular, Wang [2026] proposes an effective sample size defined by

$$n_{\text{eff}} := \sup\left\{t \leq n : \mathcal{L}_Q \preceq_{\mathsf{H}} \frac{n}{t}\left(\mathcal{L}_P + \tilde{\lambda}\mathcal{I}\right)\right\},$$

with a prescribed regularization level $\tilde{\lambda}$. Equivalently, n/n_{eff} is the smallest $c \geq 1$ such that $\mathcal{L}_Q \preceq_{\mathsf{H}} c(\mathcal{L}_P + \tilde{\lambda}\mathcal{I})$. Suppose that $\tilde{\lambda} = \lambda_n$, the truncation $w_n(X) \geq \underline{\rho}$ for some $\underline{\rho} \in (0, 1]$ $\mathbb{P}$-almost surely, and $\tau_n \geq 1$. Since $\underline{\rho}(\mathcal{L}_P + \lambda_n\mathcal{I}) \preceq_{\mathsf{H}} \mathcal{L}_{P^w} + \lambda_n\mathcal{I} \preceq_{\mathsf{H}} \tau_n(\mathcal{L}_P + \lambda_n\mathcal{I})$, our Assumption 4.4 holds with $\mathfrak{C}_1$ satisfying $n/(n_{\text{eff}}\tau_n) \leq \mathfrak{C}_1^2 \leq n/(n_{\text{eff}}\underline{\rho})$. The effective sample size n_{eff} measures source-to-target compatibility in the regularized unweighted-source geometry, whereas $\mathfrak{C}_1$ measures the remaining mismatch after incorporating truncated importance weighting.

Remark 2 (other distribution shift measures)**.** *There exist other measures of source–target mismatch serving different purposes, which do not directly supply the domination inequality in Assumption 4.4. Hypothesis-class discrepancies compare distributions through the largest difference of expected losses over a chosen class; for example, the discrepancy distance of Mansour et al. [2009] is used to establish generalization bounds for domain adaptation. While the loss-reliant discrepancies provide convenient risk transfer from source to target distributions, they cannot reflect the impact of the truncated weights or the regularization level.*

Distribution-matching measures such as kernel mean matching [Huang et al., 2007] choose weights to match the kernel mean embeddings in target and weighted source distributions, i.e., controlling the first-order differences of the form $\mathbb{E}_{\mathbb{Q}} f(X) - \mathbb{E}_{\mathbb{P}}[w_n(X)f(X)]$. Local nonparametric measures, including the transfer exponent of Kpotufe and Martinet [2018] and the ball-probability ratio measure of Pathak et al. [2022], compare how source and target mass scale on metric balls and are designed to characterize local transfer rates for Lipschitz or Hölder function classes. In contrast, Assumption 4.4 goes beyond the source and target distributions: it is a kernel-, truncation-, and regularization-dependent domination condition for the specific RKHS geometry in which the IWKRR estimator resides.

In addition, we require a non-degenerate covariance condition for valid inference, expressed as compatibility between integral operators. This lower bound condition on the variance is typical in high-dimensional statistical inference where estimators are usually biased. See, e.g., Assumption 4 in Xia and Yuan [2021]. This non-degenerate covariance condition is imposed to establish anti-concentration of the norm of Gaussian element in $\mathsf{L}^2(\mathbb{Q})$.

Assumption 4.5 (non-degenerate covariance operator)**.** *There exists a constant $\underline{\rho} > 0$ such that $\rho(X) \geq \underline{\rho}$ $\mathbb{P}$-almost surely and $\tau_n \geq \underline{\rho}$. Moreover, there exists a constant $\mathfrak{C}_2 \geq 1$ such that $(\mathcal{L}_{P^w} + \lambda_n\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w} + \lambda_n\mathcal{I})^{-1} \succeq_{\mathsf{H}} \mathfrak{C}_2^{-2}(\mathcal{L}_Q + \lambda_n\mathcal{I})^{-1}\mathcal{L}_Q(\mathcal{L}_Q + \lambda_n\mathcal{I})^{-1}$ for the chosen τ_n, λ_n.*

The uniform lower bound on the density ratio is operationally less restrictive than a uniform upper bound, especially when source data are abundant. For instance, source observations in the low-ratio region $A_\delta = \{x : \rho(x) < \delta\}$ may be screened out entirely for a small $\delta > 0$. This region has target probability $\mathbb{Q}_X(A_\delta) = \mathbb{E}_{\mathbb{P}_X}\big[\rho(X)\mathbf{1}\{X \in A_\delta\}\big] \le \delta$. Alternatively, we can randomly retain source observations in this region with probability $a_\delta(x) := \min\{1, \rho(x)/\delta\}$. Denote by $\mathbb{P}_X^{(\delta)}$ the covariate distribution of the retained source sample. The resulting density ratio satisfies $\big(d\mathbb{Q}_X/d\mathbb{P}_X^{(\delta)}\big)(x) = Z_\delta\big(\rho(x)/a_\delta(x)\big) \ge Z_\delta\delta$, $\mathbb{P}_X^{(\delta)}$-almost surely, where $Z_\delta := \mathbb{E}_{\mathbb{P}_X}[a_\delta(X)]$. Thus, partial thinning of source-rich yet target-irrelevant regions can enforce a uniform lower bound while retaining some observations from those regions. By contrast, a uniform upper bound on $\rho(x)$ rules out target-important regions that are poorly represented by the source distribution, which is a more consequential restriction for target-domain inference.

Assumption 4.5 lower bounds the regularized covariance operator $(\mathcal{L}_{P^w} + \lambda_n\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w} + \lambda_n\mathcal{I})^{-1}$ by $(\mathcal{L}_Q + \lambda_n\mathcal{I})^{-1}\mathcal{L}_Q(\mathcal{L}_Q + \lambda_n\mathcal{I})^{-1}$, which prevents the Gaussian approximation from becoming degenerate in the target geometry and thereby supports anti-concentration. In contrast, the regularized transfer coverage condition in Assumption 4.4 enables transfer of estimation and bootstrap errors to the target norm. Neither condition implies the other in general.

4.2 General coverage guarantee of bootstrap confidence balls

Recall the bias-variance decomposition of IWKRR estimator $\sqrt{n}(\widehat{f} - f_*) = \sqrt{n}(\widehat{f} - f_{P^w}) + \sqrt{n}(f_{P^w} - f_*)$. The bootstrap confidence balls constructed by Algorithm 1 are valid if the quantiles of $\|B\|^2_{\mathsf{L}^2(\mathbb{Q})}$, conditioned on $\mathcal{D}$, approximate the quantiles of $n\|\widehat{f} - f_*\|^2_{\mathsf{L}^2(\mathbb{Q})}$. Broadly speaking, we need to study three approximations to establish the connection between these two quantiles:

- *Bias control of IWKRR.* There exists $\Delta_{\text{bias}} > 0$ such that the deterministic bias satisfies

$$\sqrt{n}\|f_{P^w} - f_*\|_{\mathsf{L}^2(\mathbb{Q})} \le \Delta_{\text{bias}}. \tag{4.1}$$

- *Gaussian approximation of IWKRR.* For any $\eta \in (0,1)$, there exist $\Delta_G > 0$ and $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ such that

$$P\Big(\big\|\sqrt{n}(\widehat{f} - f_{P^w}) - Z\big\|_{\mathsf{L}^2(\mathbb{Q})} \le \Delta_G\Big) \ge 1 - \eta. \tag{4.2}$$

- *Bootstrap approximation.* For any $\eta \in (0,1)$, there exist $\Delta_B > 0$ and $Z' \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$, conditionally coupled with B, such that, with probability at least $1 - \eta$ over the data $\mathcal{D}$,

$$P\big(\|B - Z'\|_{\mathsf{L}^2(\mathbb{Q})} \le \Delta_B \,\big|\, \mathcal{D}\big) \ge 1 - \eta. \tag{4.3}$$

The above approximation error terms Δ_G, Δ_B may depend on the probability tolerance η. The total approximation error is $\Delta := \Delta_G + \Delta_B + \Delta_{\text{bias}}$, which plays the primary role in the coverage guarantee of bootstrap confidence balls. For ease of exposition, we first present the coverage guarantees assuming that the inequalities (4.1)-(4.3) hold. Explicit bounds for Δ_G, Δ_B, and Δ_{bias} will be established in Section 4.3 for specific source-target and kernel spectral conditions.

Note that if the quantiles of $\|\widehat{f} - f_*\|^2_{\mathsf{L}^2(\mathbb{Q})}$ is discontinuous, establishing the $\mathsf{L}^2(\mathbb{Q})$-norm confidence balls may be infeasible. To rule out these ill-posed cases, we require that the distribution of

$\|\widehat{f} - f_*\|^2_{\mathsf{L}^2(\mathbb{Q})}$ does not concentrate too sharply near its quantiles. Due to the Gaussian approximation (4.2), it suffices to show that the density function of $\|Z\|_{\mathsf{L}^2(\mathbb{Q})}$ is uniformly bounded. Recall the covariate operator $\mathcal{V}$ defined in Section 3.1.

Lemma 4.1. *Suppose Assumptions 4.1 - 4.5 hold, $\tau_n \geq \tau_0$, $0 < \lambda_n \leq 1$, and $\mathcal{L}_Q$ has either polynomial-decay eigenvalues $\mu_i \asymp i^{-\beta}$ for some $\beta > 1$ or exponential-decay eigenvalues $\mu_i \asymp \exp(-\beta i^{\gamma})$ for some $\beta, \gamma > 0$. Let $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ with $\mathcal{V} = \mathbb{E}[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i]$. Then the density of $\|Z\|_{\mathsf{L}^2(\mathbb{Q})}$ is uniformly upper bounded by $C_1 \tau_n^{1/2}$, where the constant $C_1 > 0$ only depend on the noise and source-target distributions, kernel spectral condition, and compatibility constants.*

The general coverage guarantee of bootstrap confidence balls is presented in the following theorem.

Theorem 4.1 (general coverage guarantee)**.** *Let $\alpha \in (0,1)$ be a miscoverage level. Suppose the conditions of Lemma 4.1 hold and inequalities (4.1) - (4.3) hold with a probability tolerance satisfying $0 < \eta < \min\{\alpha, 1-\alpha\}$. Denote by $\hat{t}_\alpha$ the bootstrap quantile such that $P(\|B\|_{\mathsf{L}^2(\mathbb{Q})} \geq \hat{t}_\alpha \mid \mathcal{D}) = \alpha$, and set $\hat{c}_{\alpha,n} = \hat{t}_\alpha / n^{1/2}$. The miscoverage of the confidence ball $\mathsf{CS}_\alpha(\widehat{f})$ defined in (3.3) satisfies*

$$\left|\mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha\right| \lesssim \eta + \sqrt{\tau_n}\Delta.$$

Therefore, if $\eta_n \to 0$ and $\sqrt{\tau_n}\{\Delta_{G,n} + \Delta_{B,n} + \Delta_{\mathrm{bias},n}\} \to 0$ as $n \to \infty$, then the coverage probability of the bootstrap confidence balls converges to the nominal level.

4.3 Finite-sample bounds for coverage probability

We establish explicit bounds for the terms Δ_G, Δ_B, and Δ_{bias}, which characterize the coverage guarantee of confidence balls.

4.3.1 Bias control of IWKRR

Proposition 4.1. *Suppose that Assumptions 4.2 and 4.4 hold with $\tau_n \geq \tau_0$. Then,*

$$\sqrt{n}\|f_{P^w} - f_*\|_{\mathsf{L}^2(\mathbb{Q})} \leq \Delta_{\mathrm{bias}} := \max\{\kappa^{2s-1}, 1\}\, \mathfrak{C}_1 \sqrt{n}\, \lambda_n^{\min\{s+1/2,1\}} \|g_*\|_{\mathsf{H}}.$$

Thus the bias bound inequality (4.1) holds with above choice of Δ_{bias}, which diminishes asymptotically if $n\lambda_n^{\min\{2s+1,2\}} \to 0$ as $n \to \infty$.

4.3.2 Gaussian approximation of IWKRR

To establish the inequality (4.2), we must: (i) bound the approximation error of Bahadur representation (3.1); (ii) bound the Gaussian coupling of the normalized sum of the influence elements. The Gaussian coupling proof uses techniques similar to those in Singh and Vijaykumar [2023], adapted to the target $\mathsf{L}^2(\mathbb{Q})$ geometry and the covariance operator induced by truncated importance weighting.

Proposition 4.2 (Bahadur representation)**.** *Suppose Assumptions 4.1, 4.2, and 4.4 hold with $\tau_n \geq \tau_0$, and $\tau_n \to \infty, \lambda_n \to 0$ as $n \to \infty$. If $n > \max\left\{(\tau_n\kappa^2 + 1)\log(e + \mathcal{N}_{\lambda_n})/(C_2\lambda_n), \tau_n/(\lambda_n \mathcal{N}_{\lambda_n})\right\}$, where C_2 is a constant depending on $\log(1/\eta)$, then with probability at least $1-\eta$,*

$$\left\|\sqrt{n}(\widehat{f} - f_{P^w}) - \frac{1}{\sqrt{n}}\sum_{i=1}^n U_i\right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \Delta_{G,\mathrm{Bah}} := \mathfrak{C}_1 \tau_n \overline{\sigma}\, \kappa^3 \big(1 + \|g_*\|_{\mathsf{H}}\big) \sqrt{\frac{\mathcal{N}_{\lambda_n} \log(e + \mathcal{N}_{\lambda_n})}{\lambda_n n}} \log^2\left(\frac{2}{\eta}\right)$$

where $U_i = (\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1}\big(w_n(X_i)\varepsilon_i k_{X_i} + (w_n(X_i)k_{X_i} \otimes k_{X_i} - \mathcal{L}_{P^w})(f_* - f_{P^w})\big)$. *Moreover,* $\|U_i\|_{\mathsf{L}^2(\mathbb{Q})} \le \bar{U} := \tau_n \mathfrak{C}_1 \lambda_n^{-1/2}(\kappa^2 \|f_*\|_{\mathsf{H}} + \kappa\bar{\sigma})$.

We note that the upper bound $\bar{U}$ for $\|U_i\|_{\mathsf{L}^2(\mathbb{Q})}$ will appear frequently in the subsequent theorems. The Bahadur representation error $\Delta_{G,\mathrm{Bah}}$ diminishes asymptotically if $n\lambda_n \gg \tau_n^2 \mathcal{N}_{\lambda_n} \log(e + \mathcal{N}_{\lambda_n})$.

Recall the spectral tail $\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) = \sum_{j=m+1}^{\infty} \mu_j(\mathcal{V}, \mathsf{L}^2(\mathbb{Q}))$, where $\mu_j(\mathcal{V}, \mathsf{L}^2(\mathbb{Q}))$ denotes the j-th eigenvalue of $\mathcal{V}$ in the $\mathsf{L}^2(\mathbb{Q})$ space. The spectral tail characterizes the smallest error of approximating $\mathcal{V}$ by a finite-rank operator.

Proposition 4.3 (Gaussian coupling in target $\mathsf{L}^2(\mathbb{Q})$)**.** *There exists* $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ *such that, with probability at least* $1 - \eta$,

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^{n} U_i - Z \right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \Delta_{G,\mathrm{coup}} := \inf_{m \ge 1} \left\{ \log^{1/2}(6/\eta) \sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) + \frac{m^2 \bar{U} \log(m^2/\eta)}{\sqrt{n}} \right\}$$

The Gaussian coupling error $\Delta_{G,\mathrm{coup}}$ is determined by the decay rate of the eigenvalues of $\mathcal{V}$ in $\mathsf{L}^2(\mathbb{Q})$. By Propositions 4.2 and 4.3, the Gaussian approximation inequality (4.2) holds with $\Delta_G = \Delta_{G,\mathrm{Bah}} + \Delta_{G,\mathrm{coup}}$.

4.3.3 Bootstrap approximation

To establish (4.3), we must: (i) bound the difference between the oracle bootstrap function B_{ora} and the computable bootstrap function B in $\mathsf{L}^2(\mathbb{Q})$-norm; (ii) bound the Gaussian coupling of B_{ora}. The regularized transfer coverage Assumption 4.4 is required in the first step because both the residual and the empirical regularized operator must be transferred from the weighted-source geometry to target $\mathsf{L}^2(\mathbb{Q})$ geometry.

Proposition 4.4 (difference between oracle and computable bootstrap functions)**.** *Suppose Assumptions 4.1, 4.2, and 4.4 hold with* $\tau_n \ge \tau_0$, *and* $\tau_n \to \infty, \lambda_n \to 0$ *as* $n \to \infty$. *If* $n > \max\big\{(\tau_n\kappa^2 + 1)\log(e + \mathcal{N}_{\lambda_n})/(C_2\lambda_n), \tau_n/(\lambda_n \mathcal{N}_{\lambda_n})\big\}$, *where* C_2 *is a constant depending on* $\log(1/\eta)$, *then there exists a data event* $\mathcal{E}$ *with* $\mathbb{P}(\mathcal{E}) \ge 1 - \eta$ *such that, conditioned on* $\mathcal{E}$ *and data* $\mathcal{D}$, *with probability at least* $1 - \eta$,

$$\|B - B_{\mathsf{ora}}\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \Delta_{B,\mathrm{feas}} := \mathfrak{C}_1\, \overline{\sigma}\, \kappa^{2s+4}\big(1 + \|g_*\|_{\mathsf{H}}\big) \frac{\tau_n \mathcal{N}_{\lambda_n}}{\sqrt{n\lambda_n}} \cdot \log^3(2/\eta),$$

where

$$B = \frac{1}{n} \sum_{1 \le i \ne j \le n} h_{ij} \cdot \frac{\widehat{V}_i - \widehat{V}_j}{\sqrt{2}} \quad \text{and} \quad B_{\mathsf{ora}} = \frac{1}{n} \sum_{1 \le i \ne j \le n} h_{ij} \cdot \frac{V_i - V_j}{\sqrt{2}},$$

with $V_i = w_n(X_i)\big(Y_i - f_{P^w}(X_i)\big)\,(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1} k_{X_i}$ *and* $\widehat{V}_i = w_n(X_i)\big(Y_i - \widehat{f}(X_i)\big)\,(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1} k_{X_i}$. *Moreover,* $B_{\mathsf{ora}} \mid \mathcal{D} \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \widehat{\mathcal{V}}_{\mathsf{ora}})$ *and* $B \mid \mathcal{D} \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \widehat{\mathcal{V}})$, *where* $\widehat{\mathcal{V}}_{\mathsf{ora}} = \mathbb{E}_{\mathbb{P}_n}[V_i \otimes_{\mathsf{L}^2(\mathbb{Q})} V_i] - \mathbb{E}_{\mathbb{P}_n}[V_i] \otimes_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}_{\mathbb{P}_n}[V_i]$ *and* $\widehat{\mathcal{V}} = \mathbb{E}_{\mathbb{P}_n}[\widehat{V}_i \otimes_{\mathsf{L}^2(\mathbb{Q})} \widehat{V}_i] - \mathbb{E}_{\mathbb{P}_n}[\widehat{V}_i] \otimes_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}_{\mathbb{P}_n}[\widehat{V}_i]$, *respectively.*

The error $\Delta_{B,\mathrm{feas}}$ diminishes asymptotically if $n\lambda_n \gg \tau_n^2 \mathcal{N}_{\lambda_n}^2$. Denote the trace $\mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V}) = \sigma^2_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V}) = \sum_{j=1}^{\infty} \mu_j(\mathcal{V}, \mathsf{L}^2(\mathbb{Q}))$.

Proposition 4.5 (bootstrap coupling in $\mathsf{L}^2(\mathbb{Q})$)**.** *Let g be the isonormal Gaussian process in $\mathsf{L}^2(\mathbb{Q})$. Then there exists a random element $G \overset{\text{d.}}{=} \widehat{\mathcal{V}}_{ora}^{1/2} g \big| \mathcal{D}$ such that, conditioned on $\mathcal{D}$, with total probability at least $1-6\eta$,*

$$\left\| \mathcal{V}^{1/2} g - G \right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \Delta_{B,\text{coup}} := \inf_{m \geq 1} \left[m^{1/4} \left(\frac{\bar{U}^2 \text{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})}{n} + \frac{\bar{U}^4}{n^2} \right)^{1/4} + \sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \right] \log^{3/2}(2/\eta),$$

where the covariance operator $\mathcal{V} = \mathbb{E}\big[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i\big]$.

Putting together Propositions 4.4 and 4.5, we can couple the Bootstrap function B with a Gaussian element in $\mathsf{L}^2(\mathbb{Q})$, conditioned on data $\mathcal{D}$, which is formalized in the following corollary.

Corollary 4.1 (bootstrap approximation in $\mathsf{L}^2(\mathbb{Q})$)**.** *Suppose the conditions of Propositions 4.4 and 4.5 hold. There exist a universal constant $c > 0$ and a random element $Z' \overset{\text{d.}}{=} \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V}) \big| \mathcal{D}$, conditionally coupled with B, such that with probability at least $1-\eta$ over the data,*

$$P\big(\|B - Z'\|_{\mathsf{L}^2(\mathbb{Q})} \leq \Delta_{B,\text{feas}} + \Delta_{B,\text{coup}} \,\big|\, \mathcal{D} \big) \geq 1 - \eta,$$

where $\Delta_{B,\text{feas}}$ and $\Delta_{B,\text{coup}}$ denote the bounds in Propositions 4.4 and 4.5, respectively, evaluated with tolerance $\eta_{\text{tol}} = c\eta^2$.

By Corollary 4.1, the bootstrap approximation inequality (4.3) holds with $\Delta_B = \Delta_{B,\text{feas}} + \Delta_{B,\text{coup}}$, where $\Delta_{B,\text{feas}}$ and $\Delta_{B,\text{coup}}$ are evaluated at tolerance $c\eta^2$.

4.3.4 Examples

We now provide explicit error rates for the coverage probability of bootstrap confidence sets. By Propositions 4.1-4.5 and Corollary 4.1, the key roles in the error rates are played by: (i) target-source distribution mismatch; (ii) the spectral decay rate of the covariance operator $\mathcal{V}$, or equivalently, of the integral operator $\mathcal{L}_Q$.

For ease of exposition, we ignore the constants $\mathfrak{C}_1, \mathfrak{C}_2, \underline{\rho}$, and assume that Assumptions 4.1 - 4.3 hold with $s \geq 1/2$ and $\tau_n \geq \tau_0$. Similarly, the quantities $\gamma, W, \sigma, \bar{\rho}$, and Ω, appearing in Remark 1, will also be treated as constants. The probability parameter is fixed at $\eta \asymp n^{-1}$ so that $\log(1/\eta) \asymp \log n$. Recall that μ_k denotes the k-th eigenvalue of $\mathcal{L}_Q$.

Corollary 4.2 (bounded density ratio)**.** *Suppose that $\rho(X) \leq \overline{\rho}$ $\mathbb{P}$-almost surely, and let $\tau_n = C_0$ for a sufficiently large fixed constant $C_0 \geq \max\{\overline{\rho}, \tau_0\}$.*

(1) if $\mu_k \asymp k^{-\beta}$ for some $\beta > 5$ and all $k \geq 1$, by choosing the regularization parameter $\lambda_n \asymp n^{-\frac{2(\beta+1)}{3(\beta+3)}}$, we get

$$\left| \mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f}) \big) - \alpha \right| \lesssim n^{-\frac{(\beta-5)(\beta+10)}{6(\beta+3)(2\beta+5)}} \log^3 n;$$

(2) if $\mu_k \asymp \exp(-ck^\zeta)$ for some $c, \zeta > 0$ and all $k \geq 1$, by choosing the regularization parameter $\lambda_n \asymp n^{-3/5}$, we get

$$\left| \mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f}) \big) - \alpha \right| \lesssim n^{-1/10} (\log n)^{\frac{3}{2} + \frac{1}{2\zeta}};$$

Corollary 4.3 (sub-exponential tail)**.** *Suppose that the density ratio satisfies the sub-exponential tail condition in Definition 4.1.*

(1) if $\mu_k \asymp k^{-\beta}$ for some $\beta > 5$ and all $k \geq 1$, by choosing the truncation level $\tau_n \asymp n^{\frac{\beta-5}{9(\beta+3)(2\beta+5)}}$ and the regularization parameter $\lambda_n \asymp n^{-\frac{2(\beta+1)}{3(\beta+3)}}$, we get

$$\left|\mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha\right| \lesssim n^{-\frac{(\beta-5)(\beta+9)}{6(\beta+3)(2\beta+5)}} \log^3 n;$$

(2) if $\mu_k \asymp \exp(-ck^\zeta)$ for some $c, \zeta > 0$ and all $k \geq 1$, by choosing the truncation level $\tau_n \asymp n^{1/150}$ and the regularization parameter $\lambda_n \asymp n^{-3/5}$, we get

$$\left|\mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha\right| \lesssim n^{-\frac{9}{100}} \log^3 n;$$

If the density ratio $\rho(X)$ has heavy tails in $\mathbb{P}$, our methods can still provide valid inference as long as the eigenvalues of $\mathcal{L}_Q$ decay sufficiently fast and the density ratio has a bounded higher moment. For ease of exposition, we only present the theoretical guarantees for the case when $\mathcal{L}_Q$ has exponentially decaying eigenvalues. If $\mathcal{L}_Q$ has polynomial-decay eigenvalues, the explicit error rate for coverage probability is complicated (see Appendix D for more details).

Corollary 4.4 (bounded θ-th moment)**.** *Suppose that $\mathbb{E}_{\mathbb{P}}[\rho(X)^\theta] \leq \Omega_\theta < \infty$ for some $\theta > 7$ as in Definition 4.2. If $\mu_k \asymp \exp(-ck^\zeta)$ for some $c, \zeta > 0$ and all $k \geq 1$, then, by choosing*

$$\lambda \asymp n^{-\frac{3(\theta-1)}{5\theta-2}} \quad \text{and} \quad \tau_n \asymp n^{\frac{3}{5\theta-2}} (\log n)^{\frac{1}{\zeta(\theta-1)}},$$

we obtain

$$\left|\mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha\right| \lesssim n^{-\frac{\theta-7}{2(5\theta-2)}} (\log n)^{\frac{3}{2}+\frac{1}{2\zeta}+\frac{5}{4\zeta(\theta-1)}}.$$

4.4 Hypothesis tests and power analysis

For a prescribed function f_0, we reject the null hypothesis in (1.1) whenever $f_0 \notin \mathsf{CS}_\alpha(\widehat{f})$, or equivalently whenever $\|\widehat{f} - f_0\|_{\mathsf{L}^2(\mathbb{Q})} > \hat{c}_{\alpha,n}$. The coverage guarantees in Sections 4.2-4.3 control the size of the inverted test under the null hypothesis.

We now provide the power guarantee under fixed alternatives. The key observation is that, under H_1, the target-domain separation $\delta := \|f_* - f_0\|_{\mathsf{L}^2(\mathbb{Q})}$ is strictly positive. Therefore, the inverted test is powerful whenever the estimation error and the scaled critical value are both negligible compared with this fixed separation.

Theorem 4.2 (power guarantee)**.** *Suppose H_1 holds with a fixed separation $\delta = \|f_* - f_0\|_{\mathsf{L}^2(\mathbb{Q})} > 0$ in the target domain. If $\|\widehat{f} - f_*\|_{\mathsf{L}^2(\mathbb{Q})} + \hat{c}_{\alpha,n} \xrightarrow{\mathbb{P}} 0$ as $n \to \infty$, then*

$$\mathbb{P}\big(\|\widehat{f} - f_0\|_{\mathsf{L}^2(\mathbb{Q})} > \hat{c}_{\alpha,n}\big) \to 1, \qquad \text{as } n \to \infty.$$

5 Numerical Experiments

5.1 Simulation design

We numerically examine the finite-sample properties of our proposed method from three perspectives: (i) the calibration of bootstrap radius; (ii) the empirical coverage of the resulting confidence ball; (iii) the power of the induced test under fixed alternatives. The source covariates are

generated from $\mathcal{N}(0,4)$, while the target distribution is $\mathcal{N}(1,1)$. The true regression function is $f_*(x) = \exp(-0.5x^2)$, and the independent noise $\varepsilon_i \sim \mathcal{N}(0, 0.09)$. We use the Gaussian radial kernel to examine the performance of bootstrap radius calibration and coverage guarantee. For power evaluation, we compare three kernel functions with a fixed length scale $\ell = 1$. All regularization and truncation parameters are selected by validation as described below. These kernels are

$$k_{\mathrm{G}}(x,x') = \exp(-r^2/2), \qquad k_{3/2}(x,x') = (1+\sqrt{3}r)\exp(-\sqrt{3}r), \qquad k_{1/2}(x,x') = \exp(-r),$$

with $r = |x - x'|$, where the latter two are the Matérn kernels with smoothness parameters $\nu = 3/2$ and $\nu = 1/2$, respectively. Note that $k_{1/2}(\cdot,\cdot)$ is also known as the Laplacian kernel. All three kernels satisfy $k(x,x) = 1$.

The target-domain norm $\|\cdot\|_{\mathsf{L}^2(\mathbb{Q})}$ is approximated by Monte Carlo integration using 1200 target samples. The IWKRR estimator uses normalized importance weights and truncated weights $w_n(x) = \min\{\rho(x), \tau_n\}$.

Tuning the regularization and truncation parameters. For calibration, coverage at each sample size, and power, we independently draw a source-labelled tuning sample D^{tune} and a source-labelled validation sample D^{val}, each of size n, where $n = 300$ for calibration and power and $n \in \{200, 300\}$ for coverage. The parameters are chosen from

$$\lambda \in n^{-0.6}\{1/128, 1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1, 2, 4\},$$
$$\tau \in \left(\frac{n}{300}\right)^{0.06}\{0.0625, 0.125, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 1.75, 2, 2.5\}.$$

For each pair (λ, τ), the candidate model is fitted on D^{tune}, whereas its target prediction risk is estimated on D^{val} using the self-normalized, untruncated importance-weighted criterion

$$\widehat{R}_k^{\mathrm{val}}(\lambda,\tau) = \frac{\sum_{i=1}^n \rho(X_i^{\mathrm{val}})\{Y_i^{\mathrm{val}} - \widehat{f}_{k,\lambda,\tau}(X_i^{\mathrm{val}})\}^2}{\sum_{i=1}^n \rho(X_i^{\mathrm{val}})}.$$

If numerical ties occur, we prefer larger λ and smaller τ. The selected pair is held fixed across all subsequent Monte Carlo repetitions: one pair is used for calibration, one pair for each n in coverage, and one pair for each kernel in power across all 20 independent experiments.

5.2 Bootstrap calibration

We use 1000 independent draws to simulate the bootstrap distribution. Figure 2 compares the conditional bootstrap distribution with the empirical distribution of the true scaled target error. The true distribution is estimated using Monte Carlo simulation from 1000 independent training samples, each with a sample size $n = 300$. The bootstrap distribution is computed conditional on one representative fitted data set.

Table 1 reports the calibration summary, whereas Figure 2 provides a smoother display of the corresponding empirical and conditional bootstrap distributions. The table's bootstrap quantiles are close to the empirical sampling quantiles at both levels, with absolute relative errors between three and five percent.

Table 1: Bootstrap estimates of upper quantiles at $\alpha \in \{0.10, 0.05\}$ and $n = 300$.

α	Sampling quantile	Conditional bootstrap quantile	Relative error
0.10	1.109	1.071	-0.034
0.05	1.222	1.162	-0.049

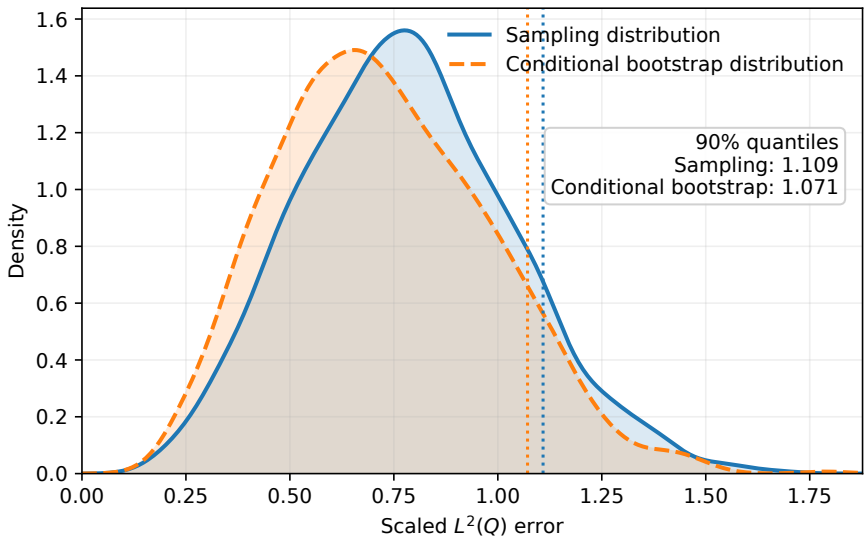


Figure 2: Bootstrap distribution of the $\mathsf{L}^2(\mathbb{Q})$ error. The figure compares the true sampling distribution of $\sqrt{n}\|\widehat{f} - f_*\|_{\mathsf{L}^2(\mathbb{Q})}$ with the conditional bootstrap distribution of $\|B\|_{\mathsf{L}^2(\mathbb{Q})}$.

5.3 Empirical coverage and size

We evaluate the empirical coverage of the feasible confidence ball $\mathsf{CS}_\alpha(\widehat{f})$ with the critical value $\hat{c}_{\alpha,n}$ estimated from 250 multiplier-bootstrap repetitions. Under the null hypothesis $f_0 = f_*$, we record the relative frequency of rejecting the null hypothesis using the inverted test $\mathbf{1}\{\|\widehat{f} - f_0\|_{\mathsf{L}^2(\mathbb{Q})} > \hat{c}_{\alpha,n}\}$. For each sample size, the rejection probability is estimated from 1000 independent repetitions. Table 2 shows modest finite-sample over-rejection and under-coverage at $n = 200$, whereas at $n = 300$ the empirical size and coverage are close to their nominal levels.

Table 2: Empirical size of the inverted test and coverage of the confidence ball under $H_0 : f_0 = f_*$.

n	α	Empirical size	Empirical coverage
200	0.10	0.138	0.862
200	0.05	0.084	0.916
300	0.10	0.099	0.901
300	0.05	0.054	0.946

5.4 Power performance

We evaluate the empirical power of the induced test under fixed alternatives. Let $f_{0,\delta}(x) = f_*(x) + \delta\, h(x)$ with $h(x) = g(x)\|g\|^{-1}_{\mathsf{L}^2(\mathbb{Q})}$, where $g(x) := k_{\mathrm{G}}(x,1) - k_{\mathrm{G}}(x,0) = \exp\{-(x-1)^2/2\} - \exp(-x^2/2)$ and $\|g\|_{\mathsf{L}^2(\mathbb{Q})} = 3^{-1/4}(1 - e^{-1/3})^{1/2}$. Since g is a difference of Gaussian kernel sections and is a Schwartz function, it belongs to each of the three RKHSs considered. Similarly, $f_{0,\delta}$ belongs to all three RKHSs and $\|f_{0,\delta} - f_*\|_{\mathsf{L}^2(\mathbb{Q})} = \delta$.

The significance level is fixed at $\alpha = 0.10$ with $n = 300$. The critical value $\hat{c}_{\alpha,n}$ is estimated using 300 bootstrap draws. For each separation δ and kernel function, the empirical rejection probability is estimated from 100 independent test repetitions, whose empirical variation is evaluated via 20 independent experiments. The results are displayed in Figure 3. The rejection probabilities increase with the target-domain separation for all three scenarios, reaching approximately one at $\delta = 0.12$. At $\delta = 0$, the average rejection probabilities for the Gaussian, Matérn $\nu = 3/2$, and Matérn $\nu = 1/2$ scenarios are 0.1030, 0.0985, and 0.1075, respectively. These values are empirical sizes, which are close to the nominal level 0.10.

6 Application to Survey-Weighted Household Finance

6.1 Data and survey weights

The Survey of Consumer Finances (SCF) is a triennial cross-sectional survey of U.S. families that records income, balance sheets, credit use, and demographic characteristics. Our experiment focuses on the implicate 1 of 2022 public summary extract released by Board of Governors of the Federal Reserve System [2023], which contains 4,595 records representing distinct families. Because the SCF records are not randomly sampled, each observation comes with a survey weight indicating how many U.S. families that observation represents. These survey weights are normalized to have mean one, denoted by ρ_i satisfying $n^{-1}\sum_{i=1}^n \rho_i = 1$, which help convert averages over the unweighted

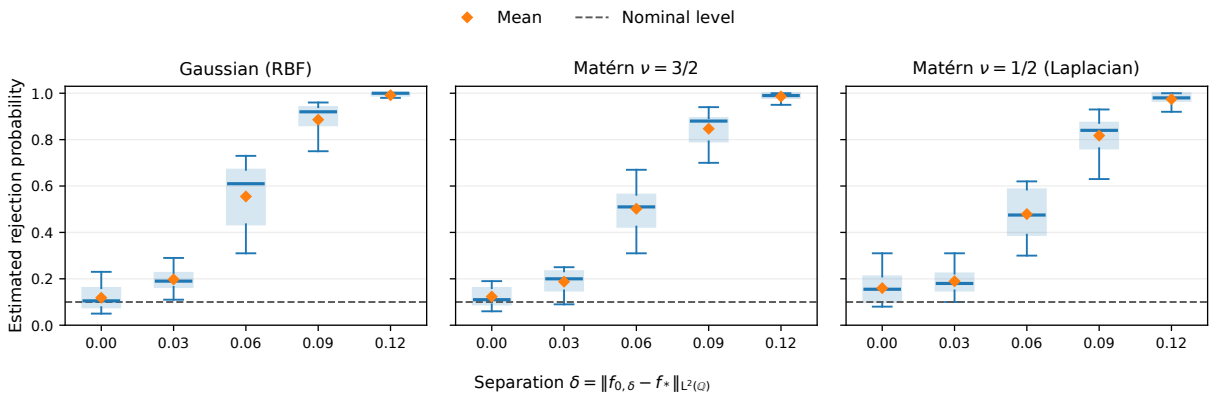


Figure 3: Empirical power under fixed alternatives $\|f_{0,\delta} - f_*\|_{\mathsf{L}^2(\mathbb{Q})} = \delta$ for three kernel functions. From left to right, the panels correspond to the Gaussian RBF, Matérn $\nu = 3/2$, and Matérn $\nu = 1/2$ (Laplacian) kernels, all with length scale $\ell = 1$. The parameters (λ, τ) are tuned using independent source-labelled tuning and validation samples of size 300; the selected pair for each kernel is held fixed across all 20 simulation experiments. The rejection critical values are estimated using 300 multiplier-bootstrap draws. The boxplots summarize variation across the 20 experiments, and the diamonds mark the corresponding means. The dashed horizontal line marks the nominal level $\alpha = 0.10$.

records into averages for the full U.S. family population represented by the SCF. We regard the unweighted records as a sample from the source distribution $\mathbb{P}$ and the survey-weighted records as a sample from the target distribution $\mathbb{Q}$, under which the financial conclusions are sought.

Figure 4 illustrates the empirical distribution of the normalized weights and the effect of reweighting on two specific covariates. Two measures are provided in panel A describing the variation of normalized weights, where the coefficient of variation $\mathrm{CV} := \big(\sum_{i=1}^n (\rho_i - 1)^2/n\big)^{1/2}$ and effective size size $\mathrm{ESS} = n^2/\big(\sum_{i=1}^n \rho_i^2\big)$. More summary statistics are reported in Appendix E.1.

Our experiments primarily study four questions: (i) how does the recorded characteristics affect the household net worth; (ii) is the probability of holding stock equity nondecreasing in usual income; (iii) do households allocate the same share of wealth to stock equity; (iv) does the late-payment probability change across years? Due to page constraints, we present the two studies in this section and other results in the Appendix.

6.2 Study 1: predicting household net worth

The response is $Y = \mathrm{asinh}\big(\texttt{NETWORTH} \times 10^{-5}\big)$ and target regression function $f_*(x) = \mathbb{E}(Y \mid X = x)$. Numerical variables are standardized, and categorical variables are encoded, yielding a 33-dimensional covariate variable X. The variable construction details are reported in Appendix. We propose four candidate functions for f_0 and test against $H_0 : \|f_* - f_0\|_{\mathsf{L}^2(\mathbb{Q})} = 0$, among which the ordinary KRR $f_{0,\mathrm{KRR}}$ trained on $\mathbb{P}$, i.e., unweighted household records, is of primary interest.

The data is randomly split into three disjoint subsets with size: $|\mathcal{D}_0| = 1{,}400, |\mathcal{D}_I| = 1{,}800, |\mathcal{D}_Q| = 1{,}395$. The subset $\mathcal{D}_0$ is reserved for training the candidate function. Three constant-valued candidate functions are $f_{0,\mathrm{tgt}}(x) \equiv \sum_{i\in\mathcal{D}_0} \rho_i Y_i / \big(\sum_{i\in\mathcal{D}_0} \rho_i\big)$, $f_{0,\mathrm{src}}(x) \equiv \sum_{i\in\mathcal{D}_0} Y_i/|\mathcal{D}_0|$, and $f_{0,\mathrm{zero}}(x) \equiv 0$.

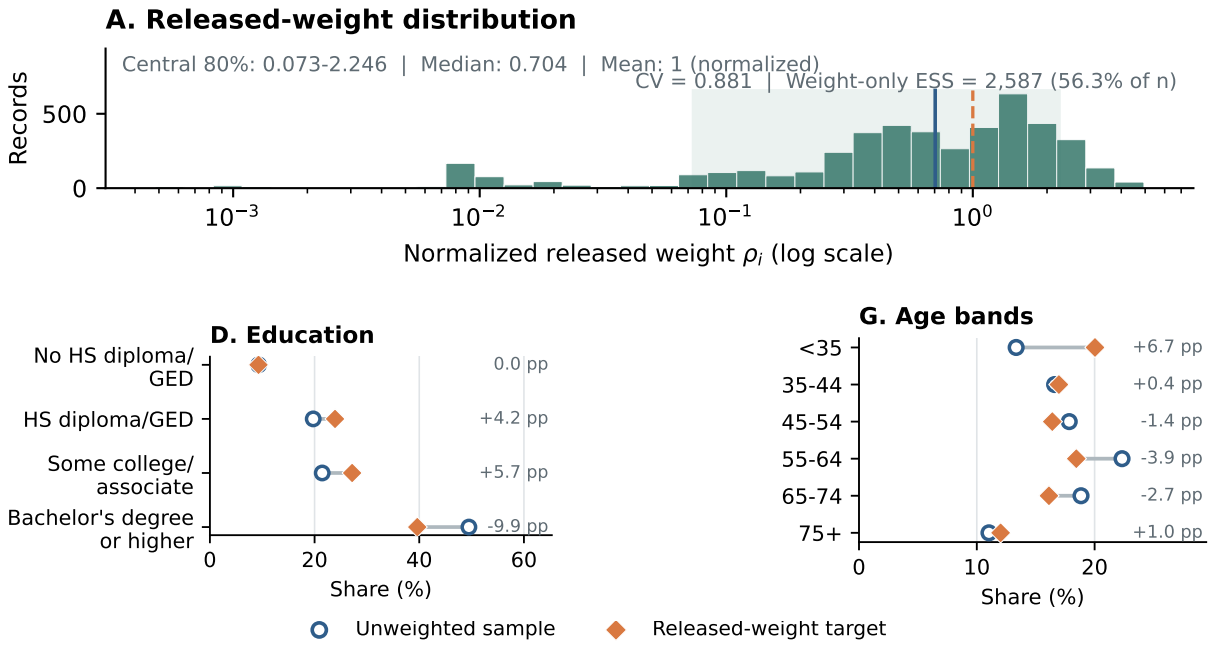


Figure 4: Panel A shows the normalized released weights on a logarithmic scale; the shaded region spans the 10th through 90th percentiles, and the solid and dashed vertical lines mark the median and normalized mean, respectively. Panels D and G compare unweighted and weighted sample percentages for education and age bands, respectively.

For $f_{0,\mathrm{KRR}}$, we use the Gaussian RBF kernel with $\gamma = 1/33$ with λ_n tuned by cross validation. The subset $\mathcal{D}_I$ is used for fitting IWKRR $\widehat{f}$ and implementing the multiplier bootstrap. The tuning parameters λ_n and τ_n are selected by cross validation using $\mathcal{D}_0$. The subset $\mathcal{D}_Q$ is used for evaluation, based on which we define the empirical $\mathsf{L}^2(\mathbb{Q})$-norm: $\|g\|_{\widehat{\mathbb{Q}}}^2 = \sum_{i\in\mathcal{D}_Q} \rho_i g^2(X_i)/\big(\sum_{i\in\mathcal{D}_Q}\rho_i\big)$.

By $B = 5{,}000$ multiplier-bootstrap draws, we get the 90%-percentile $\hat{c}_{0.10} = 0.1165$ as an estimate of the quantile for $\|\widehat{f} - f_*\|_{\mathsf{L}^2(\mathbb{Q})}$. By the triangle inequality $\big|\|f_* - f_0\|_{\widehat{\mathbb{Q}}} - \|\widehat{f} - f_0\|_{\widehat{\mathbb{Q}}}\big| \le \|\widehat{f} - f_*\|_{\widehat{\mathbb{Q}}}$, we construct the 90% confidence interval for $\|f_* - f_0\|_{\mathsf{L}^2(\mathbb{Q})}$ by $\|\widehat{f} - f_0\|_{\widehat{\mathbb{Q}}} \pm \hat{c}_{0.10}$, where the left endpoint is rounded to zero if negative. The confidence interval is conservative by nature.

The bootstrap distribution, test values $\|\widehat{f} - f_0\|_{\widehat{\mathbb{Q}}}$ for four candidate functions, and the corresponding 90% confidence intervals are illustrated in Figure 5. All four intervals exclude zero. In particular, the source-only KRR candidate $f_{0,\mathrm{KRR}}$ is rejected at the nominal level. Further details are provided in Appendix E.2.

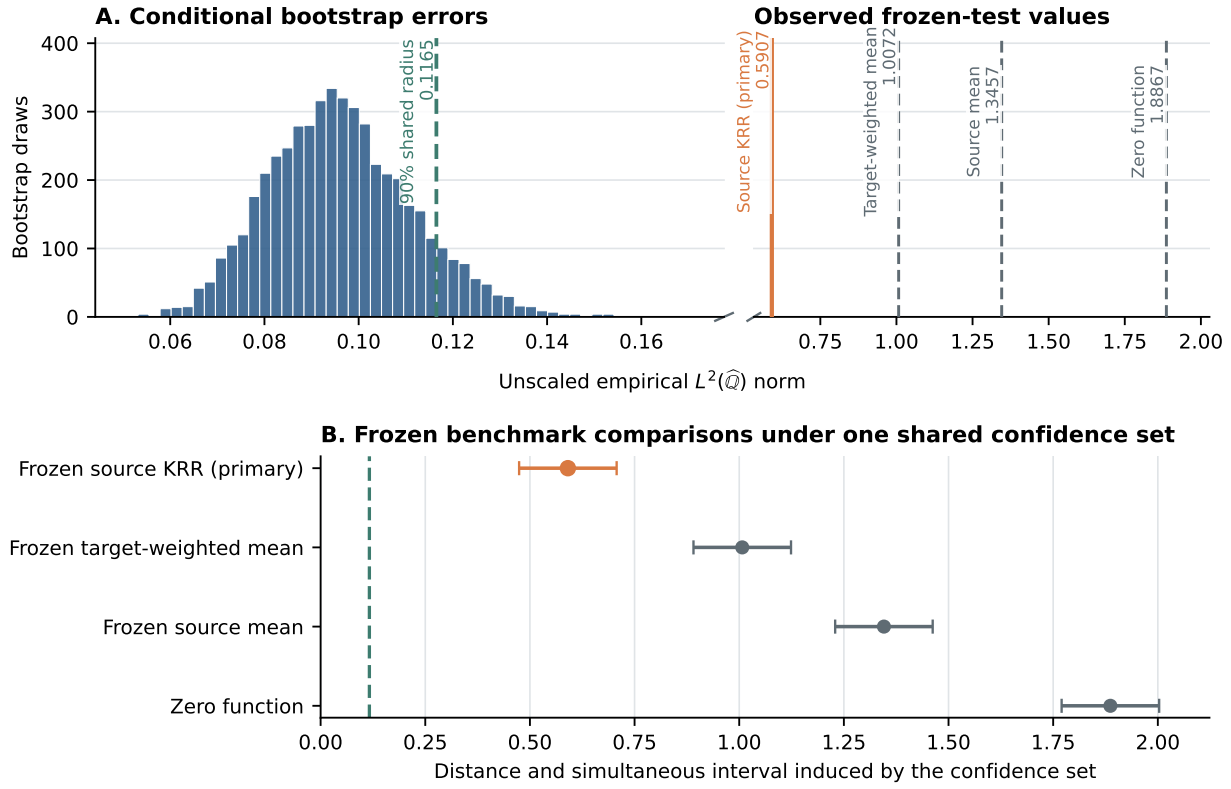


Figure 5: testing candidate functions for estimating household net worth based on IWKRR $\widehat{f}$. The four candidate functions include the source-only KRR $f_{0,\mathrm{KRR}}$, the target-weighted mean $f_{0,\mathrm{tgt}}$, the source mean $f_{0,\mathrm{src}}$, and the trivial function $f_{0,\mathrm{zero}}$. Panel A displays the bootstrap distribution over 5,000 draws, 90% confidence-set radius $\hat{c}_{0.1}$, and the test values $\|\widehat{f} - f_0\|$ of four candidate functions. Panel B shows the 90% CI for $\|f_* - f_0\|_{\mathsf{L}^2(\mathbb{Q})}$ for four candidate functions, based on which all are rejected at the nominal level $\alpha = 0.10$.

6.3 Study 2: does higher income increase the probability of holding equity?

We are interested in the relationship between household income and the probability of holding stock equity. Let the response variable be $Y = \mathbf{1}\{\texttt{HEQUITY} = 1\}$, the single covariate $Z = \log\{1+\texttt{NORMINC}\}$, and unknown true regression function $f_*(z) = \mathbb{Q}(Y = 1 \mid Z = z)$, where NORMINC denotes usual or normal income. In particular, we want to test whether $f_*(z)$ is monotonic in z. Let $\mathcal{M}_\uparrow$ be the class of $[0,1]$-valued nondecreasing functions on the income domain, where functions equal $\mathbb{Q}$-almost surely are identified. The class $\mathcal{M}_\uparrow$ is a closed subset of $\mathsf{L}^2(\mathbb{Q})$. We aim to test against $H_0 : f_* \in \mathcal{M}_\uparrow$. This is a composite test. Alternatively, we test against $H_0 : d_{\mathbb{Q}}(f_*, \mathcal{M}_\uparrow) = 0$, where the projection distance $d_{\mathbb{Q}}(h, \mathcal{M}_\uparrow) := \inf_{g\in\mathcal{M}_\uparrow} \|h - g\|_{\mathsf{L}^2(\mathbb{Q})}$.

We random split the data into the training set $\mathcal{D}_t$ of size 1,800 and the evaluation set $\mathcal{D}_e$ of size 2,795. The IWKRR estimator $\widehat{f}$ is learned using the Gaussian RBF kernel on the training set where the parameters (λ_n, τ_n) are selected by cross validation. The empirical target distribution $\widehat{\mathbb{Q}}$ is computed using the evaluation set by $\widehat{\mathbb{Q}}(Z \le z) := \sum_{i\in\mathcal{D}_e} \rho_i \mathbf{1}(Z_i \le z) / \sum_{i\in\mathcal{D}_e} \rho_i$. The empirical distance is $d(\widehat{f}, \mathcal{M}_\uparrow) = \inf_{g\in\mathcal{M}_\uparrow} \|\widehat{f} - g\|_{\mathsf{L}^2(\widehat{\mathbb{Q}})} = 0.0151$, and the 90%-percentile based on $B = 5,000$ multiplier bootstrap is $\hat{c}_{0.10} = 0.0252$. Note that $\hat{c}_{0.10}$ is an estimate of the 90%-quantile of $\|\widehat{f} - f_*\|_{\mathsf{L}^2(\mathbb{Q})}$. By the inequality $\left|d_{\mathbb{Q}}(\widehat{f}, \mathcal{M}_\uparrow) - d_{\mathbb{Q}}(f_*, \mathcal{M}_\uparrow)\right| \le \|\widehat{f} - f_*\|_{\mathsf{L}^2(\mathbb{Q})}$, we construct the 90% confidence interval for $d_{\mathbb{Q}}(f_*, \mathcal{M}_\uparrow)$ by $d_{\mathbb{Q}}(\widehat{f}, \mathcal{M}_\uparrow) \pm \hat{c}_{0.10}$, resulting into the interval $[0, 0.0403]$, where the left endpoint is rounded to zero if negative. Since zero is included in the interval, we do not reject the null hypothesis at the significance level $\alpha = 0.10$. Figure 6 reports the multiplier-bootstrap distribution, the empirical distance $d(\widehat{f}, \mathcal{M}_\uparrow)$, and the critical value $\hat{c}_{0.1}$. More details can be found in the Appendix.

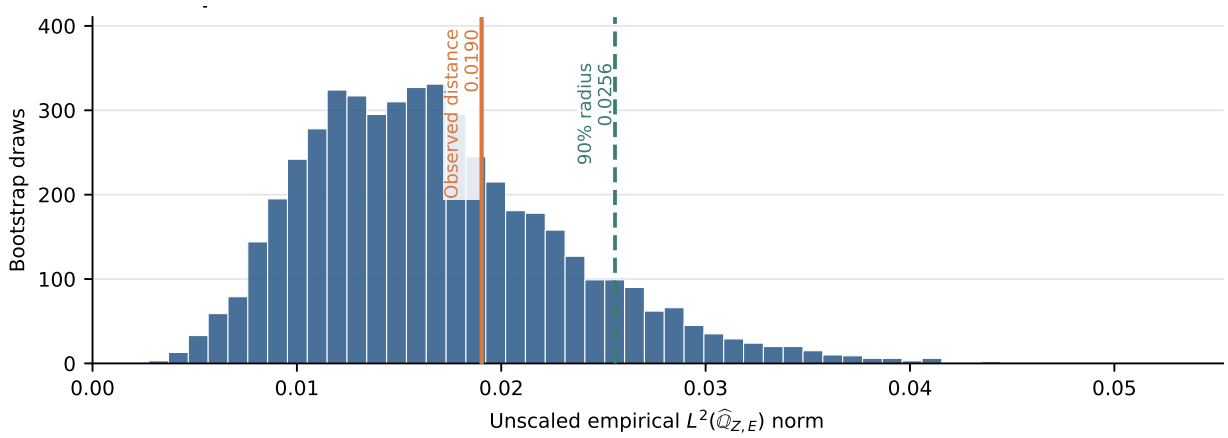


Figure 6: testing whether the probability of holding stock equity is monotone with respect to the usual income. The histogram plot shows the empirical bootstrap distribution based on 5,000 draws. The solid orange line is the observed distance $d(\widehat{f}, \mathcal{M}_\uparrow) = 0.0151$, and the dashed green line is the 90% bootstrap percentile $\hat{c}_{0.10} = 0.0252$. The null hypothesis is not rejected at level 0.10.

7 Discussion

Unknown density ratio. For simplicity and technical convenience, we assumed known density ratio

$\rho(x)$ throughout the paper. While this assumption may hold in semi-supervised learning where unlabelled datasets are abundant, it can be restrictive in many applications. There are several existing methods to estimate the density ratio if it is unknown. Suppose that a sample of source covariates $X_1, \cdots, X_n \overset{\text{i.i.d.}}{\sim} \mathbb{P}$ and target covariates $\widetilde{X}_1, \cdots, \widetilde{X}_N \overset{\text{i.i.d.}}{\sim} \mathbb{Q}$ are available. For example, the least squares density-ratio estimator [Kanamori et al., 2009] approximates $\rho(x)$ by minimizing $\mathbb{E}_{\mathbb{P}_n}[\rho^2(X)] - 2\mathbb{E}_{\mathbb{Q}_N}[\rho(X)]$; the Kullback-Leibler importance estimation procedure (KLIEP, Sugiyama et al. [2008]) maximizes the target likelihood $\mathbb{E}_{\mathbb{Q}_N} \log \rho(X)$ subject to the normalization constraint $\mathbb{E}_{\mathbb{P}_n}\rho(X) = 1$; the kernel mean matching (KMM, Huang et al. [2007], Yu and Szepesvári [2012]) approach aims to minimize the discrepancy between mean feature maps $\|\mathbb{E}_{\mathbb{P}_n}\rho(X)\phi(X) - \mathbb{E}_{\mathbb{Q}_N}\phi(X)\|_{\mathsf{H}}$ where ϕ denotes the canonical feature map. These empirical estimates of $\rho(x)$ are applicable to our procedures. However, their effect on inference remains unclear and challenging to characterize, because the main issue is not only consistency of $\widehat{\rho}$, but also its impact to the first-order approximation, bootstrap approximation, as well as the finite-sample coverage guarantees. We shall leave this in potential future work.

Sub-Gaussian kernel function. The main theorems are established for the bounded kernel where $\sup_{x\in\mathfrak{X}} k(x,x) \leq \kappa^2$. Our methods and theories are still valid if the kernel evaluating function $k_X = k(X, \cdot)$ is sub-Gaussian as an H-valued element in the source distribution. In Appendix A, we provide the theoretical guarantees for the Gaussian and bootstrap approximation under sub-Gaussian kernel functions [Wang, 2026].

Density ratio moment condition and kernel spectral decay rate. The explicit error rates for the coverage probability established in Corollary 4.2 - 4.4 require a bounded high-order moment of the density ratio and sufficiently fast-decaying eigenvalues of the integral operator $\mathcal{L}_Q$. These conditions are stronger than the typical ones in existing literature on the consistency of (IW)KRR. The stronger conditions arise because valid inference procedures require accurate distributional approximation such as the Bahadur representation involving $\sqrt{n}$-scale linearization, Gaussian and bootstrap approximations, and anti-concentration for coverage guarantee. Higher moments of the density ratio make the truncation-induced discrepancy between the target and weighted-source integral operators decay faster. Moreover, faster spectral decay reduces the effective dimension and the spectral-tail error in the infinite-dimensional Gaussian and bootstrap approximations. Together, these conditions place the truncation level, effective dimension, and spectral tail in a joint regime under which all inferential remainders are asymptotically negligible, thereby yielding valid coverage. It is possible that the moment order and spectral-decay requirements in those corollaries are improvable, which we leave in potential future work.

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A Extension I: sub-Gaussian Kernel Function

In addition to the bounded-kernel condition assumed throughout the paper, this section considers instead the sub-Gaussian feature regularity and derives the corresponding Gaussian and bootstrap coupling bounds and their specific spectral-rate consequences.

Definition A.1 (Sub-Gaussian in Hilbert Space)**.** *For a square-integrable random element X in a Hilbert space G, with its expectation understood as a Bochner mean, we say X is G-b-sub-Gaussian if*

$$\log \mathbb{E}_{\mathbb{P}}\left[\exp\langle X-\mathbb{E}X,t\rangle_{\mathsf{G}}\right] \le \frac{b^2}{2}\mathbb{E}_{\mathbb{P}}\left(\langle X-\mathbb{E}X,t\rangle_{\mathsf{G}}^2\right), \quad \forall t\in\mathsf{G}.$$

Assumption A.1 (Sub-Gaussian Feature)**.** *For some $b_0 \ge 1$, the feature element $k_X = k(X,\cdot)$ is H-b_0-sub-Gaussian. Equivalently, for every $t \in \mathsf{H}$,*

$$\begin{aligned}\log \mathbb{E}_{\mathbb{P}}\left[\exp\langle k_X-\mathbb{E}_{\mathbb{P}}k_X,t\rangle_{\mathsf{H}}\right] &= \log \mathbb{E}_{\mathbb{P}}\left[\exp\{t(X)-\mathbb{E}_{\mathbb{P}}[t(X)]\}\right]\\ &\le \frac{b_0^2}{2}\mathbb{E}_{\mathbb{P}}\left[\{t(X)-\mathbb{E}_{\mathbb{P}}[t(X)]\}^2\right].\end{aligned}$$

For $\tau \ge \tau_0$, let $w_\tau(x) := \min\{\rho(x),\tau\}$. Since importance weighting is used, we additionally assume that, for every $\tau \ge \tau_0$ and $t \in \mathsf{H}$,

$$\begin{aligned}\log \mathbb{E}_{\mathbb{P}}\left[\exp\langle w_\tau(X)k_X-\mathbb{E}_{\mathbb{P}}[w_\tau(X)k_X],t\rangle_{\mathsf{H}}\right] &= \log \mathbb{E}_{\mathbb{P}}\left[\exp\{w_\tau(X)t(X)-\mathbb{E}_{\mathbb{P}}[w_\tau(X)t(X)]\}\right]\\ &\le \frac{b_0^2}{2}\mathbb{E}_{\mathbb{P}}\left[w_\tau(X)^2t(X)^2\right].\end{aligned}$$

The weighted-feature condition is imposed on the truncation family and does not follow from sub-Gaussianity of k_X alone.

We first verify that the influence elements U_i in the Bahadur representation (3.1) satisfy the required sub-Gaussian condition.

Lemma A.1. *Under Assumptions 4.1, 4.2, and A.1, if $\tau_n \ge \tau_0$, then U_i is $\mathsf{L}^2(\mathbb{Q})$-b-sub-Gaussian with*

$$b = C_{sg}^3 \cdot \frac{(\kappa\|f_*\|+\overline{\sigma})}{\underline{\sigma}} b_0,$$

where C_{sg} is a universal constant as in Vershynin [2018].

The sub-Gaussian condition yields the following new Gaussian and bootstrap coupling bounds, whereas the Bahadur representation and feasible-bootstrap approximation remain as in the main analysis.

Proposition A.1 (Gaussian coupling under sub-Gaussian regularity)**.** *Suppose $U_1,\dots,U_n$ are i.i.d., centered, $\mathsf{L}^2(\mathbb{Q})$-b-sub-Gaussian random elements with covariance operator $\mathcal{V}$. For any $\eta \in (0,1)$ and any integer $m \in \{1,\dots,n\}$, if $n \ge \max\{2,\log(2/\eta)\}$, then there exists $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})$ such that, with probability at least $1-\eta$,*

$$\left\|\left(\frac{1}{\sqrt{n}}\sum_{i=1}^n U_i\right) - Z\right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim b\sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})\sqrt{\log\left(\frac{6}{\eta}\right)} + \|\mathcal{V}\|_{\mathsf{L}^2(\mathbb{Q})}^{\frac{1}{2}} b^3 \left\{\frac{m^{3/2}\log(n)^2}{\sqrt{n}}\right\}(3/\eta)^{1/\log(mn)}.$$

Proposition A.2 (Bootstrap coupling under sub-Gaussian regularity). *Suppose $U_1, \dots, U_n$ are i.i.d., centered, $\mathsf{L}^2(\mathbb{Q})$-b-sub-Gaussian random elements. Let g be the isonormal Gaussian process in $\mathsf{L}^2(\mathbb{Q})$. Then there exists a random element $G \overset{\mathrm{d.}}{=} \widehat{\mathcal{V}}_{\mathsf{ora}}^{1/2} g \big| \mathcal{D}$ such that, conditioned on $\mathcal{D}$, with total probability at least $1 - 6\eta$,*

$$\left\| \mathcal{V}^{1/2} g - G \right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \inf_{m \geq 1} \left[m^{1/4} \left\{ \frac{b^4 \sigma^4_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})}{n} \right\}^{1/4} + \sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \right] \left(\log\left(\frac{2}{\eta}\right) \right)^{3/2}.$$

where the covariance operator $\mathcal{V} = \mathbb{E}[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i]$.

Combining these two coupling bounds with the unchanged bias, Bahadur, feasible-bootstrap, and anti-concentration arguments from the main analysis yields sub-Gaussian counterparts of Corollaries 4.2–4.4. Throughout the following corollaries, fix $\alpha \in (0,1)$, take $\eta \asymp n^{-1}$, and let $n \to \infty$. Suppose Assumptions 4.1, 4.2, 4.3, 4.5, and A.1 hold, with source exponent $s \geq 1/2$ and $\tau_n \geq \tau_0$ for all sufficiently large n. In each corollary, the stated density-ratio and truncation conditions verify Assumption 4.4 with $\mathfrak{C}_1$ bounded uniformly in n. All other constants in these assumptions are independent of n.

Corollary A.1 (bounded density ratio under sub-Gaussian regularity). *Suppose that $\rho(X) \leq \overline{\rho}$ $\mathbb{P}$-almost surely, and let $\tau_n = C_0$ for a sufficiently large fixed constant $C_0 \geq \max\{\overline{\rho}, \tau_0\}$.*

(1) If $\mu_k \asymp k^{-\beta}$ for some $\beta > 3$ and all $k \geq 1$, then, by choosing

$$\lambda_n \asymp n^{-\frac{2\beta}{3\beta+2}},$$

we obtain

$$\left| \mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha \right| \lesssim n^{-\frac{\beta-3}{2(3\beta+2)}} \log^3 n.$$

(2) If $\mu_k \asymp \exp(-ck^\zeta)$ for some $c, \zeta > 0$ and all $k \geq 1$, then, by choosing $\lambda_n \asymp n^{-2/3}$, we obtain

$$\left| \mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha \right| \lesssim n^{-1/6} (\log n)^{3+1/\zeta}.$$

Corollary A.2 (sub-exponential density ratio under sub-Gaussian regularity). *Suppose that the density ratio satisfies the sub-exponential tail condition in Definition 4.1.*

(1) If $\mu_k \asymp k^{-\beta}$ for some $\beta > 3$ and all $k \geq 1$, then, by choosing

$$\tau_n \asymp n^{\frac{\beta-3}{6(3\beta+2)}}, \qquad \lambda_n \asymp n^{-\frac{2\beta}{3\beta+2}},$$

we obtain

$$\left| \mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha \right| \lesssim n^{-\frac{\beta-3}{4(3\beta+2)}} \log^3 n.$$

(2) If $\mu_k \asymp \exp(-ck^\zeta)$ for some $c, \zeta > 0$ and all $k \geq 1$, then, by choosing

$$\tau_n \asymp n^{1/150}, \qquad \lambda_n \asymp n^{-2/3},$$

we obtain

$$\left| \mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha \right| \lesssim n^{-47/300} (\log n)^{3+1/\zeta}.$$

Corollary A.3 (bounded θ-th moment under sub-Gaussian regularity)**.** *Suppose that* $\mathbb{E}_{\mathbb{P}}[\rho(X)^\theta] \leq \Omega_\theta < \infty$ *for some* $\theta > 5$*. If* $\mu_k \asymp \exp(-ck^\zeta)$ *for some* $c, \zeta > 0$ *and all* $k \geq 1$*, then, by choosing*

$$\lambda_n \asymp n^{-\frac{8(\theta^2-1)}{3(4\theta^2+3\theta-3)}}, \qquad \tau_n \asymp n^{\frac{19\theta+1}{6(4\theta^2+3\theta-3)}},$$

we obtain

$$\left|\mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha\right| \lesssim n^{-\frac{(\theta-5)(8\theta+1)}{12(4\theta^2+3\theta-3)}} (\log n)^{3+1/\zeta}.$$

B Extension II: Bias-Corrected Multiplier Bootstrap

The asymptotic validity of the confidence ball constructed by Algorithm 1 relies on undersmoothing to render the deterministic bias

$$\sqrt{n}(f_{P^w} - f_*) = -\sqrt{n}\,\lambda_n(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1} f_*$$

negligible in $\mathsf{L}^2(\mathbb{Q})$. The procedure developed here instead estimates the corresponding bias-correction term by replacing $(\mathcal{L}_{P^w}, f_*)$ with $(\mathcal{L}_{P_n^w}, \widehat{f})$, yielding, with $\widehat{\alpha} := (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\mathbf{y}$,

$$\widehat{b} := \sqrt{n}\,\lambda_n(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1}\widehat{f} = \sqrt{n}\,\mathbf{k}_{\mathbf{x}}^\top(\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1} n\lambda_n \mathbf{W}^{-1}\widehat{\alpha}. \tag{B.1}$$

As $\widehat{b}$ estimates the negative of $\sqrt{n}(f_{P^w} - f_*)$, the bias-adjusted analogue of $\sqrt{n}(\widehat{f} - f_*)$ is $B - \widehat{b}$. With $\widehat{d} := n\lambda_n \mathbf{W}^{-1}\widehat{\alpha}$, equations (3.2) and (B.1) give

$$B - \widehat{b} = \sqrt{n}\,\mathbf{k}_{\mathbf{x}}^\top(\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\left\{\mathrm{diag}(\widehat{\boldsymbol{\varepsilon}})\mathbf{z} - \widehat{d}\right\}.$$

Algorithm 2 summarizes this practical bias-corrected variant of Algorithm 1, which may improve finite-sample calibration accuracy.

C Extension III: Residual Bootstrap

The residual bootstrap resamples with replacement from the empirical distribution of the centered fitted residuals. Conditional on the observed covariates, it generates bootstrap responses $Y_{i,\mathrm{res}}^{(b)} = \widehat{f}(X_i) + \widehat{\varepsilon}_{i,\mathrm{res}}^{(b)}$, $i = 1, \ldots, n$, and refits IWKRR with the same $w_n(x)$ and λ_n.

Step 1: Gaussian and bootstrap approximation. The residual-bootstrap coverage guarantee can be established using the same proof framework as the multiplier bootstrap. Because the point estimator is unchanged, it suffices to replace the bootstrap approximation in (4.3) by its residual-bootstrap analogue: conditionally on $\mathcal{D}$, $\sqrt{n}(\widehat{f}_{\mathrm{res}}^{(b)} - \widehat{f})$ can be approximated by the same Gaussian element as in (4.2). If the corresponding error is $\Delta_{B,\mathrm{res}}$, the proof of Theorem 4.1 yields the same coverage bound with Δ_B replaced by $\Delta_{B,\mathrm{res}}$.

To identify the required bootstrap approximation, let

$$\psi_n(x) := w_n(x)(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1} k_x, \qquad \widehat{\psi}_{i,n} := w_n(X_i)(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1} k_{X_i}.$$

The leading noise component of the influence function is

$$U_{i,\varepsilon} := \varepsilon_i \psi_n(X_i).$$

Algorithm 2 Bias-corrected multiplier bootstrap for $\mathsf{L}^2(\mathbb{Q})$-norm confidence balls

Require: Data $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$, kernel $k(\cdot,\cdot)$, density ratio $\rho(x) = q(x)/p(x)$, weight $w_n(x) = \min\{\rho(x), \tau_n\}$, confidence level $1-\alpha$, bootstrap repetitions M, regularization parameter λ_n;

1: Form $\mathbf{K}$, $\mathbf{W}$, $\mathbf{y}$, $\mathbf{k}_{\mathbf{x}}$, and $\mathbf{K}_{\mathbb{Q}}$ as in Algorithm 1.

2: Compute $\widehat{\alpha} = (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\mathbf{y}$ and $\widehat{f} = \mathbf{k}_{\mathbf{x}}^\top \widehat{\alpha}$.

3: Compute the residual vector $\widehat{\boldsymbol{\varepsilon}} = (Y_1 - \widehat{f}(X_1), \dots, Y_n - \widehat{f}(X_n))^\top$.

4: Form the dual bias-correction vector $\widehat{d} = n\lambda_n \mathbf{W}^{-1}\widehat{\alpha}$.

5: **for** $b = 1, \dots, M$ **do**

6: Sample $\mathbf{z}^{(b)} \sim N(0, \mathbf{I}_n - \mathbf{1}\mathbf{1}^\top/n)$.

7: Set the bias-corrected coefficient vector

$$\mathbf{a}_{\mathrm{bc}}^{(b)} = \sqrt{n}\,(\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\{\mathrm{diag}(\widehat{\boldsymbol{\varepsilon}})\mathbf{z}^{(b)} - \widehat{d}\}.$$

8: Compute the bootstrap statistic

$$T_{\mathrm{bc}}^{(b)} = \|\mathbf{k}_{\mathbf{x}}^\top \mathbf{a}_{\mathrm{bc}}^{(b)}\|_{\mathsf{L}^2(\mathbb{Q})} = \sqrt{(\mathbf{a}_{\mathrm{bc}}^{(b)})^\top \mathbf{K}_{\mathbb{Q}} \mathbf{a}_{\mathrm{bc}}^{(b)}}.$$

9: **end for**

10: Let $\hat{r}_\alpha^{\mathrm{bc}}$ be the empirical $(1-\alpha)100\%$-percentile of $\{T_{\mathrm{bc}}^{(b)}\}_{b=1}^M$.

11: Define $\mathsf{CS}_\alpha^{\mathrm{bc}}(\widehat{f}) = \{f \in \mathsf{H} : \sqrt{n}\,\|\widehat{f} - f\|_{\mathsf{L}^2(\mathbb{Q})} \le \hat{r}_\alpha^{\mathrm{bc}}\}$.

Ensure: the IWKRR $\widehat{f}$ and the bias-corrected critical value $\hat{r}_\alpha^{\mathrm{bc}}$.

The normal equations yield the exact decompositions

$$\widehat{f} - f_* = \frac{1}{n}\sum_{i=1}^n \varepsilon_i \widehat{\psi}_{i,n} - \lambda_n(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1} f_*, \tag{C.1}$$

$$\widehat{f}_{\mathrm{res}}^{(b)} - \widehat{f} = \frac{1}{n}\sum_{i=1}^n \widehat{\varepsilon}_{i,\mathrm{res}}^{(b)} \widehat{\psi}_{i,n} - \lambda_n(\mathcal{L}_{P_n^w} + \lambda_n \mathcal{I})^{-1} \widehat{f}. \tag{C.2}$$

The same Bahadur, undersmoothing, and operator-approximation methods used in the main analysis can be applied to the exact decompositions above to obtain the first-order representations

$$\sqrt{n}(\widehat{f} - f_*) \approx_{\mathsf{L}^2(\mathbb{Q})} \frac{1}{\sqrt{n}}\sum_{i=1}^n \varepsilon_i \psi_n(X_i),$$

$$\sqrt{n}(\widehat{f}_{\mathrm{res}}^{(b)} - \widehat{f}) \approx_{\mathsf{L}^2(\mathbb{Q})} \frac{1}{\sqrt{n}}\sum_{i=1}^n \widehat{\varepsilon}_{i,\mathrm{res}}^{(b)} \widehat{\psi}_{i,n},$$

where we ignore the bias $f_{P^w} - f_*$ for simplicity. Starting from these representations, the bootstrap-approximation and coverage arguments in the main text can be adapted to the residual bootstrap, subject to the covariance-consistency condition below.

Step 2: covariance comparison. The population covariance operator of $\frac{1}{\sqrt{n}}\sum_{i=1}^n \varepsilon_i \psi_n(X_i)$ is

$$\mathcal{V}_{\varepsilon,n} := \mathbb{E}\big[U_{i,\varepsilon} \otimes_{\mathsf{L}^2(\mathbb{Q})} U_{i,\varepsilon}\big] = \mathbb{E}\big[\mathbb{E}[\varepsilon^2 \mid X]\,\psi_n(X) \otimes_{\mathsf{L}^2(\mathbb{Q})} \psi_n(X)\big]\,. \tag{C.3}$$

Algorithm 3 Residual bootstrap for $\mathsf{L}^2(\mathbb{Q})$-norm confidence balls

Require: Data $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$, kernel $k(\cdot,\cdot)$, density ratio $\rho(x) = q(x)/p(x)$, weight $w_n(x) = \min\{\rho(x), \tau_n\}$, confidence level $1-\alpha$, bootstrap repetitions M, regularization parameter λ_n.

Form $\mathbf{K}$, $\mathbf{W}$, $\mathbf{y}$, $\mathbf{k}_\mathbf{x}$, and $\mathbf{K}_\mathbb{Q}$ as in Algorithm 1.

Compute $\widehat{\alpha} = (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\mathbf{y}$ and $\widehat{f} = \mathbf{k}_\mathbf{x}^\top \widehat{\alpha}$.

Compute the residual vector $\widehat{\boldsymbol{\varepsilon}} = (Y_1 - \widehat{f}(X_1), \ldots, Y_n - \widehat{f}(X_n))^\top$.

Center the fitted residuals by setting

$$\bar{\widehat{\varepsilon}} := \frac{1}{n}\sum_{j=1}^n \widehat{\varepsilon}_j, \qquad \widehat{\varepsilon}_j^\circ := \widehat{\varepsilon}_j - \bar{\widehat{\varepsilon}}.$$

for $b = 1, \ldots, M$ **do**

  Sample $\widehat{\varepsilon}_{1,\mathrm{res}}^{(b)}, \ldots, \widehat{\varepsilon}_{n,\mathrm{res}}^{(b)}$ independently with replacement from $\{\widehat{\varepsilon}_1^\circ, \ldots, \widehat{\varepsilon}_n^\circ\}$.

  Form the bootstrap-response vector

$$\mathbf{y}_{\mathrm{res}}^{(b)} := \big(\widehat{f}(X_1) + \widehat{\varepsilon}_{1,\mathrm{res}}^{(b)}, \ldots, \widehat{f}(X_n) + \widehat{\varepsilon}_{n,\mathrm{res}}^{(b)}\big)^\top.$$

  Refit IWKRR with the same regularized inverse:

$$\widehat{f}_{\mathrm{res}}^{(b)} := \mathbf{k}_\mathbf{x}^\top (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1} \mathbf{y}_{\mathrm{res}}^{(b)}.$$

  Define the coefficient perturbation

$$\mathbf{a}_{\mathrm{res}}^{(b)} := (\mathbf{K} + n\lambda_n \mathbf{W}^{-1})^{-1}\big(\mathbf{y}_{\mathrm{res}}^{(b)} - \mathbf{y}\big), \qquad \widehat{f}_{\mathrm{res}}^{(b)} - \widehat{f} = \mathbf{k}_\mathbf{x}^\top \mathbf{a}_{\mathrm{res}}^{(b)}.$$

  Compute the bootstrap statistic

$$T_{\mathrm{res}}^{(b)} := \|\widehat{f}_{\mathrm{res}}^{(b)} - \widehat{f}\|_{\mathsf{L}^2(\mathbb{Q})} = \|\mathbf{k}_\mathbf{x}^\top \mathbf{a}_{\mathrm{res}}^{(b)}\|_{\mathsf{L}^2(\mathbb{Q})} = \Big\{(\mathbf{a}_{\mathrm{res}}^{(b)})^\top \mathbf{K}_\mathbb{Q} \mathbf{a}_{\mathrm{res}}^{(b)}\Big\}^{1/2}.$$

end for

Let $\hat{c}_{\alpha,n}^{\mathrm{res}}$ be the empirical $(1-\alpha)100\%$-percentile of $\{T_{\mathrm{res}}^{(b)}\}_{b=1}^M$.

Define $\mathsf{CS}_\alpha^{\mathrm{res}}(\widehat{f})$ by replacing $\hat{c}_{\alpha,n}$ with $\hat{c}_{\alpha,n}^{\mathrm{res}}$ in (3.3).

Ensure: The residual-bootstrap confidence ball $\mathsf{CS}_\alpha^{\mathrm{res}}(\widehat{f})$.

Conditional on $\mathcal{D}$, the covariance operator of the leading term generated by Algorithm 3 is

$$\widehat{\mathcal{V}}_{\text{res}} := \widehat{\sigma}^2_{\text{res}}\, \mathbb{E}_n\Big[\widehat{\psi}_{i,n} \otimes_{\mathsf{L}^2(\mathbb{Q})} \widehat{\psi}_{i,n}\Big], \qquad \widehat{\sigma}^2_{\text{res}} := \mathbb{E}_n\big[(\widehat{\varepsilon}_i^{\circ})^2\big]. \tag{C.4}$$

Under conditional homoskedasticity, $\mathbb{E}[\varepsilon^2 \mid X] = \mathbb{E}[\varepsilon^2]$ almost surely, (C.3) factorizes as $\mathcal{V}_{\varepsilon,n} = \mathbb{E}[\varepsilon^2]\,\mathbb{E}[\psi_n(X) \otimes_{\mathsf{L}^2(\mathbb{Q})} \psi_n(X)]$. Write

$$\widehat{\mathcal{A}}_n := \mathbb{E}_n\Big[\widehat{\psi}_{i,n} \otimes_{\mathsf{L}^2(\mathbb{Q})} \widehat{\psi}_{i,n}\Big], \qquad \mathcal{A}_n := \mathbb{E}\big[\psi_n(X) \otimes_{\mathsf{L}^2(\mathbb{Q})} \psi_n(X)\big].$$

Covariance consistency is therefore ensured by

$$\big|\widehat{\sigma}^2_{\text{res}} - \mathbb{E}[\varepsilon^2]\big| \left\|\widehat{\mathcal{A}}_n\right\|_{\text{HS},\mathsf{L}^2(\mathbb{Q})} + \mathbb{E}[\varepsilon^2] \left\|\widehat{\mathcal{A}}_n - \mathcal{A}_n\right\|_{\text{HS},\mathsf{L}^2(\mathbb{Q})} \xrightarrow{\mathbb{P}} 0, \quad \text{as } n \to \infty.$$

Step 3: comparison with the multiplier bootstrap. Notice that $\widehat{V}_i = \widehat{\varepsilon}_i \widehat{\psi}_{i,n}$. The conditional covariance operators of the multiplier and residual bootstraps are, respectively,

$$\widehat{\mathcal{V}} = \mathbb{E}_n\Big[\widehat{\varepsilon}_i^2\big(\widehat{\psi}_{i,n} \otimes_{\mathsf{L}^2(\mathbb{Q})} \widehat{\psi}_{i,n}\big)\Big] - \mathbb{E}_n[\widehat{V}_i] \otimes_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}_n[\widehat{V}_i], \qquad \widehat{\mathcal{V}}_{\text{res}} = \widehat{\sigma}^2_{\text{res}} \mathbb{E}_n[\widehat{\psi}_{i,n} \otimes_{\mathsf{L}^2(\mathbb{Q})} \widehat{\psi}_{i,n}].$$

Thus, the multiplier bootstrap weights each $\widehat{\psi}_{i,n} \otimes_{\mathsf{L}^2(\mathbb{Q})} \widehat{\psi}_{i,n}$ by its paired squared residual $\widehat{\varepsilon}_i^2$, whereas the residual bootstrap uses the common factor $\widehat{\sigma}^2_{\text{res}}$ for all i. The multiplier bootstrap can therefore accommodate conditional heteroskedasticity, whereas the residual bootstrap requires the covariance-matching condition above.

D Extension IV: mis-coverage rate for heavy-tailed density ratio

For ease of exposition, the mis-coverage rate is presented in Corollary 4.4 only for the case when $\mathcal{L}_Q$ has exponential-decay eigenvalues. The following corollary showcases the mis-coverage rate if $\mathcal{L}_Q$ has polynomial-decay eigenvalues.

Corollary D.1 (Polynomial spectral decay under a bounded θ-th moment)**.** *Under the assumptions of Section 4.3.4, suppose that $\mathbb{E}_{\mathbb{P}}[\rho(X)^\theta] \le \Omega_\theta < \infty$ for some $\theta > 2$. If $\mu_k \asymp k^{-\beta}$ for some $\beta > 5$ and all $k \ge 1$, assume that*

$$\theta - 1 > \frac{4(\beta+1)^2(2\beta+5)}{\beta(\beta-5)(\beta+10)}.$$

Take $d = 1/\beta$ in Assumption 4.3 and choose

$$\delta = \frac{\beta(\theta-1)(\beta-5)(\beta+10) - 4(\beta+1)^2(2\beta+5)}{3(2\beta+5)(\beta+10)\{3\beta(\theta-1)(\beta+3) + 2(\beta+1)^2\}}.$$

By choosing

$$\lambda_n \asymp n^{-\frac{2\beta(\theta-1)(\beta+1)}{3\beta(\theta-1)(\beta+3)+2(\beta+1)^2}},$$
$$\tau_n \asymp n^{\frac{\beta(\theta-1)(\beta-5)(\beta+10)+2(3\beta+28)(\beta+1)^2(2\beta+5)}{3(2\beta+5)(\beta+10)\{3\beta(\theta-1)(\beta+3)+2(\beta+1)^2\}}},$$

we obtain

$$\big|\mathbb{P}\big(f_* \notin \mathsf{CS}_\alpha(\widehat{f})\big) - \alpha\big| \lesssim n^{-\frac{(\beta+9)\{\beta(\theta-1)(\beta-5)(\beta+10)-4(\beta+1)^2(2\beta+5)\}}{2(\beta+10)(2\beta+5)\{3\beta(\theta-1)(\beta+3)+2(\beta+1)^2\}}} \log^3 n.$$

In particular, the displayed rate converges to zero under any of the following sufficient regimes:

$$\theta \ge 9,\ \beta \ge 113; \qquad \theta > 9,\ \beta > 5 + \frac{144}{5(\theta-9)}; \qquad \theta \ge 10,\ \beta \ge 20.$$

Another sufficient regime is $\theta \ge 11$ and $\beta \ge 15$.

E Additional Results of SCF Data Analysis

This appendix provides additional data, implementation, and diagnostic details for Studies 1 and 2 reported in Section 6, and presents additional results from Studies 3 and 4.

E.1 Data description.

Studies 1–3 use the 2022 implicate-1 sample described in Section 6.1. Study 4 additionally uses implicate 1 of the 2019 public summary extract, comprising 5,777 distinct families.

To illustrate how the released weights change covariate distribution, we partition the 33 scalar coordinates into a collection $\mathcal{B}$ of 12 demographic and household covariate blocks. For a coordinate Z observed on the n records, define the unweighted moments $\widehat{\mu}_P(Z) := n^{-1}\sum_{i=1}^n Z_i$ and $\widehat{\sigma}_P^2(Z) := n^{-1}\sum_{i=1}^n\{Z_i - \widehat{\mu}_P(Z)\}^2$, and the released-weight moments $\widehat{\mu}_Q(Z) := n^{-1}\sum_{i=1}^n \rho_i Z_i$ and $\widehat{\sigma}_Q^2(Z) := n^{-1}\sum_{i=1}^n \rho_i\{Z_i - \widehat{\mu}_Q(Z)\}^2$. The coordinate-level signed standardized difference and the score of each block $b \in \mathcal{B}$ are

$$D(Z) := \frac{\widehat{\mu}_Q(Z) - \widehat{\mu}_P(Z)}{\{[\widehat{\sigma}_P^2(Z) + \widehat{\sigma}_Q^2(Z)]/2\}^{1/2}}, \qquad S_b := \max_{Z\in b}|D(Z)|.$$

We rank the 12 block scores S_b, and Figure 7 displays the six highest-ranked blocks.

E.2 Study 1: predicting household net worth

Variables and preprocessing. Table 3 describes the variables used in the 33-dimensional design summarized in Section 6.2. The work-status block is given by the four-level public-file variable `OCCAT1`.

Table 3: Additional details of variables used for Study 1.

Object	SCF variable	Meaning	Processing
Response	`NETWORTH`	Total family net worth	asinh(`NETWORTH`/100000), truncated to zero for negative net worth.
Target weight	`WGT`	# of U.S. families represented	normalized to mean one
Continuous coordinates	`AGE` `KIDS`	Age of the reference person; Number of children	standardized.
Ordered block	`EDCL`	Four levels of education	$\mathbf{1}\{\texttt{EDCL} \geq k\}$, $k = 2, 3, 4$.

Unordered blocks	`HHSEX`
	`MARRIED`
	`LF`
	`LIFECL`
	`FAMSTRUCT`
	`RACECL`
	`OCCAT1`
	`INDCAT`
	`HHOUSES`
		Sex of the reference person;
		Married or not;
		Working versus not working;
		Six categories of life cycle;
		Five-class family structure;
		(non)White, (non)Hispanic;
		Four-category work status;
		Three industry classes;
		own residence or not
			one-hot encodings yielding 28 coordinates, three education indicators, and two continuous coordinates; total number: 33.

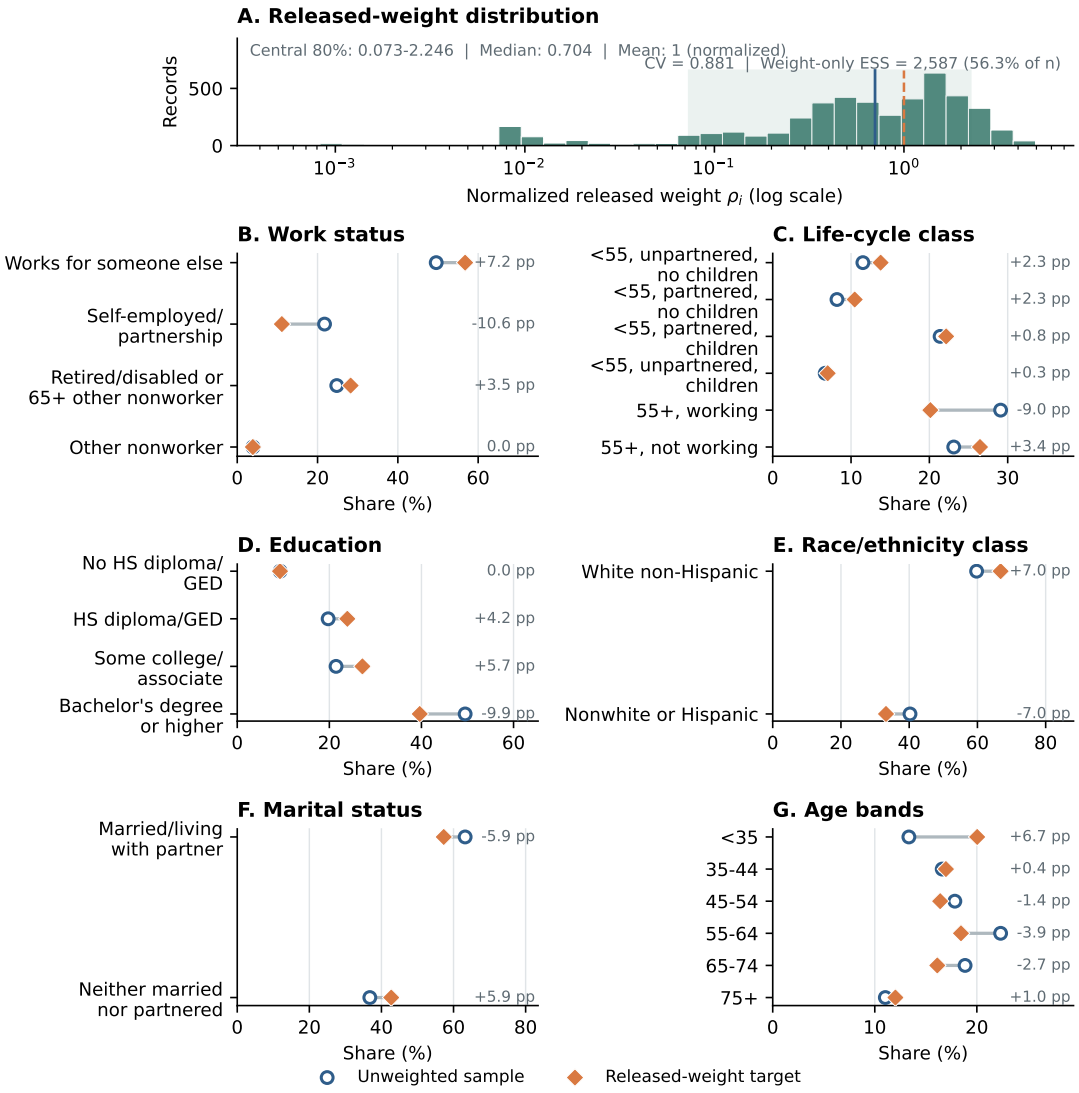


Figure 7: Distribution of survey weights and their impact in 2022 implicate-1 records. Panel A summarizes the normalized weights; the shaded band is the 10–90% percentile range, and the vertical lines mark the median and normalized mean. Panels B–G compare unweighted and weighted shares for the six blocks with the largest S_b; labels are target-minus-sample percentage-point differences.

KRR and parameters tuning on $\mathcal{D}_0$. Within the 980/420 inner split of $\mathcal{D}_0$, the 980 training families determine the continuous-variable standardization constants and categorical encoding. This transformation is applied directly to the 420 validation families. Writing $n_{\text{tr}} = 980$, the ordinary KRR search uses

$$\lambda_n(a) = a n_{\text{tr}}^{-0.6}, \qquad a \in \mathcal{A}_\lambda := \{0.0625, 0.125, 0.25, 0.5, 1, 1.5, 2, 3, 4\},$$

and are compared by unweighted validation mean squared error. The selected multiplier $a = 0.0625$ is evaluated at $n = |\mathcal{D}_0| = 1{,}400$, giving $\lambda_{\text{KRR}} = 0.000809475$.

IWKRR and multiplier bootstrap on $\mathcal{D}_I$. The same 980/420 split of $\mathcal{D}_0$ is used to tune the IWKRR parameters. Writing $n_{\text{tr}} = 980$, the joint search uses

$$\lambda_n(a) = a n_{\text{tr}}^{-0.6}, \quad a \in \mathcal{A}_\lambda, \qquad \tau_n(b) = b n_{\text{tr}}^{0.06}, \quad b \in \mathcal{A}_\tau := \{0.25, 0.5, 1, 1.5, 2\},$$

together with $\tau = 10$ as the truncation cap. After validation, the selected multipliers are fixed at $a = 0.25$ and $b = 1$. Substituting $n = |\mathcal{D}_I| = 1{,}800$ into the two sequences gives $\lambda_{\text{IW}} = 0.002784688$ and $\tau_n = 1.567893$. For the final IWKRR fit, the weights are

$$w_i = \rho_i \wedge \tau_n, \qquad \bar{w}_i = \frac{|\mathcal{D}_I| w_i}{\sum_{\ell \in \mathcal{D}_I} w_\ell},$$

and the final inference-sample fit solves

$$\widehat{f}_{\text{IW}} = \arg\min_{f \in \mathsf{H}} \left[\frac{1}{|\mathcal{D}_I|} \sum_{i \in \mathcal{D}_I} \bar{w}_i \{Y_i - f(X_i)\}^2 + \lambda_{\text{IW}} \|f\|_{\mathsf{H}}^2 \right].$$

The multiplier bootstrap is then carried out using all 1,800 records in $\mathcal{D}_I$.

Confidence set construction and p-values. The confidence-set radius in Section 6.2 corresponds to the 90% critical value 4.942249 on the $\sqrt{1{,}800}$ scale. Let R_b denote the target-norm bootstrap error in draw b. For a candidate f_0, the corresponding plus-one bootstrap p-value is

$$\widehat{p}(f_0) = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left\{ R_b \geq \|\widehat{f}_{\text{IW}} - f_0\|_{\widehat{\mathbb{Q}}} \right\}}{B + 1}.$$

All bootstrap statistics are smaller than the four observed distances, so $\widehat{p}(f_0) = 1/(B+1) \approx 0.0002$ for all the four proposed candidate f_0.

Response distribution. Figure 8 compares the empirical distribution of household net worth with and without re-weighting.

Covariate distributions before and after weighting. Figure 9 reports the leading change of coordinates from each of the 12 blocks. For example, the label "Work status: Self-employed/partnership" means that, among the one-hot coordinates in the work-status block, the self-employed/partnership indicator has the largest $|D(Z)|$. Figure 10 reports $|D(Z)|$ for every coordinate. These standardized differences illustrate the change of covariates' distribution under reweighting.

E.3 Study 2: does higher income increase the probability of holding equity?

The training set $\mathcal{D}_t$ is divided into a fitting subset $\mathcal{D}_{t,\text{fit}}$ of size 1,260 and a validation subset $\mathcal{D}_{t,\text{val}}$ of size 540 for parameter tuning. The RBF kernel with $\gamma = 1$ is used after standardizing the income variable.

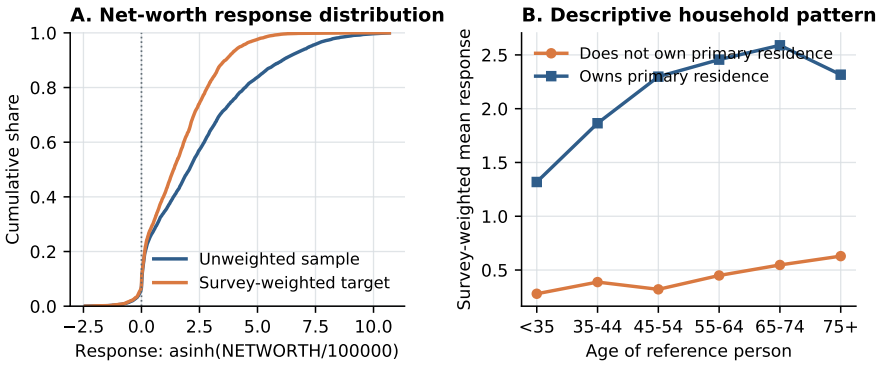


Figure 8: empirical distribution of household net worth with and without re-weighting in Study 1, where the full-sample means are 2.497 and 1.587, respectively.

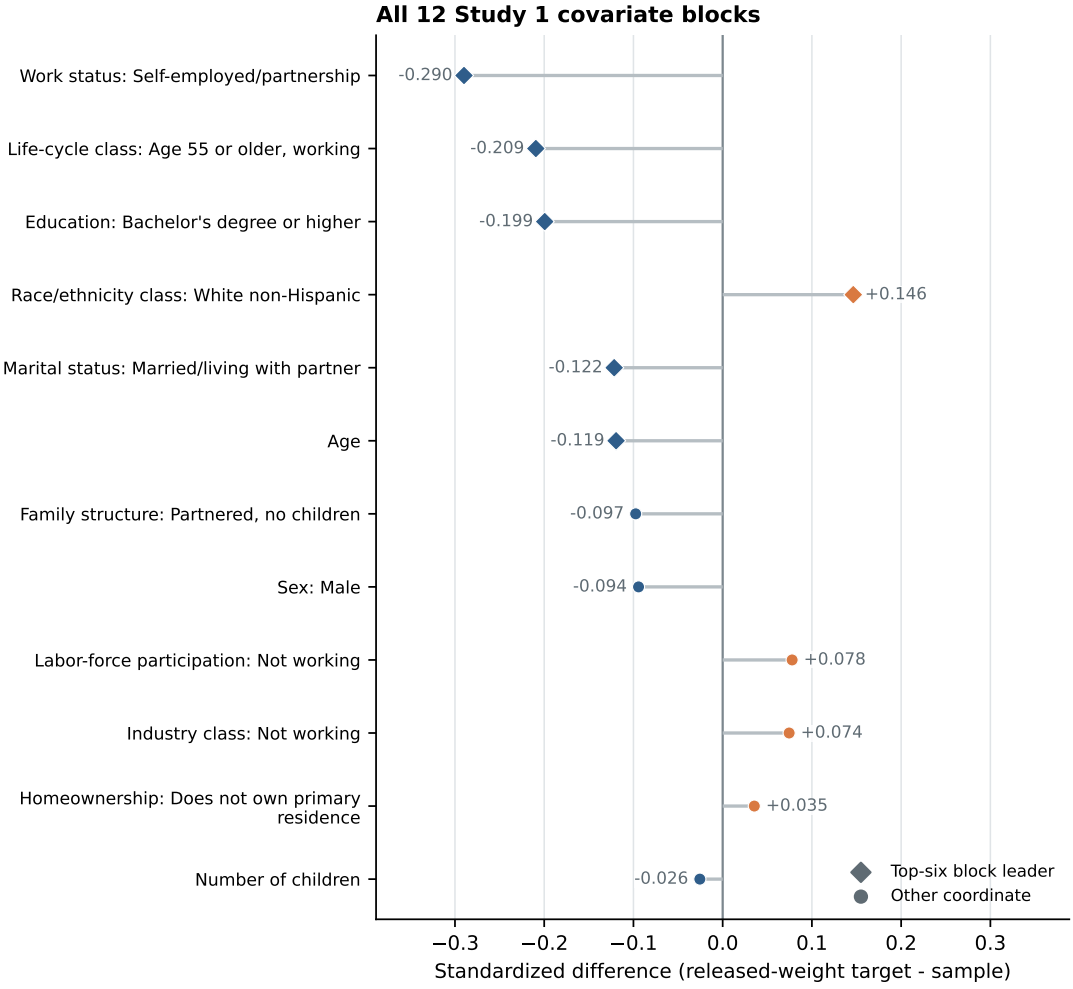


Figure 9: block-level source–target comparison of covariates used in Study 1. The coordinate with the largest $|D(Z)|$ within each block is presented. The top-six covariates are already reported in Figure 7.

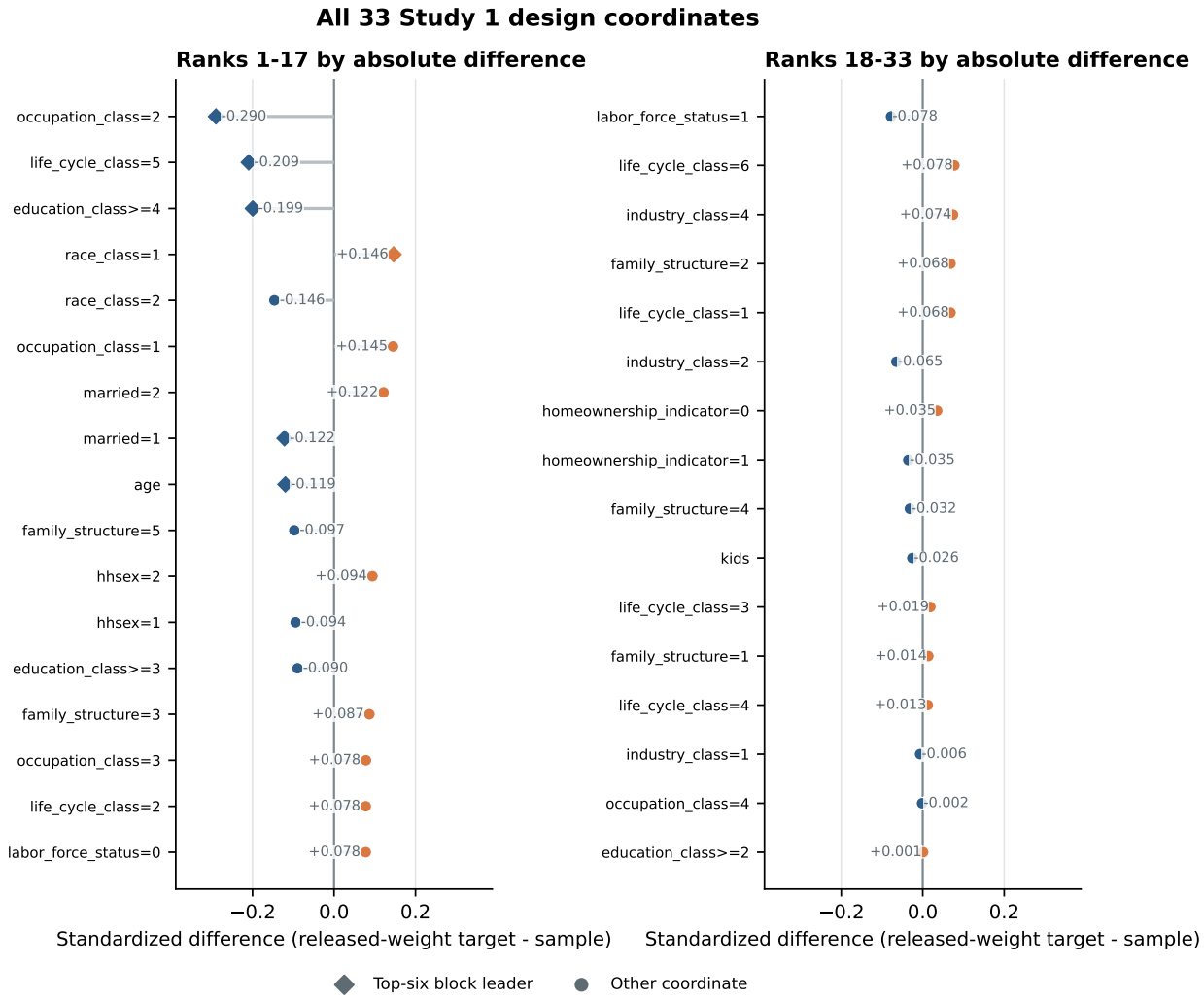


Figure 10: Coordinate-level source–target comparison for all 33 covariates used in Study 1, ordered by $|D(Z)|$. The top-six covariates are already reported in Figure 7.

Parameter tuning. Writing $n_t = |\mathcal{D}_t| = 1{,}800$, the validation search uses

$$\lambda_n(a) = a n_t^{-0.6}, \quad a \in \mathcal{A}_\lambda := \{0.015625, 0.03125, 0.0625, 0.125, 0.25, 0.5, 1, 1.5, 2, 3, 4\},$$

$$\tau_n(b) = b n_t^{0.06}, \quad b \in \mathcal{A}_\tau := \{0.0625, 0.125, 0.25, 0.5, 1, 1.5, 2\},$$

as well as the weak-truncation candidate $\tau_n = 10$. The final selection is $(\lambda_n, \tau_n) = (0.005569, 0.1960)$, corresponding to $(a, b) = (0.5, 0.125)$. There are 1,480 records in $\mathcal{D}_t$ have $\rho_i > \tau_n$, which carry 98.9% of the total weight in $\mathcal{D}_t$ if no truncation.

Projection and evaluation. Let $\mathbb{Q}_Z$ denote the marginal law of Z under $\mathbb{Q}$, and let $\mathcal{I}_e$ denote the index set of the 2,795 records in $\mathcal{D}_e$. The empirical target measure uses the untruncated ρ_i values, normalized to sum to one over $\mathcal{I}_e$:

$$\pi_i := \frac{\rho_i}{\sum_{j \in \mathcal{I}_e} \rho_j}, \quad i \in \mathcal{I}_e, \qquad \widehat{\mathbb{Q}} := \sum_{i \in \mathcal{I}_e} \pi_i \delta_{Z_i}.$$

Let $z_{(1)} < \cdots < z_{(K)}$ be the distinct values in $\{Z_i : i \in \mathcal{I}_e\}$; here $K = 802$. For $I_k = \{i \in \mathcal{I}_e : Z_i = z_{(k)}\}$, define

$$W_k = \sum_{i \in I_k} \pi_i, \qquad \bar{f}_k = W_k^{-1} \sum_{i \in I_k} \pi_i \widehat{f}(Z_i).$$

The pool-adjacent-violators algorithm (PAVA) computes the bounded weighted projection onto the nondecreasing class:

$$(\widehat{g}_1, \ldots, \widehat{g}_K) = \arg \min_{0 \le g_1 \le \cdots \le g_K \le 1} \sum_{k=1}^{K} W_k (\bar{f}_k - g_k)^2,$$

and assigns $\widehat{g}(Z_i) = \widehat{g}_k$ for every $i \in I_k$.

The fitted IWKRR and its monotonic projection are presented in Figure 11. Among the 5,000 multiplier bootstrap draws, 2,697 exceed the observed distance, yielding the p-value 0.5395.

E.4 Study 3: equity-share allocation over financial wealth

We are interested in whether, among families holding stock equity (`HEQUITY` $= 1$) and having positive financial assets (`FIN` > 0), the share of financial assets invested in stock equity (`EQUITY`/`FIN`) is constant across levels of financial wealth (`FIN`). Define

$$S = \mathbf{1}\{\texttt{HEQUITY} = 1,\ \texttt{FIN} > 0\}, \qquad \mathbb{Q}^{\text{alloc}} = \mathbb{Q}(\cdot \mid S = 1),$$

$$Y = \frac{\texttt{EQUITY}}{\texttt{FIN}}, \qquad Z = \log\{1 + \texttt{FIN}\}, \qquad f^*(z) = \mathbb{E}_{\mathbb{Q}^{\text{alloc}}}(Y \mid Z = z).$$

Let $\mathbb{Q}_Z^{\text{alloc}}$ denote the marginal law of Z under $\mathbb{Q}^{\text{alloc}}$ and

$$\mathcal{C} = \{g_c : g_c(z) \equiv c,\ c \in [0, 1]\}$$

be the class of constant-valued functions. We test the composite null $H_0 : f^* \in \mathcal{C}$, $\mathbb{Q}_Z^{\text{alloc}}$-almost surely. Equivalently,

$$H_0 : d_{\mathbb{Q}^{\text{alloc}}}(f^*, \mathcal{C}) = 0, \qquad d_{\mathbb{Q}^{\text{alloc}}}(h, \mathcal{C}) := \inf_{g \in \mathcal{C}} \|h - g\|_{\mathsf{L}^2(\mathbb{Q}_Z^{\text{alloc}})}.$$

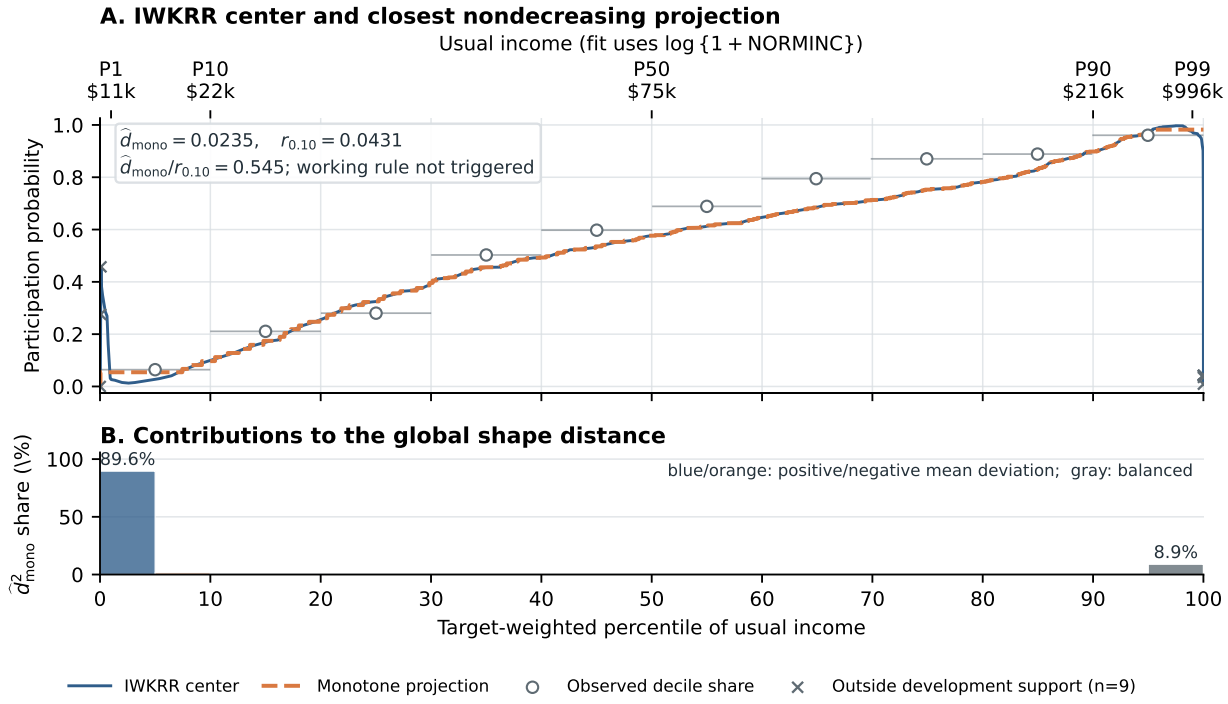


Figure 11: probability of holding stock-equity as a function of usual income. Panel A shows the IWKRR center and its closest nondecreasing projection. Horizontal segments and open circles give the income range and observed participation mean, respectively. Panel B partitions the difference between $\widehat{f}$ and its monotonic projection into 20 intervals, each containing approximately 5% of the evaluation target weight.

There are 2,895 eligible records. A random permutation assigns 800 records to the training set and the remaining to the held-out evaluation set. After tuning, we select $\lambda_n = 0.001132$, $\tau_n = 10$, and refit the IWKRR on the full training set.

On the held-out survey-weighted evaluation set, let $\widehat{\mathbb{Q}}^{\text{alloc}}$ denote the normalized released-weight empirical measure of Z. Projecting the resulting IWKRR $\widehat{f}$ onto $\mathcal{C}$ gives the nearest constant 0.4432, and

$$d(\widehat{f}, \mathcal{C}) = 0.08703, \qquad \hat{c}_{0.10} = 0.043620.$$

The constructed confidence interval is $d(f^*, \mathcal{C}) \in [0.043406, 0.130646]$, and the bootstrap p-value is 0.0002, leading to the rejection of the null hypothesis. The empirical bootstrap distribution is displayed in Figure 12. The fitted IWKRR $\widehat{f}$ is illustrated in Figure 13.

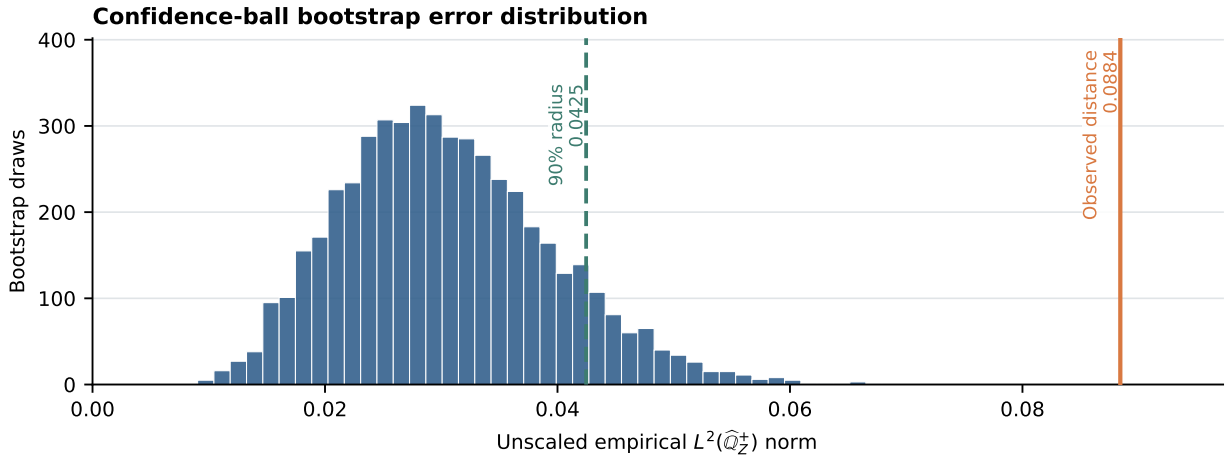


Figure 12: empirical bootstrap distribution for Study 3 based on 5,000 bootstrap draws. The dashed green line is the 90% percentile $\hat{c}_{0.10} = 0.043620$, and the orange line is the observed statistic $d(\widehat{f}, \mathcal{C}) = 0.087026$, yielding the plus-one p-value 0.0002. We reject the null hypothesis that households allocate the same share of wealth to equity stocks.

E.5 Study 4: does late-payment occurrence change across years?

We study whether late-payment occurrence in the full target population changed between the 2019 and 2022 SCF waves. Let $\mathbb{Q}_{2019}$ and $\mathbb{Q}_{2022}$ denote the survey-weighted target distributions in the two waves. For each $t \in \{2019, 2022\}$, the SCF public summary extract contains the Federal Reserve–constructed binary late-payment variable

$$Y_t = \texttt{LATE}_t = \mathbf{1}\{\texttt{X3004}_t = 5\}.$$

Here, $Y_t = 1$ indicates that the family sometimes fell behind or missed payments during the preceding year. The value $Y_t = 0$ includes both families with $\texttt{X3004}_t = 1$, indicating payments made on or ahead of schedule, and families with $\texttt{X3004}_t = 0$, for whom the question was inapplicable because they held none of the listed loan types.

For each $t \in \{2019, 2022\}$, let X_t denote the wave-t vector of the same twelve SCF variables used in Study 1. Let $\mathbb{Q}_{t,X}$ denote the marginal distribution of X_t under $\mathbb{Q}_t$, and define the target

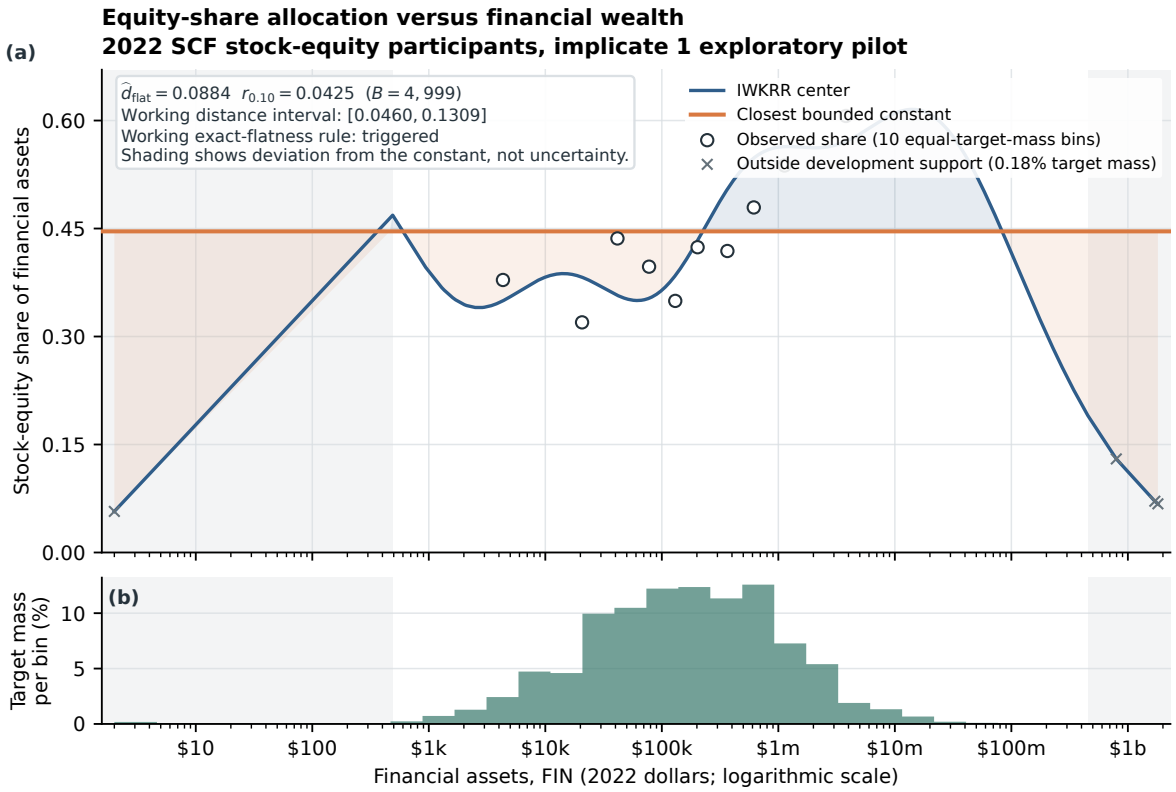


Figure 13: equity-share allocation over financial wealth. The upper panel shows the IWKRR $\widehat{f}$, its closest constant-valued function, and descriptive equity shares in ten approximately equal-target-mass wealth groups. The lower panel shows the held-out target mass over financial wealth.

regression function

$$f_t^*(x) = \mathbb{Q}_t(Y_t = 1 \mid X_t = x).$$

We test against the null hypothesis

$$H_0 : \|f_{2022}^* - f_{2019}^*\|_{\mathsf{L}^2(\mathbb{Q}_{2022,X})} = 0.$$

Let $\widehat{f}_t$ be the wave-specific IWKRR estimator, and let $\widehat{\mathbb{Q}}_{2022,E}$ be the normalized released-weight empirical covariate measure on a held-out 2022 evaluation set. Define

$$\widehat{D} = \|\widehat{f}_{2022} - \widehat{f}_{2019}\|_{\mathsf{L}^2(\widehat{\mathbb{Q}}_{2022,E})}.$$

The triangle inequality gives

$$\left|\widehat{D} - \|f_{2022}^* - f_{2019}^*\|_{\mathsf{L}^2(\widehat{\mathbb{Q}}_{2022,E})}\right| \le \|(\widehat{f}_{2022} - f_{2022}^*) - (\widehat{f}_{2019} - f_{2019}^*)\|_{\mathsf{L}^2(\widehat{\mathbb{Q}}_{2022,E})}.$$

For $b = 1, \ldots, B$, independently generate centered-Gaussian residual-multiplier perturbations $\widehat{e}_t^{(b)}$ of the two fitted functions and compute

$$R_b = \|\widehat{e}_{2022}^{(b)} - \widehat{e}_{2019}^{(b)}\|_{\mathsf{L}^2(\widehat{\mathbb{Q}}_{2022,E})}.$$

If $\widehat{c}_\alpha^{\text{joint}}$ is the $(1-\alpha)100\%$-percentile of $R_1, \ldots, R_B$, the corresponding interval for $\|f_{2022}^* - f_{2019}^*\|_{\mathsf{L}^2(\widehat{\mathbb{Q}}_{2022,E})}$ is

$$\left[\{\widehat{D} - \widehat{c}_\alpha^{\text{joint}}\}_+, \widehat{D} + \widehat{c}_\alpha^{\text{joint}}\right],$$

and its plus-one upper-tail p-value is

$$\widehat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{R_b \ge \widehat{D}\}}{B+1}.$$

In each wave, 1,200 records form the wave-specific training set. The parameters (λ_n, τ_n) are tuned separately within each wave, using the Gaussian RBF kernel with $\gamma = 1/33$. The final selections are:

$$(\lambda_n, \tau_n) = (0.003551659, 1.530209) \quad \text{for 2019}, \qquad (\lambda_n, \tau_n) = (0.028413272, 0.765105) \quad \text{for 2022}.$$

The joint multiplier bootstrap uses $B = 5{,}000$ draws. The fitted distance and 90th-percentile are

$$\widehat{D} = 0.046569, \qquad \widehat{c}_{0.10}^{\text{joint}} = 0.049029.$$

The corresponding interval is $[0, 0.095598]$, and the plus-one upper-tail p-value is 0.1562, leading to the non-rejection of the null hypothesis. Figure 14 showcases the survey-weighted distribution of the difference $\widehat{f}_{2022} - \widehat{f}_{2019}$ on the held-out 2022 evaluation set.

F Proofs of Main Results

We collect the proofs of main theorems and propositions that appear in the main text and Appendix A. Proofs of auxiliary results are collected in Appendix G. For notational simplicity, we may write λ and $w(x)$ in short for λ_n and $w_n(x)$, respectively. The notation $\|\cdot\|$ is reserved for $\|\cdot\|_{\mathsf{H}}$.

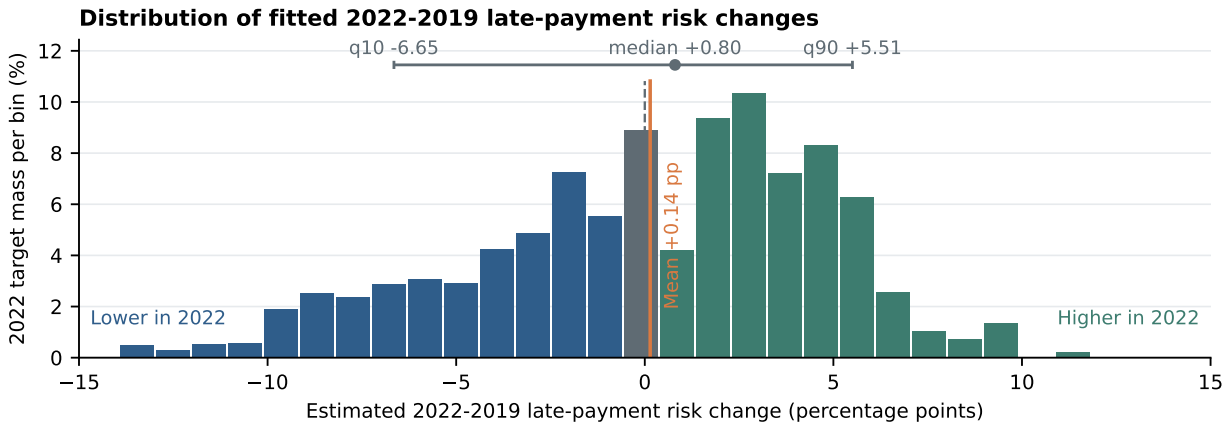


Figure 14: survey-weighted distribution of the difference $\widehat{f}_{2022} - \widehat{f}_{2019}$ on the held-out 2022 evaluation set.

F.1 Proof of Lemma 4.1

Lemma F.1. *For fixed $k \in \mathbb{N}_+$ and sufficiently small $\lambda > 0$,*

$$\sum_{i=1}^{\infty} \left(\frac{\mu_i}{\mu_i + \lambda} \right)^k \asymp s^* + \frac{1}{\lambda^k} \sum_{i=s^*+1}^{\infty} \mu_i^k,$$

where $s^* = \min\{s : \mu_s \le \lambda\}$.

Lemma F.2. *For fixed $k \in \mathbb{N}_+$ and sufficiently small $\lambda > 0$, the following rates hold. If $\mu_i \asymp i^{-\beta}$ for some $\beta > 1$, then*

$$\sum_{i=1}^{\infty} \left(\frac{\mu_i}{\mu_i + \lambda} \right)^k \asymp \lambda^{-1/\beta}.$$

If $\mu_i \asymp \exp(-\beta i^\gamma)$ for some $\beta > 0$ and $\gamma > 0$, then

$$\sum_{i=1}^{\infty} \left(\frac{\mu_i}{\mu_i + \lambda} \right)^k \asymp \{\log(1/\lambda)\}^{1/\gamma}.$$

Lemma F.3. *Under Assumption 4.1, for $\lambda > 0$,*

$$\mathcal{V} \succeq_{\mathsf{L}^2(\mathbb{Q})} \sigma^2 \mathbb{E}\left\{ \left[(\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} w(X_i) k_{X_i} \right] \otimes_{\mathsf{L}^2(\mathbb{Q})} \left[(\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} w(X_i) k_{X_i} \right] \right\}.$$

Lemma F.4.

$$\mathbb{E}\left\{ w(X_i) \left[(\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} k_{X_i} \right] \otimes_{\mathsf{L}^2(\mathbb{Q})} \left[\left((\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} k_{X_i} \right) \right] \right\} = (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_{P^w} (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_Q$$

Lemma F.5. *Let $\mathcal{T} : \mathsf{H} \to \mathsf{H}$ be a bounded positive self-adjoint operator in H. If $\mathcal{T}\mathcal{L}_Q : \mathsf{L}^2(\mathbb{Q}) \to \mathsf{L}^2(\mathbb{Q})$ is a compact operator in $\mathsf{L}^2(\mathbb{Q})$ and $\mathcal{L}_Q^{1/2} \mathcal{T} \mathcal{L}_Q^{1/2} : \mathsf{H} \to \mathsf{H}$ is a compact operator in H, then their nonzero eigenvalues, counted with multiplicity, coincide. In particular, after padding a finite nonzero spectrum by zeros,*

$$\mu_k(\mathcal{T}\mathcal{L}_Q, \mathsf{L}^2(\mathbb{Q})) = \mu_k(\mathcal{L}_Q^{1/2} \mathcal{T} \mathcal{L}_Q^{1/2}, \mathsf{H}), \qquad k \ge 1.$$

Lemma F.6. *Under Assumption 4.2, when $\tau_n \geq \tau_0$, we have*

$$\|f_{P^w} - f_*\| \leq \max\{\kappa^{2s-1}, 1\}\lambda^{\min\{s,1/2\}}\|g_*\|.$$

Lemma F.7. *Under Assumptions 4.1, 4.2, and 4.4, when $\tau_n \geq \tau_0$, we have*

$$\begin{aligned}
\mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V}) &\leq \tau_n \left(\overline{\sigma}^2 + \kappa^2 \max\{\kappa^{2s-1}, 1\}^2 \lambda^{2\min\{s,1/2\}} \|g_*\|^2\right) \left(\mathfrak{C}_1\right)^2 \mathrm{tr}_{\mathsf{H}}\left((\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}\right) \\
&\leq \tau_n \left(\overline{\sigma}^2 + \kappa^2 \max\{\kappa^{2s-1}, 1\}^2 \lambda^{2\min\{s,1/2\}} \|g_*\|^2\right) \left(\mathfrak{C}_1\right)^2 \mathcal{N}_\lambda, \\
\|\mathcal{V}\|_{\mathsf{L}^2(\mathbb{Q})} &\leq \tau_n \left(\overline{\sigma}^2 + \kappa^2 \max\{\kappa^{2s-1}, 1\}^2 \lambda^{2\min\{s,1/2\}} \|g_*\|^2\right) \left(\mathfrak{C}_1\right)^2.
\end{aligned}$$

Lemma F.8. *Under Assumptions 4.1 and 4.5, writing $\lambda = \lambda_n$, we have*

$$\mathcal{V} \succeq_{\mathsf{L}^2(\mathbb{Q})} \left(\mathfrak{C}_2\right)^{-2} \underline{\rho}\underline{\sigma}^2 (\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q.$$

Lemma F.9 (Theorem 2.6 of Götze et al. [2019])**.** *Let $\xi \sim \mathrm{N}_{\mathsf{G}}(0, \mathcal{V}_\xi)$, where G is a separable Hilbert space. Denoting the probability density function of $\|\xi\|_{\mathsf{G}}^2$ by $p_{\|\xi\|_{\mathsf{G}}^2}$, we have*

$$\sup_{t\geq 0} p_{\|\xi\|_{\mathsf{G}}^2}(t) \lesssim \varkappa(\mathcal{V}_\xi) \lesssim (\Lambda_{1\xi}\Lambda_{2\xi})^{-1/2}$$

where

$$\varkappa(\mathcal{V}_\xi) = \begin{cases} \Lambda_{1\xi}^{-1}, & \text{if } 3\lambda_{1\xi}^2 < \Lambda_{1\xi}^2; \\ (\lambda_{1\xi}\Lambda_{2\xi})^{-1/2}, & \text{if } 3\lambda_{1\xi}^2 \geq \Lambda_{1\xi}^2,\ 3\lambda_{2\xi}^2 < \Lambda_{2\xi}^2; \\ (\lambda_{1\xi}\lambda_{2\xi})^{-1/2}, & \text{if } 3\lambda_{1\xi}^2 \geq \Lambda_{1\xi}^2,\ 3\lambda_{2\xi}^2 \geq \Lambda_{2\xi}^2. \end{cases}$$

with $\lambda_{i\xi} = \lambda_i(\mathcal{V}_\xi, \mathsf{G})$ and $\Lambda_{k\xi}^2 = \sum_{i=k}^\infty \lambda_{i\xi}^2$ for $k = 1, 2$.

Remark 3. *Lemma F.9 states a slightly corrected version of Theorem 2.6 in Götze et al. [2019], which does not affect any of the $\lesssim$-order consequences used in this proof.*

Proof of Lemma 4.1. Let $(\nu_j, e_j)_{j\geq 1}$ be the $\mathsf{L}^2(\mathbb{Q})$-eigenpairs of $\mathcal{V}$, ordered so that $\nu_1 \geq \nu_2 \geq \cdots$, and define

$$\Lambda_{kZ}^2 := \sum_{j=k}^\infty \nu_j^2, \qquad k = 1, 2.$$

We first prove the general bound

$$\sup_{t\geq 0} p_{\|Z\|_{\mathsf{L}^2(\mathbb{Q})}}(t) \lesssim \sqrt{\mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})}(\Lambda_{1Z}\Lambda_{2Z})^{-1/2}.$$

Let

$$T := \|Z\|_{\mathsf{L}^2(\mathbb{Q})}.$$

By Lemma F.9, applied with $\mathsf{G} = \mathsf{L}^2(\mathbb{Q})$ and covariance $\mathcal{V}$, the random variable T^2 has a density satisfying

$$\sup_{u\geq 0} p_{T^2}(u) \lesssim (\Lambda_{1Z}\Lambda_{2Z})^{-1/2}.$$

For $t > 0$, the density of T satisfies

$$p_T(t) = 2t\, p_{T^2}(t^2).$$

It remains to turn this pointwise bound into a bound that is uniform in t. For a finite-rank truncation

$$Z_m = \sum_{j=1}^{m} \sqrt{\nu_j} g_j e_j,$$

the distribution function of $T_m = \|Z_m\|_{\mathsf{L}^2(\mathbb{Q})}$ can be written as

$$\mathbb{P}(T_m \le t) = \gamma_m(tK_m),$$

where γ_m is a centered Gaussian measure on $\operatorname{span}\{e_1, \dots, e_m\}$ and K_m is the unit ball in that subspace. Since Gaussian measures are log-concave and

$$a(tK_m) + (1-a)(sK_m) = \{at + (1-a)s\}K_m, \qquad a \in [0,1],$$

the distribution function $F_{T_m}(t) = \mathbb{P}(T_m \le t)$ is log-concave on $[0,\infty)$. Moreover, $T_m \to T$ almost surely. Since T has a density, $F_{T_m}(t) \to F_T(t)$ for every t, and hence F_T is also log-concave.

Put

$$A_Z := (\Lambda_{1Z}\Lambda_{2Z})^{-1/2}, \qquad m_Z := \{\operatorname{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})\}^{1/2}.$$

The Gaussian fourth-moment identity gives

$$\mathbb{E}T^4 = \{\operatorname{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})\}^2 + 2\|\mathcal{V}\|^2_{HS,\mathsf{L}^2(\mathbb{Q})} \le 3m_Z^4.$$

Applying the Paley–Zygmund inequality to T^2 therefore yields

$$\mathbb{P}\left(T \ge \frac{m_Z}{\sqrt{2}}\right) \ge \frac{1}{12}, \qquad \mathbb{E}T \ge \frac{m_Z}{12\sqrt{2}}.$$

If $0 < t \le 4\mathbb{E}T$, the squared-norm density bound gives

$$p_T(t) = 2t\, p_{T^2}(t^2) \lesssim A_Z m_Z.$$

If $t > 4\mathbb{E}T$, Markov's inequality gives $F_T(t/2) \ge 1/2$. At every differentiability point of $\log F_T$, its concavity implies

$$\frac{p_T(t)}{F_T(t)} = \{\log F_T\}'(t) \le \frac{\log F_T(t) - \log F_T(t/2)}{t/2} \le \frac{2\log 2}{t}.$$

Thus $p_T(t) \lesssim m_Z^{-1}$ for almost every t in this second regime. Finally,

$$m_Z^2 = \operatorname{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V}) \ge \Lambda_{1Z} \ge (\Lambda_{1Z}\Lambda_{2Z})^{1/2},$$

so that $m_Z^{-1} \le A_Z m_Z$. Choosing the density version satisfying these almost-everywhere bounds and combining the two regimes proves

$$\sup_{t \ge 0} p_{\|Z\|_{\mathsf{L}^2(\mathbb{Q})}}(t) \lesssim \sqrt{\operatorname{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})}(\Lambda_{1Z}\Lambda_{2Z})^{-1/2}.$$

We now bound the two covariance quantities in the preceding display. By Lemma F.7 and its proof,

$$\operatorname{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V}) \lesssim \tau_n \operatorname{tr}_{\mathsf{H}}\big((\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q\big) = \tau_n \mathcal{N}_\lambda = \tau_n \sum_{i=1}^{\infty} \frac{\mu_i}{\mu_i + \lambda},$$

where fixed constants depending on the noise, source condition, and compatibility constant are absorbed into $\lesssim$. On the other hand, Lemma F.8 gives

$$\mathcal{V} \succeq_{\mathsf{L}^2(\mathbb{Q})} \left(\mathfrak{C}_2\right)^{-2} \underline{\rho\sigma}^2 (\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q.$$

By monotonicity of eigenvalues under the Loewner order for positive compact operators, the ordered eigenvalues satisfy

$$\nu_i \gtrsim \left(\frac{\mu_i}{\mu_i + \lambda}\right)^2, \qquad i \geq 1,$$

because the eigenvalues of $(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q$ are $(\mu_i/(\mu_i + \lambda))^2$. Consequently, for $k = 1, 2$,

$$\Lambda_{kZ}^2 \gtrsim \sum_{i=k}^{\infty} \left(\frac{\mu_i}{\mu_i + \lambda}\right)^4.$$

Under either two-sided spectral condition, Lemma F.2 shows that the full sum on the right-hand side diverges as $\lambda \to 0$. Since each summand is at most one, deleting its first term does not change its order for all sufficiently small λ. On any remaining compact subinterval of $(0, 1]$, the ratio of the tail sum to the full sum is continuous and strictly positive. Thus, after enlarging the implicit constant, for $k = 1, 2$,

$$\Lambda_{kZ} \gtrsim \left\{\sum_{i=1}^{\infty} \left(\frac{\mu_i}{\mu_i + \lambda}\right)^4\right\}^{1/2}.$$

Combining the preceding bounds yields

$$\sup_{t\geq 0} p_{\|Z\|_{\mathsf{L}^2(\mathbb{Q})}}(t) \lesssim \sqrt{\tau_n}\left\{\frac{\sum_{i=1}^{\infty} \mu_i/(\mu_i + \lambda)}{\sum_{i=1}^{\infty}\{\mu_i/(\mu_i + \lambda)\}^4}\right\}^{1/2}.$$

Under either the two-sided polynomial or two-sided exponential spectral decay regime, Lemma F.2, after the same compact-interval adjustment over $0 < \lambda \leq 1$,

$$\sum_{i=1}^{\infty} \frac{\mu_i}{\mu_i + \lambda} \asymp \sum_{i=1}^{\infty} \left(\frac{\mu_i}{\mu_i + \lambda}\right)^4;$$

as $\lambda \to 0$, both sums are of order $\lambda^{-1/\beta}$ in the polynomial case and $\{\log(1/\lambda)\}^{1/\gamma}$ in the exponential case, up to constants depending only on the spectral parameters. Therefore

$$\sup_{t\geq 0} p_{\|Z\|_{\mathsf{L}^2(\mathbb{Q})}}(t) \lesssim \sqrt{\tau_n}.$$

□

F.2 Proof of Theorem 4.1

Proof of Theorem 4.1. Let

$$T := \sqrt{n}\|\hat{f} - f_*\|_{\mathsf{L}^2(\mathbb{Q})}, \qquad S := \|Z\|_{\mathsf{L}^2(\mathbb{Q})}, \qquad a_n := \sup_{t\geq 0} p_{\|Z\|_{\mathsf{L}^2(\mathbb{Q})}}(t),$$

and let t_β denote the deterministic upper β-quantile of S. By the Gaussian approximation and the bias bound, for every deterministic t,

$$P(T \geq t) \leq P(S \geq t - \Delta_G - \Delta_{\text{bias}}) + \eta,$$
$$P(T \geq t) \geq P(S \geq t + \Delta_G + \Delta_{\text{bias}}) - \eta.$$

Let $\mathcal{E}_B$ be the data event on which the conditional bootstrap approximation (4.3) holds. By the statement preceding (4.3), $P(\mathcal{E}_B) \geq 1 - \eta$. On $\mathcal{E}_B$, the conditional coupling of B and Z' gives, for every $s \geq 0$,

$$P(S \geq s + \Delta_B) \leq P\big(\|B\|_{\mathsf{L}^2(\mathbb{Q})} \geq s \mid \mathcal{D}\big) + \eta,$$
$$P\big(\|B\|_{\mathsf{L}^2(\mathbb{Q})} \geq s + \Delta_B \mid \mathcal{D}\big) \leq P(S \geq s) + \eta,$$

because $Z' \mid \mathcal{D}$ has the same law as Z. Taking $s = \hat{t}_\alpha$ in the first inequality and $s = t_{\alpha-\eta}$ in the second yields

$$t_{\alpha+\eta} - \Delta_B \leq \hat{t}_\alpha \leq t_{\alpha-\eta} + \Delta_B.$$

Consequently,

$$\begin{aligned} P(T \geq \hat{t}_\alpha) &\leq P(\mathcal{E}_B^c) + P(T \geq t_{\alpha+\eta} - \Delta_B) \\ &\leq \alpha + 3\eta + a_n\Delta, \end{aligned}$$

where the last step uses the density bound to compare the Gaussian upper-tail probabilities at $t_{\alpha+\eta} - \Delta$ and $t_{\alpha+\eta}$. Similarly,

$$\begin{aligned} P(T \geq \hat{t}_\alpha) &\geq P(T \geq t_{\alpha-\eta} + \Delta_B) - P(\mathcal{E}_B^c) \\ &\geq \alpha - 3\eta - a_n\Delta. \end{aligned}$$

Lemma 4.1 gives $a_n \lesssim \sqrt{\tau_n}$, which proves the result. □

Corollary F.1 (Asymptotic coverage)**.** *Under the conditions of Theorem 4.1, let the tuning parameters and probability tolerance depend on n. If*

$$\eta_n \to 0 \qquad \textit{and} \qquad \sqrt{\tau_n}\,\{\Delta_{G,n} + \Delta_{B,n} + \Delta_{\text{bias},n}\} \to 0,$$

then, for every fixed $\alpha \in (0,1)$,

$$P\Big(f_* \in \mathsf{CS}_\alpha(\hat{f})\Big) \to 1 - \alpha.$$

Corollary F.1 follows immediately from Theorem 4.1.

F.3 Proof of Proposition 4.1

Lemma F.10. *Under Assumption 4.2, when $\tau_n \geq \tau_0$, we have*

$$\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} f_*\| \leq \max\{\kappa^{2s-1}, 1\}\, \lambda^{\min\{s-\frac{1}{2},0\}}\, \|g_*\|.$$

Proof of Proposition 4.1. Applying Lemma F.10 with $\lambda = \lambda_n$, we obtain

$$\begin{aligned}\sqrt{n}\,\|f_{P^w} - f_*\|_{\mathsf{L}^2(\mathbb{Q})} &= \sqrt{n}\,\lambda_n \left\| \mathcal{L}_Q^{\frac{1}{2}} \left(\mathcal{L}_{P^w} + \lambda_n \mathcal{I}\right)^{-1} f_* \right\| \\ &\le \mathfrak{C}_1 \sqrt{n}\,\lambda_n \left\| \left(\mathcal{L}_{P^w} + \lambda_n \mathcal{I}\right)^{-\frac{1}{2}} f_* \right\| \\ &\le \max\left\{\kappa^{2s-1}, 1\right\} \mathfrak{C}_1 \sqrt{n}\,\lambda_n^{\min\left\{s+\frac{1}{2},1\right\}} \|g_*\|_{\mathsf{H}} = \Delta_{\text{bias}}.\end{aligned}$$

This proves the proposition. □

F.4 Proof of Proposition 4.2

Let $\boldsymbol{\varepsilon} = (\varepsilon_1, \ldots, \varepsilon_n)^\top$. The following extension of Bernstein's inequality for Hilbert–Schmidt operators is useful.

Lemma F.11 (Minsker [2017], Theorem 3.1)**.** *Let G be a separable Hilbert space, let $R > 0$, and let $\{\xi_i\}_{i=1}^n$ be independent, self-adjoint Hilbert–Schmidt random operators on G such that*

$$\mathbb{E}[\xi_i] = 0 \quad \text{and} \quad \|\xi_i\| \le R \quad \text{a.s.}$$

Suppose there exist $V, W > 0$ such that

$$\left\| \mathbb{E}\Big[\Big(\sum_{i=1}^n \xi_i\Big)^2\Big] \right\| \le V$$

$$\operatorname{Tr}\left(\mathbb{E}\Big[\Big(\sum_{i=1}^n \xi_i\Big)^2\Big] \right) \le VW$$

Then $\forall\, t \ge V^{1/2} + R/3$,

$$\mathbb{P}\left(\left\| \sum_{i=1}^n \xi_i \right\| \ge t \right) \le 4W \exp\left(-\frac{t^2}{2\big(V + Rt/3\big)} \right).$$

Lemma F.12. *For $0 < \eta < 1$ and $0 < \lambda \le 1$, with probability at least $1 - \eta$,*

$$\begin{aligned}&\left\| \left(\mathcal{L}_{P^w} + \lambda \mathcal{I}\right)^{-\frac{1}{2}} \left(\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w}\right) \left(\mathcal{L}_{P^w} + \lambda \mathcal{I}\right)^{-\frac{1}{2}} \right\| \\ <& C_1 \left[\frac{\left(\tau_n \kappa^2 + 1\right) \log(e + \mathcal{N}_\lambda)}{n\lambda} + \sqrt{\frac{\left(\tau_n \kappa^2 + 1\right) \log(e + \mathcal{N}_\lambda)}{n\lambda}} \right] \log \frac{2}{\eta}.\end{aligned}$$

If further $\tau_n \to \infty$ and $\lambda \to 0$ as $n \to \infty$, and

$$\frac{\left(\tau_n \kappa^2 + 1\right) \log(e + \mathcal{N}_\lambda)}{n\lambda} < 1,$$

then with probability at least $1 - \eta$,

$$\left\| \left(\mathcal{L}_{P^w} + \lambda \mathcal{I}\right)^{-\frac{1}{2}} \left(\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w}\right) \left(\mathcal{L}_{P^w} + \lambda \mathcal{I}\right)^{-\frac{1}{2}} \right\| \lesssim \sqrt{\frac{\left(\tau_n \kappa^2 + 1\right) \log(e + \mathcal{N}_\lambda)}{n\lambda}} \log \frac{2}{\eta}.$$

Lemma F.13. *For $0 < \eta < 1$ and $0 < \lambda \le 1$, if*

$$\frac{(\tau_n \kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{n\lambda} < C_2 := \min\left\{1, \frac{1}{\left(4C_1 \log\frac{2}{\eta}\right)^2}\right\}$$

where C_1 is the constant in Lemma F.12, then on the same $1-\eta$ probability event of Lemma F.12,

$$\left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}\right\| \le 2.$$

Lemma F.14. *Under Assumption 4.1, for $0 < \eta < 1$ and $\lambda > 0$, with probability at least $1-\eta$,*

$$\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right)\| \lesssim 2\overline{\sigma}\log\left(\frac{2}{\eta}\right)\max\{\sqrt{\frac{\tau_n\mathcal{N}_\lambda}{n}}, \frac{2\tau_n\kappa}{n\sqrt{\lambda}}\}$$

Proof of Proposition 4.2. For brevity, write $\lambda = \lambda_n$, $w = w_n$, and $\mathcal{N}_\lambda = \mathcal{N}_{\lambda_n}$ throughout this proof.

$$\begin{aligned}
\hat{f} - f_{P^w} &= (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\mathbf{y}\right) - (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P^w} f_* \\
&= (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right) \\
&\quad + (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P_n^w} f_* - (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P^w} f_* \\
&= (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right) \\
&\quad + (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P_n^w} f_* - (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P^w} f_* \\
&\quad + \left[(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1} - (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\right](\mathcal{L}_{P^w} f_*) \\
&= (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right) \\
&\quad + (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}(\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w}) f_* \\
&\quad - (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}(\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w}) f_{P^w} \\
&= (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\left[\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right) + (\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w})(f_* - f_{P^w})\right] \\
&\quad + \left[(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1} - (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\right]\left[\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right) + (\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w})(f_* - f_{P^w})\right]
\end{aligned}$$

Denote

$$u = \left[(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1} - (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\right]\left[\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right) + (\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w})(f_* - f_{P^w})\right].$$

Since

$$\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right) = \frac{1}{n}\sum_{i=1}^n w(X_i)\varepsilon_i k_{X_i}, \qquad \mathcal{L}_{P_n^w} - \mathcal{L}_{P^w} = \frac{1}{n}\sum_{i=1}^n\left(w(X_i)k_{X_i}\otimes k_{X_i} - \mathcal{L}_{P^w}\right),$$

the first term in the preceding decomposition is $n^{-1}\sum_{i=1}^n U_i$. Thus

$$\hat{f} - f_{P^w} = \frac{1}{n}\sum_{i=1}^n U_i + u.$$

With the bound

$$\left\|\mathcal{L}_Q^{\frac{1}{2}}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\right\| \le \mathfrak{C}_1,$$

we can bound $\|u\|_{\mathsf{L}^2(\mathbb{Q})}$

$$\begin{aligned}\|u\|_{\mathsf{L}^2(\mathbb{Q})} &= \left\|\mathcal{L}_Q^{\frac{1}{2}}u\right\| \\ &\le \mathfrak{C}_1 \cdot \left\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}u\right\|.\end{aligned}$$

$$\begin{aligned}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}-(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1} &= \Big[(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})-\mathcal{I}\Big](\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1} \\ &= (\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}-\mathcal{L}_{P_n^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1} \\ &= (\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}-\mathcal{L}_{P_n^w})(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}.\end{aligned}$$

Combining this identity with

$$\begin{aligned}f_* - f_{P^w} &= f_* - (\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}f_* \\ &= \lambda(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}f_*,\end{aligned}$$

rewrite

$$\begin{aligned}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}u &= (\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w}-\mathcal{L}_{P_n^w})(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top\mathbf{W}\boldsymbol{\varepsilon}\right) \\ &\quad - (\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P_n^w}-\mathcal{L}_{P^w})(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P_n^w}-\mathcal{L}_{P^w})(f_*-f_{P^w}) \\ &= \Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w}-\mathcal{L}_{P_n^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\Big] \\ &\qquad \cdot \Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\Big] \\ &\qquad \cdot \left[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top\mathbf{W}\boldsymbol{\varepsilon}\right)\right] \\ &\quad - \lambda\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P_n^w}-\mathcal{L}_{P^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\Big] \\ &\qquad \cdot \Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\Big] \\ &\qquad \cdot \Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P_n^w}-\mathcal{L}_{P^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\Big] \\ &\qquad \cdot \Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}f_*\Big].\end{aligned}$$

Apply the stochastic bounds below with suitably split failure probabilities; this changes their logarithmic factors only by universal constants. Combining Lemmas F.12, F.13, F.14, and F.10, and then using both sample-size conditions in Proposition 4.2, gives, with probability at least $1-\eta$,

$$\|u\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \mathfrak{C}_1\tau_n\overline{\sigma}\,\kappa^3\big(1+\|g_*\|_{\mathsf{H}}\big)\frac{1}{n}\sqrt{\frac{\mathcal{N}_{\lambda_n}\log(e+\mathcal{N}_{\lambda_n})}{\lambda_n}}\log^2\left(\frac{2}{\eta}\right).$$

Consequently,

$$\left\| \sqrt{n}(\hat{f} - f_{P^w}) - \frac{1}{\sqrt{n}} \sum_{i=1}^{n} U_i \right\|_{\mathsf{L}^2(\mathbb{Q})} = \sqrt{n}\, \|u\|_{\mathsf{L}^2(\mathbb{Q})}$$

$$\lesssim \mathfrak{C}_1 \tau_n \overline{\sigma}\, \kappa^3 \big(1 + \|g_*\|_{\mathsf{H}}\big) \sqrt{\frac{\mathcal{N}_{\lambda_n} \log(e + \mathcal{N}_{\lambda_n})}{\lambda_n n}} \log^2\left(\frac{2}{\eta}\right) = \Delta_{G,\mathrm{Bah}}.$$

Finally, using the definition of U_i, the transfer bound, and $\|(\mathcal{L}_{P^w} + \lambda_n \mathcal{I})^{-1/2}\| \le \lambda_n^{-1/2}$, we obtain

$$\|U_i\|_{\mathsf{L}^2(\mathbb{Q})} = \|\mathcal{L}_Q^{1/2} U_i\|$$

$$\le \frac{\tau_n \mathfrak{C}_1}{\sqrt{\lambda_n}} \left(\kappa^2 \|f_*\|_{\mathsf{H}} + \kappa \overline{\sigma}\right) = \bar{U},$$

which is the asserted uniform bound. □

F.5 Proof of Proposition 4.3

Let $\{(\nu_j, e_j)\}_{j \ge 1}$ be an eigensystem of $\mathcal{V}$ in $\mathsf{L}^2(\mathbb{Q})$, ordered so that $\nu_1 \ge \nu_2 \ge \cdots$, and define

$$\Pi_m := \sum_{j=1}^{m} e_j \otimes_{\mathsf{L}^2(\mathbb{Q})} e_j.$$

Proof of Proposition 4.3. It suffices to prove the bound inside the braces for an arbitrary fixed m, with a Gaussian element Z_m that may depend on m. The displayed right-hand side is attained over $m \ge 1$; indeed, if $\bar{U} > 0$, the finite-dimensional coupling term diverges as $m \to \infty$, while if $\bar{U} = 0$, any m is admissible. Applying the resulting construction at a minimizer gives the stated infimum. The proof follows the same projection-coupling strategy as in Singh and Vijaykumar [2023], with all projections and norms taken in $\mathsf{L}^2(\mathbb{Q})$. Let

$$A_m : \mathsf{L}^2(\mathbb{Q}) \to \mathbb{R}^m, \qquad A_m f = (\langle f, e_1 \rangle_{\mathsf{L}^2(\mathbb{Q})}, \ldots, \langle f, e_m \rangle_{\mathsf{L}^2(\mathbb{Q})}),$$

so that $A_m^* z = \sum_{s=1}^{m} z_s e_s$ and $A_m^* A_m = \Pi_m$. The covariance of $A_m U_i$ is the diagonal matrix $\mathrm{diag}(\nu_1, \ldots, \nu_m)$, and $\|A_m U_i\|_{\mathbb{R}^m} \le \|U_i\|_{\mathsf{L}^2(\mathbb{Q})} \le \bar{U}$.

By the finite-dimensional Gaussian coupling theorem of Zaitsev [1987], there exist independent Gaussian vectors $\zeta_i \in \mathbb{R}^m$, each having the same covariance as $A_m U_i$, such that

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \{\Pi_m U_i - A_m^* \zeta_i\} \right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \frac{\bar{U} m^2 \log\left(\frac{m^2}{\eta}\right)}{\sqrt{n}}$$

with probability at least $1 - \eta/3$. Define

$$Z_i = A_m^* \zeta_i + \sum_{s=m+1}^{\infty} \sqrt{\nu_s}\, h_{i,s} e_s.$$

Then the Z_i's are independent centered Gaussian elements in $\mathsf{L}^2(\mathbb{Q})$ with covariance $\mathcal{V}$. Set $Z = n^{-1/2} \sum_{i=1}^{n} Z_i$. Then $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$.

It remains to control the two tails. Since $\mathbb{E}U_i = 0$,

$$\mathbb{E}\left\|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(I-\Pi_m)U_i\right\|_{\mathsf{L}^2(\mathbb{Q})}^2 = \mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}\{(I-\Pi_m)\mathcal{V}\} = \sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V}).$$

The boundedness of U_i and the Hilbert-space Bernstein inequality [Pinelis and Sakhanenko, 1986, Corollary 1] imply, with probability at least $1-\eta/3$,

$$\left\|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(I-\Pi_m)U_i\right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})\sqrt{\log\left(\frac{6}{\eta}\right)} + \frac{\bar{U}\log\left(\frac{6}{\eta}\right)}{\sqrt{n}}.$$

Similarly, by Borell's inequality for the Gaussian element $\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(I-\Pi_m)Z_i$, whose second moment is $\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})$, with probability at least $1-\eta/3$,

$$\left\|\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(I-\Pi_m)Z_i\right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})\sqrt{\log\left(\frac{6}{\eta}\right)}.$$

Combining these three bounds by the triangle inequality and a union bound gives the claim, since the term $\bar{U}\log\left(\frac{6}{\eta}\right)/\sqrt{n}$ is absorbed by the displayed finite-dimensional coupling term for $m \geq 1$. □

F.6 Proof of Proposition 4.4

Let

$$V_i = w(X_i)\big(Y_i - f_{P^w}(X_i)\big)\,(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i},$$
$$\hat{V}_i = w(X_i)\big(Y_i - \hat{f}(X_i)\big)\,(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}k_{X_i}.$$

Let

$$B = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\,\frac{\hat{V}_i - \hat{V}_j}{\sqrt{2}},$$
$$B_{\text{oracle}} = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\,\frac{V_i - V_j}{\sqrt{2}}.$$

For notational convenience, write

$$\mathcal{L}_{P_i^w} := w(X_i)\,k_{X_i}\otimes k_{X_i},$$
$$\mathcal{L}_{P^w,\lambda} := \mathcal{L}_{P^w} + \lambda\mathcal{I},$$
$$\mathcal{L}_{P_n^w,\lambda} := \mathcal{L}_{P_n^w} + \lambda\mathcal{I},$$
$$\epsilon_i^{P^w} := Y_i - f_{P^w}(X_i)$$
$$\hat{\epsilon}_i := Y_i - \hat{f}(X_i).$$

Then

$$V_i = w(X_i)\,\epsilon_i^{P^w}\,\mathcal{L}^{-1}_{P^w,\lambda}k_{X_i}$$
$$\hat{V}_i = w(X_i)\,\hat{\epsilon}_i\,\mathcal{L}^{-1}_{P_n^w,\lambda}k_{X_i}.$$

Let $\mathbb{E}_h[\cdot]$ be the expectation with respect to the random variables $\{h_{ij}\}_{i,j=1}^n$ under the conditional law given $\mathcal{D} = \{(X_i,Y_i)\}_{i=1}^n$. Note that the randomness comes from data and $\{h_{ij}\}_{i,j=1}^n$. We first find a bound under $\mathbb{P}(\cdot\mid\mathcal{D})$, and then bound it further.

Lemma F.15. *If we denote*

$$\alpha_{ij} = \frac{1}{\sqrt{2}}\Big(w(X_i)\,\epsilon_i^{P^w}\,k_{X_i} - w(X_j)\,\epsilon_j^{P^w}\,k_{X_j}\Big),$$
$$\hat{\alpha}_{ij} = \frac{1}{\sqrt{2}}\Big(w(X_i)\,\widehat{\varepsilon}_i\,k_{X_i} - w(X_j)\,\widehat{\varepsilon}_j\,k_{X_j}\Big),$$

then

$$B - B_{ora} = \Delta_{\text{inv}} + \Delta_{\text{res}},$$

where

$$\Delta_{\text{inv}} = \left(\mathcal{L}^{-1}_{P_n^w,\lambda} - \mathcal{L}^{-1}_{P^w,\lambda}\right)\left(\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\,\hat{\alpha}_{ij}\right),$$
$$\Delta_{\text{res}} = \mathcal{L}^{-1}_{P^w,\lambda}\left(\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\,(\hat{\alpha}_{ij} - \alpha_{ij})\right).$$

Lemma F.16. *For $0 < \eta < 1$, under Assumptions 4.1 and 4.2, suppose $\tau_n \geq \tau_0$ for all sufficiently large n, $\tau_n \to \infty$, and $\lambda = \lambda_n \to 0$. If, for all sufficiently large n,*

$$n > \max\left\{\frac{(\tau_n\kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda}\right\}$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then, for all sufficiently large n, with probability at least $1 - \eta$,

$$\|\hat{f} - f_{P^w}\| \lesssim \sqrt{\frac{\tau_n\mathcal{N}_\lambda}{n\lambda}}\,\overline{\sigma}\,\kappa^{2s+1}\big(1 + \|g_*\|_{\mathsf{H}}\big)\log(\frac{2}{\eta}),$$
$$\|\hat{f} - f_*\| \lesssim \left[\sqrt{\frac{\tau_n\mathcal{N}_\lambda}{n\lambda}} + \lambda^{\min\{s,\frac{1}{2}\}}\right]\overline{\sigma}\,\kappa^{2s+1}\big(1 + \|g_*\|_{\mathsf{H}}\big)\log(\frac{2}{\eta}).$$

Lemma F.17. *For $0 < \eta < 1$, under Assumptions 4.1, 4.2, and 4.4, suppose $\tau_n \geq \tau_0$ for all sufficiently large n, $\tau_n \to \infty$, and $\lambda = \lambda_n \to 0$. If, for all sufficiently large n,*

$$n > \max\left\{\frac{(\tau_n\kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda}\right\}$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then, for all sufficiently large n, with probability at least $1 - \eta$ over the data, conditionally on the data with probability at least $1 - \eta$,

$$\|\Delta_{\text{inv}}\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \mathfrak{C}_1\overline{\sigma}\,\kappa^{2s+4}\big(1 + \|g_*\|_{\mathsf{H}}\big)\frac{\tau_n\mathcal{N}_\lambda}{\sqrt{n\lambda}}\left(\log\left(\frac{2}{\eta}\right)\right)^3.$$

Lemma F.18. *For $0 < \eta < 1$, under Assumptions 4.1, 4.2, and 4.4, suppose $\tau_n \geq \tau_0$ for all sufficiently large n, $\tau_n \to \infty$, and $\lambda = \lambda_n \to 0$. If, for all sufficiently large n,*

$$n > \max\left\{\frac{(\tau_n\kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda}\right\}$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then, for all sufficiently large n, with probability at least $1-\eta$ over the data, conditionally on the data with probability at least $1-\eta$,

$$\|\Delta_{\mathrm{res}}\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \mathfrak{C}_1 \overline{\sigma}\, \kappa^{2s+3}\big(1+\|g_*\|_{\mathsf{H}}\big)\frac{\tau_n \mathcal{N}_\lambda}{\sqrt{n\lambda}}\left(\log\left(\frac{2}{\eta}\right)\right)^2.$$

Proof of Proposition 4.4. Conditional on $\mathcal{D}$, the processes B_{ora} and B are centered Gaussian elements in $\mathsf{L}^2(\mathbb{Q})$, because the multipliers h_{ij} are independent standard Gaussian variables. The anti-symmetric covariance calculation gives the covariance operators $\hat{\mathcal{V}}_{\mathrm{oracle}}$ and $\hat{\mathcal{V}}_{\mathrm{feas}}$ displayed in Proposition 4.4. The high-probability bound for $\|B-B_{\mathsf{ora}}\|_{\mathsf{L}^2(\mathbb{Q})}$ follows by combining Lemmas F.15, F.17, and F.18. In these lemmas, the data-dependent events are intersected first, and the conditional multiplier events are then intersected on that data event. Replacing η in each constituent bound by a fixed fraction of η and applying the union bound changes only the universal constants inside the logarithms. □

F.7 Proof of Proposition 4.5

Lemma F.19 (Conditional Strassen–Dudley coupling lemma)**.** *The following simplified form is a direct consequence of the conditional Strassen–Dudley theorem of Monrad and Philipp [1991, Theorem 4]. Let (S, d) be a complete separable metric space, let $\mathcal{A}$ be a countably generated sub-σ-field, and suppose the underlying probability space supports a $[0,1]$-uniform random variable independent of $\mathcal{A}\vee\sigma(X)$. If X, X', Y' are S-valued random elements such that*

$$X \mid \mathcal{A} \overset{d}{=} X' \mid \mathcal{A}, \qquad P\{d(X',Y') > r\} \le \varepsilon$$

for some $r, \varepsilon \ge 0$, then there exists an S-valued random element Y such that

$$Y \mid \mathcal{A} \overset{d}{=} Y' \mid \mathcal{A} \qquad \text{and} \qquad P\{d(X,Y) > r\} \le \varepsilon.$$

Let $\{(\nu_j, e_j)\}_{j\ge1}$ be the eigensystem of $\mathcal{V}$ introduced above, and retain the notation $\Pi_m = \sum_{j=1}^m e_j \otimes_{\mathsf{L}^2(\mathbb{Q})} e_j$.

Proof of Proposition 4.5. The proof uses the covariance-comparison and anti-symmetric multiplier-bootstrap techniques of Singh and Vijaykumar [2023], but all covariance operators, projections, Hilbert-Schmidt norms, and Gaussian elements below are taken in $\mathsf{L}^2(\mathbb{Q})$.

First, the anti-symmetric multiplier process has the desired empirical covariance. Since $V_i = U_i + \mu$, where

$$\mu = (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(f_* - f_{P^w}),$$

we have $V_i - V_j = U_i - U_j$. Conditional on the data,

$$\frac{1}{n}\sum_{i=1}^n\sum_{j=1}^n h_{ij}\frac{V_i - V_j}{\sqrt{2}}$$

is a centered Gaussian element in $\mathsf{L}^2(\mathbb{Q})$. Its conditional covariance is

$$\mathbb{E}_h\left[\left(\frac{1}{n}\sum_{i,j} h_{ij}\frac{U_i-U_j}{\sqrt{2}}\right)\otimes_{\mathsf{L}^2(\mathbb{Q})}\left(\frac{1}{n}\sum_{k,\ell} h_{k\ell}\frac{U_k-U_\ell}{\sqrt{2}}\right)\right] = \mathbb{E}_n[U_i\otimes_{\mathsf{L}^2(\mathbb{Q})}U_i] - \mathbb{E}_n[U_i]\otimes_{\mathsf{L}^2(\mathbb{Q})}\mathbb{E}_n[U_i]$$
$$= \hat{\mathcal{V}}_{\mathsf{ora}}.$$

Thus the conditional law of the anti-symmetric multiplier process is the same as that of $\widehat{\mathcal{V}}_{\mathsf{ora}}^{1/2} g$.

Since $\mathcal{V}$ is trace class, writing $g_i := g(e_i)$ for the coordinates of the isonormal process in Proposition 4.5 gives

$$\mathcal{V}^{1/2} g = \sum_{i=1}^{\infty} \sqrt{\nu_i}\, g_i e_i \in \mathsf{L}^2(\mathbb{Q}) \quad \text{a.s.}, \qquad \mathbb{E}\Big[\big(\mathcal{V}^{1/2} g\big) \otimes_{\mathsf{L}^2(\mathbb{Q})} \big(\mathcal{V}^{1/2} g\big) \Big] = \mathcal{V}.$$

We next compare the Gaussian elements with covariances $\mathcal{V}$ and $\widehat{\mathcal{V}}_{\mathsf{ora}}$. Fix $m \geq 1$, and write $\Pi_m^{\perp} = I - \Pi_m$. Define

$$\Delta_{\mathrm{HS}} = \|\widehat{\mathcal{V}}_{\mathsf{ora}} - \mathcal{V}\|_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})}, \qquad \Delta_{\mathrm{tail}} = \Big| \mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})} \{ \Pi_m^{\perp} (\widehat{\mathcal{V}}_{\mathsf{ora}} - \mathcal{V}) \Pi_m^{\perp} \} \Big|.$$

The covariance-comparison argument gives a random element G, with the same conditional distribution as $\widehat{\mathcal{V}}_{\mathsf{ora}}^{1/2} g$, such that conditional on the data, with probability at least $1 - 3\eta$,

$$\|\mathcal{V}^{1/2} g - G\|_{\mathsf{L}^2(\mathbb{Q})} \leq \{1 + \sqrt{2 \log\Big(\frac{1}{\eta}\Big)}\} \Big\{ m^{1/4} \Delta_{\mathrm{HS}}^{1/2} + \Delta_{\mathrm{tail}}^{1/2} + 2\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \Big\}. \tag{F.1}$$

Indeed, decompose $\mathcal{V}^{1/2} g$ into its Π_m and $\Pi_m^{\perp}$ parts. On the leading m-dimensional subspace, the Powers–Størmer inequality gives

$$\|(\Pi_m \mathcal{V} \Pi_m)^{1/2} - (\Pi_m \widehat{\mathcal{V}}_{\mathsf{ora}} \Pi_m)^{1/2}\|_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})} \leq m^{1/4} \|\mathcal{V} - \widehat{\mathcal{V}}_{\mathsf{ora}}\|_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})}^{1/2}.$$

Borell's inequality then controls the corresponding Gaussian norm. The $\Pi_m^{\perp} \mathcal{V} \Pi_m^{\perp}$ tail contributes $\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V})$. The remaining $\Pi_m^{\perp} \widehat{\mathcal{V}}_{\mathsf{ora}} \Pi_m^{\perp}$ tail is coupled to the full $\widehat{\mathcal{V}}_{\mathsf{ora}}^{1/2} g$ by Lemma F.19 and is bounded by Borell's inequality through

$$\|(\Pi_m^{\perp} \widehat{\mathcal{V}}_{\mathsf{ora}} \Pi_m^{\perp})^{1/2}\|_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})} \leq \Delta_{\mathrm{tail}}^{1/2} + \sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}).$$

This proves (F.1).

It remains to bound Δ_{HS} and Δ_{tail} under the target-norm bound on U_i. Hilbert-space Bernstein bounds (Lemma F.11) for the empirical covariance operator imply, with probability at least $1 - 3\eta$,

$$\Delta_{\mathrm{HS}} \lesssim \Big(\log\Big(\frac{2}{\eta}\Big) \Big)^2 \left\{ \left(\frac{\bar{U}^2 \sigma_{\mathsf{L}^2(\mathbb{Q})}^2(0, \mathcal{V})}{n} \right)^{1/2} + \frac{\bar{U}^2}{n} \right\},$$

and

$$\Delta_{\mathrm{tail}} \lesssim \log\Big(\frac{2}{\eta}\Big) \left\{ \left(\frac{\bar{U}^2 \sigma_{\mathsf{L}^2(\mathbb{Q})}^2(m, \mathcal{V})}{n} \right)^{1/2} + \frac{\bar{U}^2}{n} \right\}.$$

Substituting these bounds into (F.1), using

$$\Delta_{\mathrm{tail}}^{1/2} \lesssim \Big(\log\Big(\frac{2}{\eta}\Big) \Big)^{1/2} \left[\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) + \left\{ \frac{\bar{U}^2 \sigma_{\mathsf{L}^2(\mathbb{Q})}^2(0, \mathcal{V})}{n} + \frac{\bar{U}^4}{n^2} \right\}^{1/4} \right],$$

and absorbing logarithmic factors into $\Big(\log\Big(\frac{2}{\eta}\Big) \Big)^{3/2}$, gives the claim after taking the infimum over $m \geq 1$. The total probability is at least $1 - 6\eta$ by a union bound over the covariance-estimation event and the conditional Gaussian-coupling event. □

F.8 Proof of Corollary 4.1

Proof of Corollary 4.1. Fix $\eta \in (0,1)$, and let $\eta_{\mathrm{tol}} = c\eta^2$, where $c > 0$ is a sufficiently small universal constant. In this proof, $\Delta_{B,\mathrm{feas}}$ and $\Delta_{B,\mathrm{coup}}$ are evaluated with tolerance η_{tol}. Write

$$W = B_{\mathsf{ora}}, \qquad W' = B, \qquad r_{\mathrm{feas}} = \Delta_{B,\mathrm{feas}}, \qquad r_{\mathrm{coup}} = \Delta_{B,\mathrm{coup}}.$$

Let $\mathcal{E}_{\mathrm{feas}}$ be the data event from Proposition 4.4, evaluated with tolerance η_{tol}. Then $P(\mathcal{E}_{\mathrm{feas}}^c) \le \eta_{\mathrm{tol}}$, and, on $\mathcal{E}_{\mathrm{feas}}$, the conditional failure probability is at most η_{tol}. Consequently,

$$\begin{aligned} P\big(\|W' - W\|_{\mathsf{L}^2(\mathbb{Q})} > r_{\mathrm{feas}}\big) &\le P(\mathcal{E}_{\mathrm{feas}}^c) + \mathbb{E}\big[\mathbf{1}_{\mathcal{E}_{\mathrm{feas}}} P\big(\|W' - W\|_{\mathsf{L}^2(\mathbb{Q})} > r_{\mathrm{feas}} \,\big|\, \mathcal{D}\big)\big] \\ &\le 2\eta_{\mathrm{tol}}. \end{aligned}$$

Next, the identity $V_i = U_i + \mu$, with

$$\mu = (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_{P^w}(f_* - f_{P^w}),$$

implies $V_i - V_j = U_i - U_j$. Hence, conditional on $\mathcal{D}$, the oracle anti-symmetric bootstrap W is centered Gaussian with covariance

$$\widehat{\mathcal{V}}_{\mathsf{ora}} = \mathbb{E}_n[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i] - \mathbb{E}_n[U_i] \otimes_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}_n[U_i].$$

Thus $W \mid \mathcal{D}$ has the same law as $\widehat{\mathcal{V}}_{\mathsf{ora}}^{1/2} g$. By Proposition 4.5, applied with tolerance η_{tol}, there is a Gaussian element $Z_0 \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ and a random element G, with $G \mid \mathcal{D}$ distributed as $W \mid \mathcal{D}$, such that

$$P\big(\|Z_0 - G\|_{\mathsf{L}^2(\mathbb{Q})} > r_{\mathrm{coup}}\big) \le 6\eta_{\mathrm{tol}}.$$

By Lemma F.19, applied to the regular conditional laws given $\mathcal{D}$, we may replace the copy G by the actual oracle bootstrap W: there exists a copy Z with

$$Z \mid \mathcal{D} \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$$

such that

$$P\big(\|Z - W\|_{\mathsf{L}^2(\mathbb{Q})} > r_{\mathrm{coup}}\big) \le 6\eta_{\mathrm{tol}}.$$

Gluing this coupling with the original pair (W, W') and using the triangle inequality gives

$$P\big(\|Z - W'\|_{\mathsf{L}^2(\mathbb{Q})} > r_{\mathrm{coup}} + r_{\mathrm{feas}}\big) \le 8\eta_{\mathrm{tol}}.$$

Let

$$A(\mathcal{D}) = P\big(\|Z - W'\|_{\mathsf{L}^2(\mathbb{Q})} > r_{\mathrm{coup}} + r_{\mathrm{feas}} \,\big|\, \mathcal{D}\big).$$

Then $\mathbb{E}A(\mathcal{D}) \le 8\eta_{\mathrm{tol}}$. Markov's inequality gives

$$P\{A(\mathcal{D}) > \eta\} \le \frac{8\eta_{\mathrm{tol}}}{\eta}.$$

Choosing $c \le 1/8$ yields $8\eta_{\mathrm{tol}}/\eta \le \eta$. Therefore, with probability at least $1 - \eta$ over the data,

$$P\big(\|B - Z\|_{\mathsf{L}^2(\mathbb{Q})} \le \Delta_{B,\mathrm{feas}} + \Delta_{B,\mathrm{coup}} \,\big|\, \mathcal{D}\big) \ge 1 - \eta.$$

Renaming Z as Z' proves the corollary. □

F.9 Proof of Corollary 4.2

Lemma F.20. *Suppose that the eigenvalues of $\mathcal{L}_Q$ satisfy $\mu_k \le \alpha k^{-\beta}$ for all $k \ge 1$, where $\alpha > 0$ and $\beta > 1$. Then, for every $\lambda > 0$,*

$$\mathcal{N}_\lambda \le \alpha^{1/\beta} \frac{\beta}{\beta - 1} \lambda^{-1/\beta}.$$

Lemma F.21. *Suppose that the eigenvalues of $\mathcal{L}_Q$ satisfy $\mu_k \le \alpha k^{-\beta}$ for all $k \ge 1$, where $\alpha > 0$ and $\beta > 1$. Then, for every $m \in \mathbb{N}_+$,*

$$\sigma(m, \mathcal{L}_Q) \le \left(\frac{\alpha}{\beta - 1} \right)^{1/2} m^{1/2 - \beta/2}.$$

Lemma F.22. *For $\beta > 0$ and $\gamma > 0$,*

$$\lim_{a \to \infty} \frac{\int_a^\infty \exp(-\beta x^\gamma)\, dx}{\exp(-\beta a^\gamma)/(\beta \gamma a^{\gamma - 1})} = 1.$$

Lemma F.23. *Suppose that the eigenvalues of $\mathcal{L}_Q$ satisfy $\mu_k \le \alpha \exp(-\beta k^\gamma)$ for all $k \ge 1$, where $\alpha > 0$, $\beta > 0$, and $\gamma > 0$. Then there exist $\lambda_0 \in (0, 1)$ and $C_{\alpha,\beta,\gamma} > 0$ such that, for every $0 < \lambda \le \lambda_0$,*

$$\mathcal{N}_\lambda \le C_{\alpha,\beta,\gamma} \{\log(1/\lambda)\}^{1/\gamma}.$$

Lemma F.24. *Suppose that the eigenvalues of $\mathcal{L}_Q$ satisfy $\mu_k \le \alpha \exp(-\beta k^\gamma)$ for all $k \ge 1$, where $\alpha > 0$, $\beta > 0$, and $\gamma > 0$. Then there exists $C_{\alpha,\beta,\gamma} > 0$ such that, for every $m \in \mathbb{N}_+$,*

$$\sigma(m, \mathcal{L}_Q) \le C_{\alpha,\beta,\gamma} m^{1/2 - \gamma/2} \exp(-\beta m^\gamma / 2).$$

Lemma F.25. *Under Assumptions 4.1 and 4.2, when $\tau_n \ge \tau_0$, for every $m \in \mathbb{N}_+$ and $\lambda > 0$, we have*

$$\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \le \frac{\tau_n}{\lambda} \left(\overline{\sigma}^2 + \kappa^2 \max\{\kappa^{2s-1}, 1\}^2 \lambda^{2\min\{s, 1/2\}} \|g_*\|^2 \right) \sigma^2(m, \mathcal{L}_Q).$$

Proposition F.1. *Suppose $\mu_k \le \alpha k^{-\beta}$ for some $\alpha > 0$ and $\beta > 1$. Under Assumptions 4.1, 4.2, and 4.4, when $\tau_n \ge \tau_0$, if $\tau_n \to \infty$, $\lambda \to 0$, and*

$$n > \max\left\{ \frac{(\tau_n \kappa^2 + 1) \log(e + \mathcal{N}_\lambda)}{C_2 \lambda}, \frac{\tau_n}{\lambda \mathcal{N}_\lambda} \right\},$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then with probability at least $1 - \eta$, there exist i.i.d. Gaussian elements $Z_i \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ such that

$$\begin{aligned} &\left\| \sqrt{n}(\hat{f} - f_{P^w}) - \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \right\|_{\mathsf{L}^2(\mathbb{Q})} \\ &\quad \lesssim \frac{\tau_n}{\sqrt{n}} \lambda^{-1/2 - 1/(2\beta)} \log\left(\frac{1}{\lambda}\right) \left(\log\left(\frac{2}{\eta}\right) \right)^2 + \sqrt{\frac{\tau_n}{\lambda}} \left(\frac{\tau_n}{n} \right)^{\frac{\beta - 1}{2(\beta + 3)}} \log(n) \log\left(\frac{2}{\eta}\right). \end{aligned}$$

Proof of Proposition F.1. Apply Propositions 4.2 and 4.3 with a fixed fraction of the tolerance η. By Lemma F.20,

$$\mathcal{N}_\lambda \lesssim \lambda^{-1/\beta}.$$

Consequently, the Bahadur remainder satisfies

$$\Delta_{G,\mathrm{Bah}} \lesssim \frac{\tau_n}{\sqrt{n}} \lambda^{-\frac{1}{2}-\frac{1}{2\beta}} \log\left(\frac{1}{\lambda}\right) \left(\log\left(\frac{2}{\eta}\right)\right)^2.$$

Lemmas F.25 and F.21 give

$$\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \lesssim \sqrt{\frac{\tau_n}{\lambda}}\, m^{-\frac{\beta-1}{2}}.$$

Moreover, Proposition 4.2 gives $\bar{U} \lesssim \tau_n/\sqrt{\lambda}$. Hence Proposition 4.3 yields

$$\Delta_{G,\mathrm{coup}} \lesssim \inf_{m\geq 1} \left\{ \sqrt{\frac{\tau_n}{\lambda}}\, m^{-\frac{\beta-1}{2}} \left(\log\left(\frac{2}{\eta}\right)\right)^{1/2} + \frac{\tau_n}{\sqrt{n\lambda}} m^2 \log\left(\frac{m^2}{\eta}\right) \right\}.$$

Take

$$m = \left\lceil \left(\frac{n}{\tau_n}\right)^{1/(\beta+3)} \right\rceil.$$

The sample-size condition implies $n/\tau_n \to \infty$, and substitution gives

$$\Delta_{G,\mathrm{coup}} \lesssim \sqrt{\frac{\tau_n}{\lambda}} \left(\frac{\tau_n}{n}\right)^{\frac{\beta-1}{2(\beta+3)}} \log(n) \log\left(\frac{2}{\eta}\right).$$

The triangle inequality and a union bound combine the Bahadur and coupling events. Finally, the coupled Gaussian element can be represented as $n^{-1/2}\sum_{i=1}^n Z_i$ by adjoining an independent Gaussian bridge, with the Z_i's i.i.d. with covariance $\mathcal{V}$. This proves the claim. □

Proposition F.2. *Suppose $\mu_k \leq \alpha k^{-\beta}$ for some $\alpha > 0$ and $\beta > 1$. Under Assumptions 4.1, 4.2, and 4.4, together with Definition 4.1, when $\tau_n \geq \tau_0$, if $\tau_n \to \infty$, $\lambda \to 0$, and*

$$n > \max\left\{ \frac{(\tau_n \kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{C_2 \lambda}, \frac{\tau_n}{\lambda \mathcal{N}_\lambda} \right\},$$

where C_2 is the constant in Lemma F.13 evaluated at the component tolerance $\eta_{\mathrm{tol}} = c\eta^2$ used in Corollary 4.1, then, on a data event of probability at least $1-\eta$, there exists a coupling of B and Z, conditionally on $\mathcal{D}$, such that $Z \mid \mathcal{D} \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$. On this event, let

$$R_{\mathrm{B,poly}} := \left(\log\left(\frac{2}{\eta}\right)\right)^3 \frac{\tau_n}{\sqrt{n}} \lambda^{-1/2-1/\beta} + \left(\log\left(\frac{2}{\eta}\right)\right)^{3/2} \sqrt{\frac{\tau_n}{\lambda}} \left(\frac{\tau_n \lambda^{1-1/\beta}}{n}\right)^{\frac{\beta-1}{2(2\beta-1)}}.$$

Then

$$\mathbb{P}\left\{ \|Z - B\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim R_{\mathrm{B,poly}} \,\middle|\, \mathcal{D} \right\} \geq 1 - \eta.$$

Proof of Proposition F.2. Apply Corollary 4.1. By Proposition 4.4 and Lemma F.20,

$$\Delta_{B,\mathrm{feas}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^3 \frac{\tau_n}{\sqrt{n}} \lambda^{-\frac{1}{2}-\frac{1}{\beta}}.$$

Also, Proposition 4.2 and Lemma F.7 imply

$$\frac{\bar{U}^2 \mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})}{n} + \frac{\bar{U}^4}{n^2} \lesssim \frac{\tau_n^3 \mathcal{N}_\lambda}{n\lambda} + \frac{\tau_n^4}{n^2\lambda^2}.$$

The condition $n > \tau_n/(\lambda\mathcal{N}_\lambda)$ ensures that the second term is bounded by the first. Therefore, Proposition 4.5 and Lemmas F.25, F.20, and F.21 give

$$\Delta_{B,\mathrm{coup}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^{3/2} \inf_{m\geq 1}\left\{ m^{1/4}\left(\frac{\tau_n^3\lambda^{-1/\beta}}{n\lambda}\right)^{1/4} + \sqrt{\frac{\tau_n}{\lambda}}\, m^{-\frac{\beta-1}{2}} \right\}.$$

Choose

$$m = \left\lceil \left(\frac{n}{\tau_n \lambda^{1-1/\beta}}\right)^{1/(2\beta-1)} \right\rceil.$$

Both terms inside the braces are then bounded by

$$\sqrt{\frac{\tau_n}{\lambda}} \left(\frac{\tau_n\lambda^{1-1/\beta}}{n}\right)^{\frac{\beta-1}{2(2\beta-1)}}.$$

Combining the feasible and coupling bounds in Corollary 4.1, with tolerances adjusted by universal constants, proves the stated outer and conditional probability claim. □

Proposition F.3. *Suppose $\mu_k \leq \alpha\exp(-\beta k^\gamma)$ for some $\alpha, \beta, \gamma > 0$. Under Assumptions 4.1, 4.2, and 4.4, when $\tau_n \geq \tau_0$, if $\tau_n \to \infty$, $\lambda \to 0$, and*

$$n > \max\left\{ \frac{(\tau_n\kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda} \right\},$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then with probability at least $1-\eta$, there exist i.i.d. Gaussian elements $Z_i \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ such that

$$\begin{aligned}
&\left\| \sqrt{n}(\hat{f} - f_{P^w}) - \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i \right\|_{\mathsf{L}^2(\mathbb{Q})} \\
&\quad \lesssim \frac{\tau_n}{\sqrt{n\lambda}} \log^2\left(\frac{2}{\eta}\right) \left[\log^{1/(2\gamma)}\left(\frac{1}{\lambda}\right) \log\log\left(\frac{1}{\lambda}\right) + \log^{2/\gamma}\left(\frac{n}{\tau_n}\right) \log\log\left(\frac{n}{\tau_n}\right) \right].
\end{aligned}$$

Proof of Proposition F.3. Apply Propositions 4.2 and 4.3 with a fixed fraction of η. Lemma F.23 gives

$$\mathcal{N}_\lambda \lesssim \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/\gamma},$$

and hence

$$\Delta_{G,\mathrm{Bah}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^2 \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/(2\gamma)} \log\log\left(\frac{1}{\lambda}\right).$$

Lemmas F.25 and F.24 imply

$$\sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V}) \lesssim \sqrt{\frac{\tau_n}{\lambda}}\, m^{\frac{1}{2}-\frac{\gamma}{2}} \exp\left(-\frac{\beta m^\gamma}{2}\right).$$

Together with $\bar{U} \lesssim \tau_n/\sqrt{\lambda}$, this yields

$$\Delta_{G,\mathrm{coup}} \lesssim \inf_{m\geq 1}\left\{\sqrt{\frac{\tau_n}{\lambda}}\, m^{\frac{1}{2}-\frac{\gamma}{2}} e^{-\beta m^\gamma/2} + \frac{\tau_n}{\sqrt{n\lambda}} m^2 \log\left(\frac{m^2}{\eta}\right)\right\} \log\left(\frac{2}{\eta}\right).$$

Take

$$m = \left\lceil \left\{ \frac{1}{\beta} \log\left(\frac{n}{\tau_n}\right)\right\}^{1/\gamma} \right\rceil.$$

The sample-size condition makes this choice well defined for all sufficiently large n. Since $m^{1/2-\gamma/2} \leq 1 + m^2$, substitution gives

$$\Delta_{G,\mathrm{coup}} \lesssim \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{n}{\tau_n}\right)\right)^{2/\gamma} \log\log\left(\frac{n}{\tau_n}\right) \log\left(\frac{2}{\eta}\right).$$

The triangle inequality, a union bound, and the Gaussian-bridge representation used in the polynomial case complete the proof. □

Proposition F.4. *Suppose $\mu_k \leq \alpha \exp(-\beta k^\gamma)$ for some $\alpha, \beta, \gamma > 0$. Under Assumptions 4.1, 4.2, and 4.4, together with Definition 4.1, when $\tau_n \geq \tau_0$, if $\tau_n \to \infty$, $\lambda \to 0$, and*

$$n > \max\left\{\frac{(\tau_n\kappa^2+1)\log(e+\mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda}\right\},$$

where C_2 is the constant in Lemma F.13 evaluated at the component tolerance $\eta_{\mathrm{tol}} = c\eta^2$ used in Corollary 4.1, then, on a data event of probability at least $1-\eta$, there exists a coupling of B and Z, conditionally on $\mathcal{D}$, such that $Z \mid \mathcal{D} \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})$ and

$$\mathbb{P}\Bigg\{\|Z-B\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^3 \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/\gamma} + \left(\log\left(\frac{2}{\eta}\right)\right)^{3/2} \left(\frac{\tau_n^3}{n\lambda}\right)^{1/4} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/(4\gamma)} \left(\log\left(\frac{n}{\tau_n\lambda}\right)\right)^{1/(4\gamma)} \;\Big|\; \mathcal{D}\Bigg\} \geq 1-\eta.$$

Proof of Proposition F.4. Apply Corollary 4.1. Proposition 4.4 and Lemma F.23 give

$$\Delta_{B,\mathrm{feas}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^3 \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/\gamma}.$$

As in the polynomial case, the sample-size condition and Lemma F.7 imply

$$\frac{\bar{U}^2 \mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})}{n} + \frac{\bar{U}^4}{n^2} \lesssim \frac{\tau_n^3 \mathcal{N}_\lambda}{n\lambda}.$$

Thus Proposition 4.5 and Lemmas F.25, F.23, and F.24 yield

$$\Delta_{B,\mathrm{coup}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^{3/2} \inf_{m\geq 1}\left\{ m^{1/4} \left(\frac{\tau_n^3}{n\lambda}\right)^{1/4} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/(4\gamma)} + \sqrt{\frac{\tau_n}{\lambda}}\, m^{\frac{1}{2}-\frac{\gamma}{2}} e^{-\beta m^\gamma/2}\right\}.$$

For a sufficiently large fixed constant $C > 1/(2\beta)$, take

$$m = \left\lceil \left\{ C \log\left(\frac{n}{\tau_n \lambda}\right) \right\}^{1/\gamma} \right\rceil .$$

The first term in braces is bounded by

$$\left(\frac{\tau_n^3}{n\lambda}\right)^{1/4} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/(4\gamma)} \left(\log\left(\frac{n}{\tau_n\lambda}\right)\right)^{1/(4\gamma)} .$$

Because $C\beta/2 > 1/4$, the exponential factor in the spectral-tail term absorbs its polynomial factor, so that term is bounded by the same display for all sufficiently large n. Combining the coupling and feasible-bootstrap bounds proves the result. □

Proof of Corollary 4.2. Let $\bar{\rho} < \infty$ be an almost-sure upper bound for the density ratio and choose $C_0 \geq \max\{\bar{\rho}, \tau_0\}$. Then $w_n = \rho$, $\mathcal{L}_{P^w} = \mathcal{L}_Q$, and Assumption 4.4 holds with $\mathfrak{C}_1 = 1$. A bounded density ratio also satisfies Definition 4.1: one may take $\gamma = 1$ and $W = \sigma^2 = \max\{\bar{\rho}, 1\}$.

We next justify the use of the spectral approximation bounds with a fixed envelope. Under either two-sided spectral condition in the corollary, $\mathcal{N}_\lambda \to \infty$ as $\lambda \to 0$. Since $\tau_n = C_0$, the first, fully nonasymptotic inequality in Lemma F.12 implies the same simplified Bahadur bound used in Proposition 4.2: the linear term is absorbed by the square-root term under the stated sample-size condition, while $\tau_n\kappa^2 + 1 \lesssim C_0$ and $\log(e + \mathcal{N}_\lambda) \asymp \log \mathcal{N}_\lambda$ for all sufficiently large n. The remaining Gaussian and bootstrap arguments use only this bound, the fixed truncated-weight envelope, and their displayed sample-size conditions. Thus their nonasymptotic conclusions, and hence the specialized spectral bounds, remain valid for $\tau_n = C_0$.

First consider polynomial spectral decay. At $\eta \asymp n^{-1}$, Proposition 4.1 and Propositions F.1 and F.2 give

$$\Delta \lesssim \log^3 n \left(\Delta_0 + \Delta_1 + \Delta_2 + \Delta_3\right), \tag{F.2}$$

where

$$\begin{aligned}
\Delta_0 &= \sqrt{n}\,\lambda, \\
\Delta_1 &= n^{-\frac{\beta-1}{2(\beta+3)}} \lambda^{-\frac{1}{2}} \tau_n^{\frac{\beta+1}{\beta+3}}, \\
\Delta_2 &= n^{-\frac{1}{2}} \lambda^{-\frac{1}{2}-\frac{1}{\beta}} \tau_n, \\
\Delta_3 &= n^{-\frac{\beta-1}{2(2\beta-1)}} \lambda^{-\frac{\beta^2+\beta-1}{2\beta(2\beta-1)}} \tau_n^{\frac{3\beta-2}{2(2\beta-1)}}.
\end{aligned}$$

Here the Gaussian Bahadur term $\tau_n n^{-1/2}\lambda^{-1/2-1/(2\beta)}$ is absorbed by Δ_2, since their ratio is $\lambda^{1/(2\beta)} \to 0$. The factor $\log^3 n$ in (F.2) is obtained directly from the component bounds: the Gaussian Bahadur and coupling terms contribute at most $\log^3 n$ and $\log^2 n$, respectively, whereas the feasible-bootstrap and bootstrap-coupling terms contribute at most $\log^3 n$ and $\log^{3/2} n$, respectively.

Write $\lambda = n^{-a}$. Since $\tau_n = C_0$, the four powers of n in (F.2) are

$$\delta_0 = \frac{1}{2} - a, \quad \delta_1 = -\frac{\beta-1}{2(\beta+3)} + \frac{a}{2}, \quad \delta_2 = -\frac{1}{2} + a\left(\frac{1}{2} + \frac{1}{\beta}\right), \quad \delta_3 = -\frac{\beta-1}{2(2\beta-1)} + a\frac{\beta^2+\beta-1}{2\beta(2\beta-1)}.$$

The exponent δ_0 is decreasing in a, whereas $\delta_1, \delta_2, \delta_3$ are increasing. The balance $\delta_0 = \delta_2$ gives $a = 2\beta/(3\beta+2)$, but at this value $\delta_1 > \delta_0$ and $\delta_3 > \delta_0$, so it cannot be optimal.

Balancing $\delta_0 = \delta_1$ gives

$$a_{01} = \frac{2(\beta+1)}{3(\beta+3)}, \qquad A_{01} = \frac{-\beta+5}{6(\beta+3)}.$$

At this value, $\delta_2 < A_{01}$, and

$$\delta_1 - \delta_3 = -\frac{(\beta-1)(\beta^2-12\beta+2)}{6\beta(2\beta-1)(\beta+3)}.$$

Balancing $\delta_0 = \delta_3$ gives

$$a_{03} = \frac{\beta(3\beta-2)}{5\beta^2-\beta-1}, \qquad A_{03} = \frac{-\beta^2+3\beta-1}{2(5\beta^2-\beta-1)}.$$

At this value, $\delta_2 < A_{03}$, and

$$\delta_1 - \delta_0 = \frac{(\beta-1)(-\beta^2+12\beta-2)}{2(\beta+3)(5\beta^2-\beta-1)}.$$

Consequently, with $\beta_0 = 6+\sqrt{34}$, the sharp optimizer is

$$(a_\beta^\sharp, r_\beta^\sharp) = \begin{cases} \left(\dfrac{2(\beta+1)}{3(\beta+3)}, \dfrac{\beta-5}{6(\beta+3)}\right), & 5 < \beta \le \beta_0, \\ \left(\dfrac{\beta(3\beta-2)}{5\beta^2-\beta-1}, \dfrac{\beta^2-3\beta+1}{2(5\beta^2-\beta-1)}\right), & \beta > \beta_0. \end{cases}$$

The switch occurs when $\beta^2 - 12\beta + 2 = 0$. Thus the sharper, piecewise choice $a = a_\beta^\sharp$ yields

$$\Delta \lesssim n^{-r_\beta^\sharp} \log^3 n$$

and, by Theorem 4.1,

$$\left|\mathbb{P}\{f_* \notin \mathsf{CS}_\alpha(\widehat{f})\} - \alpha\right| \lesssim n^{-1} + n^{-r_\beta^\sharp} \log^3 n.$$

For the non-piecewise statement in the corollary, instead set

$$\widetilde{a}_\beta = \frac{2(\beta+1)}{3(\beta+3)}, \qquad \widetilde{r}_\beta = \frac{(\beta-5)(\beta+10)}{6(\beta+3)(2\beta+5)}.$$

At $a = \widetilde{a}_\beta$, the four exponents simplify to

$$\delta_0 = \delta_1 = -\frac{\beta-5}{6(\beta+3)}, \qquad \delta_2 = -\frac{(\beta-1)(\beta+4)}{6\beta(\beta+3)},$$

and

$$\delta_3 = -\frac{(\beta-2)(\beta^2+4\beta-1)}{6\beta(\beta+3)(2\beta-1)}.$$

Writing $x = \beta - 5 > 0$, direct calculation gives

$$-\delta_0 - \widetilde{r}_\beta = -\delta_1 - \widetilde{r}_\beta = \frac{(\beta-5)^2}{6(\beta+3)(2\beta+5)} \ge 0,$$

$$-\delta_2 - \widetilde{r}_\beta = \frac{x^3 + 21x^2 + 192x + 540}{6\beta(\beta+3)(2\beta+5)} > 0,$$

and

$$-\delta_3 - \widetilde{r}_\beta = \frac{97x^2 + 879x + 1980}{6\beta(\beta+3)(2\beta-1)(2\beta+5)} > 0.$$

Consequently, $\delta_j \le -\widetilde{r}_\beta$ for $j = 0, 1, 2, 3$, and hence

$$\Delta \lesssim n^{-\widetilde{r}_\beta} \log^3 n.$$

Because $\sqrt{\tau_n} = O(1)$, Theorem 4.1 with $\eta \asymp n^{-1}$ proves part (1).

Under exponential spectral decay, Proposition 4.1 and Propositions F.3 and F.4 give, after suppressing only their displayed logarithmic and tolerance factors, the polynomial-order envelope

$$\Delta \lesssim \sqrt{n}\,\lambda + \frac{\tau_n}{\sqrt{n\lambda}} + \left(\frac{\tau_n^3}{n\lambda}\right)^{1/4}. \tag{F.3}$$

Since $\tau_n = C_0$, its three polynomial-order contributions are

$$\sqrt{n}\lambda, \qquad \frac{1}{\sqrt{n\lambda}}, \qquad \left(\frac{1}{n\lambda}\right)^{1/4}.$$

The active balance is between the first and third terms, giving $\lambda \asymp n^{-3/5}$ and the power $n^{-1/10}$. At $\eta \asymp n^{-1}$, the logarithmic multiplier of the active bootstrap coupling term in Proposition F.4 is

$$(\log n)^{3/2}(\log(1/\lambda))^{1/(4\zeta)}(\log(n/\lambda))^{1/(4\zeta)} \lesssim (\log n)^{\frac{3}{2}+\frac{1}{2\zeta}}.$$

The bias term has the same power without this logarithmic multiplier, and all remaining stochastic terms have a strictly smaller power of n. A final application of Theorem 4.1 proves part (2). □

F.10 Proof of Corollary 4.3

Lemma F.26 (**Cordes inequality; see** Cordes [1987]). *Let A and B be bounded, self-adjoint, positive semidefinite operators on a Hilbert space G. Then, for every $s \in [0, 1]$,*

$$\|A^s B^s\|_{\mathsf{G}} \le \|AB\|_{\mathsf{G}}^s.$$

The positive-semidefinite case follows by applying the strictly positive version to $A + \varepsilon I$ and $B + \varepsilon I$ and letting $\varepsilon \downarrow 0$.

Lemma F.27 (Gogolashvili et al. [2023], Lemma 21). *Under Definition 4.1, for every integer $m \ge 2$ and $\tau_n > 0$,*

$$\mathbb{E}_{\mathbb{Q}}\left(1 - \frac{w(X)}{\rho(X)}\right)^2 \le \left\{\frac{1}{2}\tau_n^{-(m-1)} m! W^{m-2}\sigma^2\right\}^{1/\gamma}.$$

Lemma F.28 (Gogolashvili et al. [2023], Lemma 22). *Under Definition 4.1, for every integer $m \ge 2$, $\tau_n > 0$, and $\lambda > 0$,*

$$\left\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\right\| \le \kappa\lambda^{-1/2}\mathcal{N}_\lambda^{1/2}\left\{\frac{1}{2}\tau_n^{-(m-1)} m! W^{m-2}\sigma^2\right\}^{1/(2\gamma)}.$$

Lemma F.29 (Gogolashvili et al. [2023], Lemma 23)**.** *Under Assumption 4.3 and Definition 4.1, if, for some integer $m_0 \geq 2$ and $0 < \lambda \leq 1$,*

$$\tau_n \geq \left(2^{2\gamma-1}\kappa^{2\gamma}E_d^{2\gamma}m_0!W^{m_0-2}\sigma^2\right)^{1/(m_0-1)}\lambda^{-\frac{(1+d)\gamma}{m_0-1}},$$

then, as operators on H*,*

1. $\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\| \leq 1/2$;
2. $\|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\| \leq 2$;
3. $\|\mathcal{L}_Q^{1/2}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2}\| \leq \sqrt{2}$.

Proof of Corollary 4.3. Let $\tau_n = n^\delta$ for a fixed $\delta > 0$, and consider a tuning parameter $\lambda = n^{-a}$. The polynomial spectral condition verifies Assumption 4.3 with $d = 1/\beta$, while the exponential condition does so with any fixed $d \in (0, 1]$. Choose a fixed integer $m_0 \geq 2$ such that

$$m_0 - 1 > \frac{a(1+d)\gamma}{\delta}.$$

It follows that

$$\lambda^{-\frac{(1+d)\gamma}{m_0-1}} = n^{\frac{a(1+d)\gamma}{m_0-1}} = o(n^\delta).$$

Consequently, for all sufficiently large n, Lemma F.29 verifies Assumption 4.4 with $\mathfrak{C}_1 \leq \sqrt{2}$. We may also write $\tau_n = \lambda^{-\varrho}$, where $\varrho = \delta/a$. Because the Bernstein-type condition holds for every integer $m_0 \geq 2$, this argument is available for every fixed $\delta > 0$. Taking δ sufficiently small also ensures all sample-size restrictions in the relevant spectral approximation propositions, including $a + \delta < 1$.

For polynomial spectral decay, take

$$a = \widetilde{a}_\beta = \frac{2(\beta+1)}{3(\beta+3)}, \qquad \widetilde{r}_\beta = \frac{(\beta-5)(\beta+10)}{6(\beta+3)(2\beta+5)}.$$

Restoring $\tau_n = n^\delta$ in the four terms of (F.2), their powers of n are

$$\delta_0, \qquad \delta_1 + \delta\frac{\beta+1}{\beta+3}, \qquad \delta_2 + \delta, \qquad \delta_3 + \delta\frac{3\beta-2}{2(2\beta-1)},$$

where the preceding proof shows that $\delta_j \leq -\widetilde{r}_\beta$ for $j = 0, 1, 2, 3$. Both fractional truncation coefficients in the display are smaller than one. Hence

$$\Delta \lesssim n^{-\widetilde{r}_\beta+\delta}\log^3 n.$$

The factor n^δ is a uniform upper bound for the truncation loss of every term; the active terms have a strictly smaller truncation power. Thus any remaining fixed logarithmic factors are absorbed by this polynomial slack. Theorem 4.1 now yields

$$\left|\mathbb{P}\{f_* \notin \mathsf{CS}_\alpha(\widehat{f})\} - \alpha\right| \lesssim n^{-1} + n^{-\widetilde{r}_\beta+\frac{3}{2}\delta}\log^3 n.$$

For part (1), choose

$$\delta = \frac{\beta-5}{9(\beta+3)(2\beta+5)}, \qquad \tau_n = n^{\frac{\beta-5}{9(\beta+3)(2\beta+5)}}.$$

Then

$$\widetilde{r}_\beta - \frac{3}{2}\delta = \frac{(\beta-5)(\beta+9)}{6(\beta+3)(2\beta+5)},$$

and hence the preceding display gives the bound in part (1).

For exponential spectral decay, take $a = 3/5$. Substituting $\lambda = n^{-3/5}$ and $\tau_n = n^\delta$ into (F.3), and then multiplying the approximation error by the anti-concentration factor $\sqrt{\tau_n}$, gives the conservative bound

$$\left|\mathbb{P}\{f_* \notin \mathsf{CS}_\alpha(\widehat{f})\} - \alpha\right| \lesssim n^{-1} + n^{-\frac{1}{10}+\frac{3}{2}\delta}\log^3 n.$$

Indeed, after multiplication by $\sqrt{\tau_n}$, the active coupling term has truncation power $5\delta/4$, leaving a polynomial margin of $\delta/4$ that absorbs the fixed ζ-dependent logarithmic factors. Substituting

$$\delta = \frac{1}{150}, \qquad \tau_n = n^{1/150},$$

and using

$$\frac{1}{10} - \frac{3}{2}\delta = \frac{9}{100}$$

proves part (2). The two selected values of δ satisfy the sufficient positivity conditions

$$0 < \delta < \frac{(\beta-5)(\beta+10)}{9(\beta+3)(2\beta+5)} \qquad \text{and} \qquad 0 < \delta < \frac{1}{15},$$

respectively. These upper bounds are also smaller than $1-a$ for the corresponding choices of a, and hence imply $a + \delta < 1$. □

F.11 Proof of Corollary D.1

Definition F.1 (Bounded ϑ-th moment condition)**.** *There exist $\vartheta > 2$ and $\Omega_\vartheta < \infty$ such that*

$$\mathbb{E}_{\mathbb{P}}\left[\rho(X)^\vartheta\right] = \mathbb{E}_{\mathbb{Q}}\left[\rho(X)^{\vartheta-1}\right] \le \Omega_\vartheta.$$

Lemma F.30. *Under Definition F.1, for $\tau_n > 0$ and $\lambda > 0$,*

$$\left\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\right\| \le \kappa\lambda^{-1/2}\mathcal{N}_\lambda^{1/2}\tau_n^{-(\vartheta-1)/2}\Omega_\vartheta^{1/2}.$$

Lemma F.31. *Under Assumption 4.3 and Definition F.1, for $0 < \lambda \le 1$, if*

$$\tau_n \ge \left(4\kappa^2\Omega_\vartheta E_d^2\lambda^{-(1+d)}\right)^{1/(\vartheta-1)}, \tag{F.4}$$

then, as operators on H,

1. $\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\| \le 1/2$;
2. $\|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\| \le 2$;
3. $\|\mathcal{L}_Q^{1/2}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2}\| \le \sqrt{2}$.

The following compatibility reduction is common to Corollaries D.1 and 4.4. Set $q_\theta = \theta - 1$. Then $q_\theta > 1$. Since $d\mathbb{Q} = \rho\, d\mathbb{P}$, the moment condition in either corollary is Definition F.1 with $\vartheta = \theta$. Hence, by Lemma F.30,

$$\left\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\right\| \lesssim \lambda^{-1/2}\mathcal{N}_\lambda^{1/2}\tau_n^{-q_\theta/2}.$$

Thus the perturbation proof of Lemma F.31 verifies Assumption 4.4 whenever

$$\tau_n \gtrsim (\lambda^{-1}\mathcal{N}_\lambda)^{1/q_\theta}. \tag{F.5}$$

Once this compatibility condition holds, the proofs of the Gaussian and bootstrap approximation propositions use only the bounded truncated-weight envelope and the standing assumptions. In particular, the Bernstein-type condition appearing in the statements of the specialized bootstrap propositions is not used in their proofs. Repeating those proofs from Propositions 4.4 and 4.5, together with the corresponding effective-dimension and spectral-tail lemmas, yields the same spectral envelopes under (F.5).

Proof of Corollary D.1. Suppose that $\mu_k \asymp k^{-\beta}$. Lemma F.20 gives $\mathcal{N}_\lambda \lesssim \lambda^{-1/\beta}$, so we may take $d = 1/\beta$ in Assumption 4.3. Set

$$v_{\beta,\theta} = \frac{\beta+1}{\beta q_\theta}.$$

For $\lambda = n^{-a}$, because $n^\delta \to \infty$, the truncation level

$$\tau_n \asymp n^{a v_{\beta,\theta}+\delta}$$

satisfies (F.5) for all sufficiently large n. Setting $\delta = 0$, after multiplying the four terms in (F.2) by the anti-concentration factor $\sqrt{\tau_n}$, their powers of n are

$$\begin{aligned}
e_0^{(0)}(a) &= \frac{1}{2} - a + \frac{a v_{\beta,\theta}}{2},\\
e_1^{(0)}(a) &= -\frac{\beta-1}{2(\beta+3)} + \frac{a}{2} + a v_{\beta,\theta}\left(\frac{\beta+1}{\beta+3} + \frac{1}{2}\right),\\
e_2^{(0)}(a) &= -\frac{1}{2} + a\left(\frac{1}{2} + \frac{1}{\beta}\right) + \frac{3}{2} a v_{\beta,\theta},\\
e_3^{(0)}(a) &= -\frac{\beta-1}{2(2\beta-1)} + a\frac{\beta^2+\beta-1}{2\beta(2\beta-1)} + a v_{\beta,\theta}\left(\frac{3\beta-2}{2(2\beta-1)} + \frac{1}{2}\right).
\end{aligned}$$

For reference, the exact optimizer of these four exponents remains piecewise. Let

$$\beta_0 = 6 + \sqrt{34}, \qquad D_1 = 3\beta q_\theta(\beta+3) + 2(\beta+1)^2,$$

$$D_2 = q_\theta(5\beta^2 - \beta - 1) + (\beta+1)(3\beta-2).$$

Then

$$a^\sharp_{\beta,\theta} = \begin{cases} \dfrac{2\beta q_\theta(\beta+1)}{D_1}, & 5 < \beta \le \beta_0,\\[2ex] \dfrac{\beta q_\theta(3\beta-2)}{D_2}, & \beta > \beta_0, \end{cases}$$

$$t^{\sharp}_{\beta,\theta} = a^{\sharp}_{\beta,\theta} v_{\beta,\theta} = \begin{cases} \dfrac{2(\beta+1)^2}{D_1}, & 5 < \beta \le \beta_0, \\ \dfrac{(\beta+1)(3\beta-2)}{D_2}, & \beta > \beta_0, \end{cases}$$

$$r^{\sharp}_{\beta,\theta} = \begin{cases} \dfrac{\beta q_\theta(\beta-5) - 4(\beta+1)^2}{2D_1}, & 5 < \beta \le \beta_0, \\ \dfrac{q_\theta(\beta^2-3\beta+1) - 2(\beta+1)(3\beta-2)}{2D_2}, & \beta > \beta_0. \end{cases}$$

Indeed, the first branch balances $e_0^{(0)} = e_1^{(0)}$, the second balances $e_0^{(0)} = e_3^{(0)}$, and the crossover equation is $\beta^2 - 12\beta + 2 = 0$. These sharper piecewise quantities are retained only for comparison and are not needed in Corollary D.1.

For the non-piecewise statement, put

$$D = 3\beta q_\theta(\beta+3) + 2(\beta+1)^2, \qquad a_{\mathrm{M}} = \frac{2\beta q_\theta(\beta+1)}{D}.$$

Thus

$$\lambda = n^{-a_{\mathrm{M}}}, \qquad \tau_n \asymp n^{u_{\mathrm{P}}}, \qquad u_{\mathrm{p}} = a_{\mathrm{M}} v_{\beta,\theta} + \delta = \frac{2(\beta+1)^2}{D} + \delta,$$

which is the truncation power used in Corollary D.1. At $\delta = 0$, define

$$r_{01} = \frac{\beta q_\theta(\beta-5) - 4(\beta+1)^2}{2D},$$

$$r_{03} = \frac{q_\theta(\beta^3+2\beta^2-9\beta+2) - 4(\beta+1)^2(2\beta-1)}{2(2\beta-1)D}$$

and

$$r_{\mathrm{M}} = \frac{\beta q_\theta(\beta-5)(\beta+10) - 4(\beta+1)^2(2\beta+5)}{2(2\beta+5)D}.$$

Direct substitution gives

$$e_0^{(0)}(a_{\mathrm{M}}) = e_1^{(0)}(a_{\mathrm{M}}) = -r_{01}, \qquad e_2^{(0)}(a_{\mathrm{M}}) - e_0^{(0)}(a_{\mathrm{M}}) = -\frac{2q_\theta(2\beta-1)}{D} < 0,$$

and $e_3^{(0)}(a_{\mathrm{M}}) = -r_{03}$. Moreover,

$$r_{01} - r_{\mathrm{M}} = \frac{q_\theta \beta(\beta-5)^2}{2(2\beta+5)D} > 0,$$

$$r_{03} - r_{\mathrm{M}} = \frac{q_\theta(97\beta^2-91\beta+10)}{2(2\beta-1)(2\beta+5)D} > 0.$$

Hence every exponent is at most $-r_{\mathrm{M}}$. Restoring the factor n^δ in τ_n adds, respectively,

$$\frac{\delta}{2}, \qquad \frac{3\beta+5}{2(\beta+3)}\delta, \qquad \frac{3\delta}{2}, \qquad \frac{5\beta-3}{2(2\beta-1)}\delta$$

to the four coverage exponents. All four coefficients are at most $3/2$, so Theorem 4.1 gives

$$\left|\mathbb{P}\{f_* \notin \mathsf{CS}_\alpha(\widehat{f})\} - \alpha\right| \lesssim n^{-1} + n^{-\{r_{\mathrm{M}} - 3\delta/2\}} \log^3 n.$$

The displayed expression for r_{M} is the polynomial rate before fixing the truncation slack. Its positivity is equivalent to

$$q_\theta > \frac{4(\beta+1)^2(2\beta+5)}{\beta(\beta-5)(\beta+10)},$$

which is assumed in Corollary D.1. Now choose

$$\delta = \frac{\beta q_\theta(\beta-5)(\beta+10) - 4(\beta+1)^2(2\beta+5)}{3(2\beta+5)(\beta+10)D} = \frac{2r_{\mathrm{M}}}{3(\beta+10)}.$$

Since $\beta + 10 > 1$, this choice lies strictly between zero and the admissible upper bound $2r_{\mathrm{M}}/3$. Substitution gives

$$u_{\mathrm{p}} = \frac{\beta q_\theta(\beta-5)(\beta+10) + 2(3\beta+28)(\beta+1)^2(2\beta+5)}{3(2\beta+5)(\beta+10)D}$$

and, writing the resulting coverage exponent as r_{p},

$$r_{\mathrm{p}} := r_{\mathrm{M}} - \frac{3\delta}{2} = \frac{\beta+9}{\beta+10} r_{\mathrm{M}} = \frac{(\beta+9)\{\beta q_\theta(\beta-5)(\beta+10) - 4(\beta+1)^2(2\beta+5)\}}{2(\beta+10)(2\beta+5)D},$$

which are the truncation power and coverage exponent stated in Corollary D.1. Moreover, the assumed moment threshold and the definition of D imply

$$0 < r_{\mathrm{p}} < \frac{(\beta+9)(\beta-5)}{6(2\beta+5)(\beta+3)} < 1.$$

Consequently, the n^{-1} term in the preceding coverage bound is absorbed by $n^{-r_{\mathrm{p}}}\log^3 n$.

It remains to verify the listed transparent sufficient regimes. The numerator of r_{M} can be written as

$$(\theta-9)\beta(\beta-5)(\beta+10) + 4(\beta^2 - 112\beta - 5).$$

It is therefore positive when $\theta \geq 9$ and $\beta \geq 113$. For the flexible regime, put

$$R(\beta) = \frac{4(\beta+1)^2(2\beta+5)}{\beta(\beta-5)(\beta+10)}.$$

For every $\beta > 5$,

$$(\beta-5)\{R(\beta) - 8\} = \frac{4(-\beta^2 + 112\beta + 5)}{\beta(\beta+10)} \leq \frac{144}{5};$$

the rational function in the middle is decreasing on $(5, \infty)$. Consequently, $\theta > 9$ and $\beta > 5 + 144/\{5(\theta-9)\}$ imply $R(\beta) < \theta - 1$. Finally, the same numerator is increasing in both parameters on each of $\{\theta \geq 10, \beta \geq 20\}$ and $\{\theta \geq 11, \beta \geq 15\}$, and its values at the respective lower corners are 1620 and 1660. This proves the remaining two sufficient regimes.

Finally, direct substitution of the selected polynomial slack gives

$$1 - a_{\mathrm{M}} - u_{\mathrm{p}} = \frac{2\beta q_\theta(\beta+10)(3\beta^2+28\beta+55) + 4(\beta+1)^2(2\beta+5)}{3(2\beta+5)(\beta+10)D} > 0.$$

Under the assumed two-sided polynomial spectral decay, Lemma F.2 gives $\mathcal{N}_\lambda \asymp \lambda^{-1/\beta}$. Together with the strict margin above, this implies $\tau_n/(n\lambda\mathcal{N}_\lambda) \to 0$; the remaining fixed logarithmic factors are absorbed by the same margin. Thus all sample-size conditions in the relevant approximation propositions hold for sufficiently large n, completing the proof. □

F.12 Proof of Corollary 4.4

Proof. Suppose that $\mu_k \asymp \exp(-ck^{\zeta})$, and write

$$q_\theta = \theta - 1, \qquad a = \frac{3q_\theta}{5q_\theta + 3}, \qquad \lambda = n^{-a}.$$

By Lemma F.23,

$$\mathcal{N}_\lambda \lesssim \{\log(1/\lambda)\}^{1/\zeta}.$$

Choose a sufficiently large fixed constant $C > 0$ and set

$$\tau_n = C\left[\lambda^{-1}\{\log(1/\lambda)\}^{1/\zeta}\right]^{1/q_\theta}.$$

Then (F.5) holds and

$$\tau_n \asymp n^{\frac{3}{5\theta-2}}(\log n)^{\frac{1}{\zeta(\theta-1)}},$$

as stated in the corollary.

After multiplying the three terms in (F.3) by the anti-concentration factor $\sqrt{\tau_n}$, their polynomial powers of n are

$$e_0 = \frac{1}{2} - a + \frac{a}{2q_\theta}, \qquad e_1 = -\frac{1}{2} + \frac{a}{2} + \frac{3a}{2q_\theta}, \qquad e_2 = \frac{a-1}{4} + \frac{5a}{4q_\theta}.$$

Define

$$r_\theta = \frac{q_\theta - 6}{2(5q_\theta + 3)} = \frac{\theta - 7}{2(5\theta - 2)}.$$

Direct substitution gives

$$e_0 = e_2 = -r_\theta, \qquad e_2 - e_1 = \frac{q_\theta}{2(5q_\theta + 3)} > 0.$$

Thus the bias and bootstrap-coupling terms have the same polynomial order, while the middle term is polynomially smaller. The bias term carries the factor $(\log n)^{1/\{2\zeta q_\theta\}}$. The bootstrap-coupling term carries

$$(\log n)^{\frac{3}{2}+\frac{1}{2\zeta}+\frac{5}{4\zeta q_\theta}},$$

where the final exponent is contributed by the factor $\tau_n^{5/4}$. Hence the bootstrap-coupling term dominates logarithmically, and the fixed logarithmic factors of the polynomially smaller terms are absorbed. Theorem 4.1 consequently yields

$$\left|\mathbb{P}\{f_* \notin \mathsf{CS}_\alpha(\widehat{f})\} - \alpha\right| \lesssim n^{-1}n^{-r_\theta}(\log n)^{\frac{3}{2}+\frac{1}{2\zeta}+\frac{5}{4\zeta q_\theta}} \lesssim n^{-r_\theta}(\log n)^{\frac{3}{2}+\frac{1}{2\zeta}+\frac{5}{4\zeta q_\theta}},$$

where $0 < r_\theta < 1$ because $\theta > 7$.

Finally,

$$1 - a - \frac{a}{q_\theta} = \frac{2q_\theta}{5q_\theta + 3} > 0.$$

Hence both sample-size restrictions in the relevant approximation propositions hold for all sufficiently large n; the strict polynomial margin absorbs the additional logarithmic factors in τ_n and in the failure-probability tolerances. This completes the proof. □

F.13 Proof of Theorem 4.2

Proof of Theorem 4.2. The type II error probability is

$$P\left(\sqrt{n}\|\hat{f}-f_0\|_{\mathsf{L}^2(\mathbb{Q})} < \hat{t}_\alpha\right) = P\left(\|\hat{f}-f_0\|_{\mathsf{L}^2(\mathbb{Q})} < \frac{\hat{t}_\alpha}{\sqrt{n}}\right).$$

By the triangle inequality,

$$\|f_*-f_0\|_{\mathsf{L}^2(\mathbb{Q})} \le \|\hat{f}-f_0\|_{\mathsf{L}^2(\mathbb{Q})} + \|\hat{f}-f_*\|_{\mathsf{L}^2(\mathbb{Q})}.$$

Hence

$$\left\{\|\hat{f}-f_0\|_{\mathsf{L}^2(\mathbb{Q})} < \frac{\hat{t}_\alpha}{\sqrt{n}}\right\} \subseteq \left\{\delta < \|\hat{f}-f_*\|_{\mathsf{L}^2(\mathbb{Q})} + \frac{\hat{t}_\alpha}{\sqrt{n}}\right\}.$$

Since $\delta > 0$ is fixed and

$$\|\hat{f}-f_*\|_{\mathsf{L}^2(\mathbb{Q})} + \frac{\hat{t}_\alpha}{\sqrt{n}} \xrightarrow{P} 0,$$

the probability of the event on the right-hand side tends to zero. Therefore

$$P\left(\sqrt{n}\|\hat{f}-f_0\|_{\mathsf{L}^2(\mathbb{Q})} < \hat{t}_\alpha\right) \to 0,$$

which is equivalent to

$$P\left(\sqrt{n}\|\hat{f}-f_0\|_{\mathsf{L}^2(\mathbb{Q})} > \hat{t}_\alpha\right) \to 1.$$

□

F.14 Proof of Proposition A.1

Proof. Fix $\eta \in (0,1)$ and an integer $m \in \{1,\dots,n\}$. Again let $A_m : \mathsf{L}^2(\mathbb{Q}) \to \mathbb{R}^m$ be the coordinate projection onto the first m eigenfunctions of $\mathcal{V}$. Since U_i is $\mathsf{L}^2(\mathbb{Q})$-b-sub-Gaussian, $A_m U_i$ is b-sub-Gaussian in $\mathbb{R}^m$, and its covariance is bounded in operator norm by $\|\mathcal{V}\|_{\mathsf{L}^2(\mathbb{Q})}$. If this covariance is singular, apply the finite-dimensional sub-Gaussian coupling theorem of Buzun et al. [2022] on the range of $\Pi_m \mathcal{V} \Pi_m$, whose dimension is at most m and whose covariance is nonsingular. Every zero-variance coordinate of the centered vector $A_m U_i$ vanishes almost surely, so the resulting Gaussian vector can be extended by zero on the orthogonal complement. Thus, in either case, the theorem gives a Gaussian vector $\zeta \in \mathbb{R}^m$ with covariance $\mathrm{diag}(\nu_1,\dots,\nu_m)$ such that, with probability at least $1-\eta/3$,

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^n \Pi_m U_i - A_m^* \zeta \right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \|\mathcal{V}\|_{\mathsf{L}^2(\mathbb{Q})}^{1/2} b^3 \frac{m^{3/2}\log(n)^2}{\sqrt{n}} (3/\eta)^{1/\log(mn)}.$$

Let

$$Z = A_m^*\zeta + \sum_{s=m+1}^{\infty} \sqrt{\nu_s}\, h_s e_s,$$

where $(h_s)_{s>m}$ are independent standard Gaussian variables, independent of the data and of ζ. Then $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})$.

For the empirical tail, the $\mathsf{L}^2(\mathbb{Q})$-sub-Gaussianity of U_i gives the moment bound

$$\left\{\mathbb{E}\|(I-\Pi_m)U_i\|_{\mathsf{L}^2(\mathbb{Q})}^p\right\}^{1/p} \lesssim b\sqrt{p}\,\sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V}), \qquad p \ge 2.$$

The Hilbert-space Bernstein inequality [Pinelis and Sakhanenko, 1986, Corollary 1] therefore yields, with probability at least $1-\eta/3$,

$$\left\| \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (I - \Pi_m) U_i \right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim b\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \sqrt{\log\left(\frac{6}{\eta}\right)} + \frac{b\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \log\left(\frac{6}{\eta}\right)}{\sqrt{n}}.$$

The last term is dominated by the first when $n \geq \log\left(\frac{2}{\eta}\right)$. Finally, Borell's inequality applied to $(I - \Pi_m)Z$, whose second moment is $\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V})$, gives

$$\|(I - \Pi_m)Z\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \sqrt{\log\left(\frac{6}{\eta}\right)}$$

with probability at least $1-\eta/3$. A union bound and the triangle inequality complete the proof. □

F.15 Proof of Proposition A.2

Proof. The anti-symmetric multiplier identity gives the oracle empirical covariance $\widehat{\mathcal{V}}_{\mathsf{ora}}$, and the covariance-comparison bound (F.1) applies with this empirical covariance. Since adding a common deterministic element does not change an empirical covariance,

$$\widehat{\mathcal{V}}_{\mathsf{ora}} - \mathcal{V} = \frac{1}{n} \sum_{i=1}^{n} (U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i - \mathcal{V}) - \left(\frac{1}{n} \sum_{i=1}^{n} U_i \right) \otimes_{\mathsf{L}^2(\mathbb{Q})} \left(\frac{1}{n} \sum_{i=1}^{n} U_i \right).$$

We first make the covariance-estimation step explicit. For every orthogonal projection P on $\mathsf{L}^2(\mathbb{Q})$ and every $p \geq 2$, the standard moment bound for a Hilbert-space sub-Gaussian random element gives

$$\left\{ \mathbb{E} \|PU_i\|^p_{\mathsf{L}^2(\mathbb{Q})} \right\}^{1/p} \lesssim b\sqrt{p} \left\{ \mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(P\mathcal{V}P) \right\}^{1/2}.$$

This follows by applying the sub-Gaussian increment bound to finite-dimensional projections and then passing to the limit by monotone convergence. Indeed, on each finite-dimensional range, Gaussian integration and Definition A.1 give

$$\mathbb{E} \exp\{\theta \|PU_i\|^2_{\mathsf{L}^2(\mathbb{Q})}\} \leq \mathbb{E}_g \exp\{\theta b^2 \langle g, P\mathcal{V}Pg \rangle_{\mathsf{L}^2(\mathbb{Q})}\},$$

for every $0 < \theta < \{2b^2 \|P\mathcal{V}P\|_{\mathsf{L}^2(\mathbb{Q})}\}^{-1}$, where g is standard Gaussian on that range. The Gaussian quadratic-form bound yields the displayed moment estimate, and increasing finite-dimensional projections give the general case. Because $\|u \otimes_{\mathsf{L}^2(\mathbb{Q})} u\|_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})} = \|u\|^2_{\mathsf{L}^2(\mathbb{Q})}$ and $\|\mathcal{V}\|_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})} \leq \mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V})$, it follows that

$$\left\{ \mathbb{E} \|U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i - \mathcal{V}\|^p_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})} \right\}^{1/p} \lesssim b^2 p\, \sigma^2_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V}).$$

The preceding moment estimate satisfies the Bernstein moment condition with variance proxy of order $b^4\sigma^4_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ and scale of order $b^2\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$. Hence the Hilbert-space Bernstein inequality [Pinelis and Sakhanenko, 1986, Corollary 1], applied in the Hilbert space of Hilbert–Schmidt operators, yields

$$\left\| \frac{1}{n} \sum_{i=1}^{n} (U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i - \mathcal{V}) \right\|_{\mathrm{HS},\mathsf{L}^2(\mathbb{Q})} \lesssim b^2 \sigma^2_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V}) \left\{ \sqrt{\frac{\log(2/\eta)}{n}} + \frac{\log(2/\eta)}{n} \right\}.$$

Independence and Definition A.1 also show directly that $n^{-1}\sum_{i=1}^n U_i$ is $\mathsf{L}^2(\mathbb{Q})$-b-sub-Gaussian with covariance $\mathcal{V}/n$. Applying the preceding norm-moment bound with $p = \max\{2, \log(2/\eta)\}$ and then Markov's inequality gives

$$\left\| \frac{1}{n}\sum_{i=1}^n U_i \right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim b\sigma_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})\sqrt{\frac{\log(2/\eta)}{n}}.$$

Combining the last three displays and enlarging the logarithmic factor gives

$$\Delta_{\mathrm{HS}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^2 \frac{b^2\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})}{\sqrt{n}}.$$

For the tail, take $P = \Pi_m^{\perp}$. The centering identity above gives

$$\begin{aligned}\mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}\{P(\widehat{\mathcal{V}}_{\mathsf{ora}} - \mathcal{V})P\} = \frac{1}{n}\sum_{i=1}^n \left\{\|PU_i\|^2_{\mathsf{L}^2(\mathbb{Q})} - \mathbb{E}\|PU_i\|^2_{\mathsf{L}^2(\mathbb{Q})}\right\} \\ - \left\|P\left(\frac{1}{n}\sum_{i=1}^n U_i\right)\right\|^2_{\mathsf{L}^2(\mathbb{Q})}.\end{aligned}$$

Moreover,

$$\left\{\mathbb{E}\left|\|PU_i\|^2_{\mathsf{L}^2(\mathbb{Q})} - \mathbb{E}\|PU_i\|^2_{\mathsf{L}^2(\mathbb{Q})}\right|^p\right\}^{1/p} \lesssim b^2 p\, \sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V}).$$

Scalar Bernstein controls the first term in the tail identity. The same sub-Gaussian closure argument applied to $P(n^{-1}\sum_{i=1}^n U_i)$, whose covariance trace is $\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})/n$, controls the second term by $b^2\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})\log(2/\eta)/n$. After replacing η by a fixed fraction and taking a union bound, these bounds hold simultaneously with probability at least $1-3\eta$, and

$$\Delta_{\mathrm{tail}} \lesssim \log\left(\frac{2}{\eta}\right)\frac{b^2\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})}{\sqrt{n}}.$$

Substitution into (F.1) yields

$$\|\mathcal{V}^{1/2}g - G\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^{3/2}\left[m^{1/4}\left\{\frac{b^4\sigma^4_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})}{n}\right\}^{1/4} + \sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})\right].$$

Here the empirical-tail contribution $b\,n^{-1/4}\sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V})$ is absorbed by the first term because $m \geq 1$ and $\sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V}) \leq \sigma_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})$. Taking the infimum over $m \geq 1$ proves the claim. □

F.16 Proof of Corollary A.1

Polynomial spectral decay.

Proposition F.5. *Suppose $\mu_k \leq \alpha k^{-\beta}$ for some $\alpha > 0$ and $\beta > 1$. Under Assumptions 4.1, 4.2, 4.4, and A.1, when $\tau_n \geq \tau_0$, if $\tau_n \to \infty$ and $\lambda \to 0$ and*

$$n > \max\left\{\frac{(\tau_n\kappa^2+1)\log(e+\mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda}\right\}$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then with probability at least $1-\eta$, there exists $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})$ such that

$$\left\|\sqrt{n}(\hat{f}-f_{P^w})-Z\right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \frac{\tau_n}{\sqrt{n}}\lambda^{-\frac{1}{2}-\frac{1}{2\beta}}\log\left(\frac{1}{\lambda}\right)\left(\log\left(\frac{2}{\eta}\right)\right)^2 + \sqrt{\frac{\tau_n}{n}}\left(\frac{n}{\lambda}\right)^{\frac{3}{2(2+\beta)}}\log(n)^2\left(\frac{3}{\eta}\right)^{\frac{1}{\log(n)}}.$$

Proof of Proposition F.5. Lemmas F.7, F.25, F.20, and F.21 give

$$\mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V}) \lesssim \tau_n\lambda^{-1/\beta}, \qquad \|\mathcal{V}\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \tau_n, \qquad \sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V}) \lesssim \sqrt{\frac{\tau_n}{\lambda}}m^{-(\beta-1)/2}.$$

By Lemma A.1, the influence elements are uniformly $\mathsf{L}^2(\mathbb{Q})$-b-sub-Gaussian. Apply Proposition A.1 with

$$m = \min\left\{n, \left\lceil\left(\frac{n}{\lambda}\right)^{1/(\beta+2)}\right\rceil\right\}.$$

Both coupling terms are then bounded, up to their displayed logarithmic and tolerance factors, by

$$\sqrt{\frac{\tau_n}{n}}\left(\frac{n}{\lambda}\right)^{\frac{3}{2(\beta+2)}}.$$

The sample-size condition and $\tau_n \to \infty$ imply $n\lambda \to \infty$; hence the untruncated ceiling is at most n for all sufficiently large n. Moreover, $(3/\eta)^{1/\log(mn)} \le (3/\eta)^{1/\log n}$.

Proposition 4.2 and $\mathcal{N}_\lambda \lesssim \lambda^{-1/\beta}$ give

$$\Delta_{G,\mathrm{Bah}} \lesssim \frac{\tau_n}{\sqrt{n}}\lambda^{-\frac{1}{2}-\frac{1}{2\beta}}\log\left(\frac{1}{\lambda}\right)\left(\log\left(\frac{2}{\eta}\right)\right)^2.$$

Combining the Bahadur and coupling events after dividing η by a fixed constant proves the result. □

Proposition F.6. *Suppose $\mu_k \le \alpha k^{-\beta}$ for some $\alpha > 0$ and $\beta > 1$. Under Assumptions 4.1, 4.2, 4.4, and A.1, when $\tau_n \ge \tau_0$, if $\tau_n \to \infty$ and $\lambda \to 0$ and*

$$n > \max\left\{\frac{(\tau_n\kappa^2+1)\log(e+\mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda}\right\}$$

where C_2 is the constant in Lemma F.13 evaluated at the component tolerance $\eta_{\mathrm{tol}} = c\eta^2$ used in Corollary 4.1, then, on a data event of probability at least $1-\eta$, there exists a coupling of B and Z, conditionally on $\mathcal{D}$, such that $Z \mid \mathcal{D} \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V})$ and

$$\mathbb{P}\Bigg\{\|Z-B\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^3\frac{\tau_n}{\sqrt{n}}\lambda^{-\frac{1}{2}-\frac{1}{\beta}} + \left(\log\left(\frac{2}{\eta}\right)\right)^{\frac{3}{2}}\left(\frac{\tau_n}{\lambda}\right)^{\frac{1}{2}}\left(\frac{\tau_n\lambda^{1-\frac{2}{\beta}}}{\sqrt{n}}\right)^{\frac{\beta-1}{2\beta-1}} \;\Bigg|\; \mathcal{D}\Bigg\} \ge 1-\eta.$$

Proof of Proposition F.6. The spectral bounds used in the preceding proof imply

$$\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(0,\mathcal{V}) \lesssim \tau_n\lambda^{-1/\beta}, \qquad \sigma_{\mathsf{L}^2(\mathbb{Q})}(m,\mathcal{V}) \lesssim \sqrt{\frac{\tau_n}{\lambda}}m^{-(\beta-1)/2}.$$

In Proposition A.2, take

$$m = \left\lceil \max\left\{1, \left(\frac{\sqrt{n}}{\tau_n \lambda^{1-2/\beta}}\right)^{\frac{2}{2\beta-1}}\right\}\right\rceil.$$

The spectral-tail term is bounded by

$$\sqrt{\frac{\tau_n}{\lambda}}\left(\frac{\tau_n \lambda^{1-2/\beta}}{\sqrt{n}}\right)^{\frac{\beta-1}{2\beta-1}}.$$

The finite-dimensional term obeys the same bound because the ratio of that term to the preceding display is at most $\tau_n^{-1/2}\lambda^{1/(2\beta)} \le 1$. If the expression inside the maximum is smaller than one, the choice $m = 1$ gives the same conclusion.

On the other hand, Proposition 4.4 and $\mathcal{N}_\lambda \lesssim \lambda^{-1/\beta}$ give

$$\Delta_{B,\mathrm{feas}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^3 \frac{\tau_n}{\sqrt{n}} \lambda^{-\frac{1}{2}-\frac{1}{\beta}}.$$

The conditional Strassen–Dudley argument in Lemma F.19, followed by the same two-layer union bound as in Corollary 4.1, couples the oracle bootstrap to Z and then combines it with the feasible-bootstrap bound. The triangle inequality proves the assertion. □

Exponential spectral decay.

Proposition F.7. *Suppose $\mu_k \le \alpha \exp(-\beta k^\gamma)$ for some $\alpha, \beta, \gamma > 0$. Under Assumptions 4.1, 4.2, 4.4, and A.1, when $\tau_n \ge \tau_0$, if $\tau_n \to \infty$ and $\lambda \to 0$ and*

$$n > \max\left\{\frac{(\tau_n \kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{C_2 \lambda}, \frac{\tau_n}{\lambda \mathcal{N}_\lambda}\right\}$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then with probability at least $1 - \eta$, there exists $Z \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ such that

$$\left\|\sqrt{n}(\hat{f} - f_{P^w}) - Z\right\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^2 \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{\frac{1}{2\gamma}} \log\log\left(\frac{1}{\lambda}\right) + \left(\frac{1}{\eta}\right)^{\frac{1}{\log n}} \sqrt{\frac{\tau_n}{n}} \left[\frac{1}{\beta}\log\left(\frac{n}{\lambda}\right)\right]^{\frac{3}{2\gamma}} (\log(n))^2.$$

Proof of Proposition F.7. Lemmas F.7, F.25, F.23, and F.24 imply

$$\mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V}) \lesssim \tau_n \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/\gamma}, \qquad \|\mathcal{V}\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \tau_n,$$

and

$$\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \lesssim \sqrt{\frac{\tau_n}{\lambda}} m^{(1-\gamma)/2} \exp(-\beta m^\gamma/2).$$

Apply Proposition A.1, using Lemma A.1, with

$$m = \min\left\{n, \left\lceil\left\{\frac{1}{\beta}\log\left(\frac{n}{\lambda}\right)\right\}^{1/\gamma}\right\rceil\right\}.$$

Since $e^{-\beta m^\gamma/2} \lesssim (n/\lambda)^{-1/2}$ whenever the ceiling is at most n, and $m^{(1-\gamma)/2} \le 1 + m^{3/2}$, both coupling terms are bounded by

$$\sqrt{\frac{\tau_n}{n}} \left\{ \frac{1}{\beta} \log\Big(\frac{n}{\lambda}\Big) \right\}^{3/(2\gamma)} \log^2(n) \left(\frac{1}{\eta}\right)^{1/\log n}.$$

The sample-size condition and $\tau_n \to \infty$ imply $n\lambda \to \infty$; hence the untruncated ceiling is at most n for all sufficiently large n.

Finally, Proposition 4.2 and the exponential effective-dimension bound give

$$\Delta_{G,\mathrm{Bah}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^2 \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/(2\gamma)} \log\log\left(\frac{1}{\lambda}\right).$$

A union bound completes the proof. □

Proposition F.8. *Suppose $\mu_k \le \alpha \exp(-\beta k^\gamma)$ for some $\alpha, \beta, \gamma > 0$. Under Assumptions 4.1, 4.2, 4.4, and A.1, when $\tau_n \ge \tau_0$, if $\tau_n \to \infty$ and $\lambda \to 0$ and*

$$n > \max\left\{ \frac{(\tau_n \kappa^2 + 1)\log(e + \mathcal{N}_\lambda)}{C_2 \lambda}, \frac{\tau_n}{\lambda \mathcal{N}_\lambda} \right\}$$

where C_2 is the constant in Lemma F.13 evaluated at the component tolerance $\eta_{\mathrm{tol}} = c\eta^2$ used in Corollary 4.1, then, on a data event of probability at least $1 - \eta$, there exists a coupling of B and Z, conditionally on $\mathcal{D}$, such that $Z \mid \mathcal{D} \sim \mathrm{N}_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V})$ and

$$\mathbb{P}\Bigg\{ \|Z - B\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^3 \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{\frac{1}{\gamma}} + \left(\log\left(\frac{2}{\eta}\right)\right)^{\frac{3}{2}} \frac{\sqrt{\tau_n}}{n^{\frac{1}{4}}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{\frac{1}{2\gamma}} \left(\log\left(\frac{n}{\lambda}\right)\right)^{\frac{1}{4\gamma}} \;\Big|\; \mathcal{D} \Bigg\} \ge 1 - \eta.$$

Proof of Proposition F.8. The exponential spectral bounds give

$$\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(0, \mathcal{V}) \lesssim \tau_n \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/\gamma}$$

and

$$\sigma_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \lesssim \sqrt{\frac{\tau_n}{\lambda}} m^{(1-\gamma)/2} e^{-\beta m^\gamma/2}.$$

In Proposition A.2, take

$$m = \left\lceil \left\{ \frac{1}{\beta} \log\Big(\frac{n}{\lambda}\Big) \right\}^{1/\gamma} \right\rceil.$$

The finite-dimensional term is then bounded by

$$\frac{\sqrt{\tau_n}}{n^{1/4}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/(2\gamma)} \left(\log\left(\frac{n}{\lambda}\right)\right)^{1/(4\gamma)}.$$

The spectral-tail term is at most

$$\sqrt{\frac{\tau_n}{n}} \left(\log\left(\frac{n}{\lambda}\right)\right)^{(1-\gamma)/(2\gamma)},$$

which is smaller than the preceding display for all sufficiently large n, because a polynomial factor $n^{-1/4}$ dominates every fixed power of the logarithm.

Proposition 4.4 and the exponential effective-dimension bound give

$$\Delta_{B,\mathrm{feas}} \lesssim \left(\log\left(\frac{2}{\eta}\right)\right)^3 \frac{\tau_n}{\sqrt{n\lambda}} \left(\log\left(\frac{1}{\lambda}\right)\right)^{1/\gamma}.$$

Combining the sub-Gaussian oracle coupling with the feasible bootstrap through Lemma F.19, and applying the two-layer union bound from Corollary 4.1, proves the result. □

Proof of Corollary A.1. Choose $C_0 \geq \max\{\overline{\rho}, \tau_0\}$. Then $w_n = \rho$, $\mathcal{L}_{P^w} = \mathcal{L}_Q$, and Assumption 4.4 holds with $\mathfrak{C}_1 = 1$. We first establish the required spectral approximation bounds directly for the fixed envelope $\tau_n = C_0$. Because $w_n = \rho \leq \overline{\rho} \leq C_0$, every occurrence of the truncation envelope in the nonasymptotic Bahadur, covariance, and feasible-bootstrap bounds is controlled by C_0. Repeating the proofs of the four sub-Gaussian spectral approximation propositions with this fixed envelope, rather than invoking their asymptotic statements, gives the same displayed approximation bounds under the corresponding sample-size conditions. Thus $\tau_n \to \infty$ is not required in the bounded-ratio case. The corresponding sample-size conditions hold for all sufficiently large n under each tuning choice below.

We next establish the polynomial result. Since $s \geq 1/2$, Proposition 4.1 gives $\Delta_{\mathrm{bias}} \lesssim \sqrt{n}\lambda$. At $\eta \asymp n^{-1}$, the fixed-envelope polynomial bounds just established yield

$$\Delta \lesssim \log^3 n \left(\Delta_0^{\mathrm{sg}} + \Delta_1^{\mathrm{sg}} + \Delta_2^{\mathrm{sg}} + \Delta_3^{\mathrm{sg}}\right), \tag{F.6}$$

where

$$\begin{aligned}
\Delta_0^{\mathrm{sg}} &= \sqrt{n}\lambda, \\
\Delta_1^{\mathrm{sg}} &= \sqrt{\frac{\tau_n}{n}} \left(\frac{n}{\lambda}\right)^{\frac{3}{2(\beta+2)}}, \\
\Delta_2^{\mathrm{sg}} &= \frac{\tau_n}{\sqrt{n}} \lambda^{-\frac{1}{2}-\frac{1}{\beta}}, \\
\Delta_3^{\mathrm{sg}} &= \sqrt{\frac{\tau_n}{\lambda}} \left(\frac{\tau_n \lambda^{1-2/\beta}}{\sqrt{n}}\right)^{\frac{\beta-1}{2\beta-1}}.
\end{aligned}$$

Here the Bahadur term is absorbed by Δ_2^{sg}, and $(3/\eta)^{1/\log n} = O(1)$.

Before deriving the simpler rate stated in the corollary, we record the sharper optimization underlying (F.6). With $\tau_n = C_0$ and $\lambda = n^{-a}$, the four powers of n are

$$\frac{1}{2} - a, \qquad -\frac{1}{2} + \frac{3(1+a)}{2(\beta+2)}, \qquad -\frac{1}{2} + a\left(\frac{1}{2} + \frac{1}{\beta}\right), \qquad -\frac{\beta-1}{2(2\beta-1)} + a\frac{5\beta-4}{2\beta(2\beta-1)}.$$

For $\beta > 3$, their sharp balance is

$$(a_\beta^\sharp, r_\beta^\sharp) = \begin{cases} \left(\dfrac{\beta(3\beta-2)}{4\beta^2+3\beta-4}, \dfrac{2\beta^2-7\beta+4}{2(4\beta^2+3\beta-4)}\right), & 3 < \beta \leq 3+\sqrt{5}, \\ \left(\dfrac{2\beta}{3\beta+2}, \dfrac{\beta-2}{2(3\beta+2)}\right), & \beta > 3+\sqrt{5}. \end{cases} \tag{F.7}$$

Indeed, the first branch balances Δ_0^{sg} and Δ_3^{sg}, whereas the second balances Δ_0^{sg} and Δ_2^{sg}; the two formulas meet at $\beta = 3+\sqrt{5}$, and direct substitution shows that the remaining terms are smaller. Consequently,

$$\Delta \lesssim n^{-r_\beta^\sharp} \log^3 n.$$

More generally, if $\tau_n = n^t$ and the same $a_\beta^\sharp$ is used, the four powers of τ_n in $\Delta_0^{\mathrm{sg}}, \dots, \Delta_3^{\mathrm{sg}}$ are

$$0, \qquad \frac{1}{2}, \qquad 1, \qquad \frac{4\beta-3}{2(2\beta-1)} < 1.$$

Thus, whenever $a_\beta^\sharp + t < 1$,

$$\Delta \lesssim n^{-r_\beta^\sharp + t} \log^3 n, \qquad \sqrt{\tau_n}\Delta \lesssim n^{-r_\beta^\sharp + 3t/2} \log^3 n.$$

For the non-piecewise statement of the corollary, set

$$\widetilde{a}_\beta = \frac{2\beta}{3\beta+2}, \qquad \widetilde{r}_\beta = \frac{\beta-3}{2(3\beta+2)}.$$

At $a = \widetilde{a}_\beta$, the negatives of the four powers of n are

$$\frac{\beta-2}{2(3\beta+2)}, \qquad \frac{3\beta^2-7\beta-2}{2(\beta+2)(3\beta+2)}, \qquad \frac{\beta-2}{2(3\beta+2)}, \qquad \frac{(\beta-3)(3\beta-2)}{2(3\beta+2)(2\beta-1)}.$$

Each is at least $\widetilde{r}_\beta$ when $\beta > 3$. Therefore

$$\Delta \lesssim n^{-\widetilde{r}_\beta} \log^3 n.$$

Since $\sqrt{\tau_n} = O(1)$, Theorem 4.1 proves part (1).

Under exponential spectral decay, the fixed-envelope exponential bounds just established give the polynomial-order envelope

$$\Delta \lesssim_{\mathrm{poly}} \sqrt{n}\lambda + \frac{\tau_n}{\sqrt{n\lambda}} + \sqrt{\frac{\tau_n}{n}} + \frac{\sqrt{\tau_n}}{n^{1/4}}, \tag{F.8}$$

where $\lesssim_{\mathrm{poly}}$ suppresses only the displayed logarithmic and tolerance factors. With fixed τ_n and $\lambda = n^{-a}$, the potentially active powers are

$$\frac{1}{2} - a, \qquad \frac{a-1}{2}, \qquad -\frac{1}{4}.$$

The first two are balanced at the sharp polynomial-order choice $a = 2/3$, where they equal $-1/6$, while the last equals $-1/4$. More generally, with $\lambda = n^{-2/3}$ and $\tau_n = n^t$,

$$\Delta \lesssim n^{-1/6+t} (\log n)^{3+1/\zeta}, \qquad \sqrt{\tau_n}\Delta \lesssim n^{-1/6+3t/2} (\log n)^{3+1/\zeta}.$$

For $t = 0$, the active stochastic term is the feasible-bootstrap term, whose logarithmic factor is $(\log n)^{3+1/\zeta}$; all other stochastic terms have strictly smaller powers of n, so their fixed logarithmic factors are absorbed. Theorem 4.1 now proves part (2). □

F.17 Proof of Corollary A.2

Proof. Let $\tau_n = n^t$ and $\lambda = n^{-a}$. As in the proof of Corollary 4.3, polynomial spectral decay verifies Assumption 4.3 with $d = 1/\beta$, while exponential spectral decay does so with any fixed $d \in (0, 1]$. Under Definition 4.1, we may choose a fixed integer m_0 sufficiently large that

$$m_0 - 1 > \frac{a(1+d)\gamma}{t}.$$

Lemma F.29 then verifies Assumption 4.4 with $\mathfrak{C}_1 \le \sqrt{2}$ for all sufficiently large n.

For polynomial spectral decay, use

$$a = \widetilde{a}_\beta = \frac{2\beta}{3\beta+2}, \qquad t = \frac{\beta-3}{6(3\beta+2)} = \frac{\widetilde{r}_\beta}{3}.$$

By the calculation following (F.7), restoring $\tau_n = n^t$ enlarges the approximation error by at most n^t, and Theorem 4.1 contributes one further factor $n^{t/2}$. Consequently,

$$\left|\mathbb{P}\{f_* \notin \mathsf{CS}_\alpha(\widehat{f})\} - \alpha\right| \lesssim n^{-1} + n^{-\widetilde{r}_\beta + 3t/2} \log^3 n \lesssim n^{-\frac{\beta-3}{4(3\beta+2)}} \log^3 n.$$

The selected powers satisfy $a + t < 1$, so the sample-size conditions in the polynomial approximation propositions hold for all sufficiently large n. This proves part (1).

For exponential spectral decay, take $a = 2/3$ and $t = 1/150$. The calculation following (F.8) gives

$$\Delta \lesssim n^{-1/6+t} (\log n)^{3+1/\zeta} = n^{-4/25} (\log n)^{3+1/\zeta}.$$

After multiplication by the anti-concentration factor $\sqrt{\tau_n}$,

$$\left|\mathbb{P}\{f_* \notin \mathsf{CS}_\alpha(\widehat{f})\} - \alpha\right| \lesssim n^{-1} + n^{-1/6+3t/2} (\log n)^{3+1/\zeta} \lesssim n^{-47/300} (\log n)^{3+1/\zeta}.$$

Here $a + t < 1$, and the strictly smaller polynomial powers of the remaining terms absorb their fixed logarithmic factors. This proves part (2). □

F.18 Proof of Corollary A.3

Proof. By the common moment-compatibility reduction in (F.5), it is sufficient that

$$\tau_n \gtrsim (\lambda^{-1} \mathcal{N}_\lambda)^{1/(\theta-1)}.$$

The proof of Proposition F.8 uses only this compatibility condition, the bounded truncated-weight envelope, and the standing assumptions; the Bernstein-type condition is not used.

Write $\lambda = n^{-a}$ and $\tau_n = n^t$. Under exponential spectral decay, $\mathcal{N}_\lambda \lesssim \{\log(1/\lambda)\}^{1/\zeta}$. With

$$a = \frac{8(\theta^2-1)}{3(4\theta^2+3\theta-3)}, \qquad t = \frac{19\theta+1}{6(4\theta^2+3\theta-3)},$$

we have

$$t - \frac{a}{\theta-1} = \frac{\theta-5}{2(4\theta^2+3\theta-3)} > 0.$$

Thus the strict polynomial slack absorbs the logarithmic effective-dimension factor and verifies (F.5).

After suppressing logarithmic factors, the four approximation terms in (F.8) have powers

$$\frac{1}{2}-a, \qquad -\frac{1}{2}+\frac{a}{2}+t, \qquad -\frac{1}{2}+\frac{t}{2}, \qquad -\frac{1}{4}+\frac{t}{2}.$$

The feasible-bootstrap term is the largest. Indeed, its excess over the bias term equals

$$-1+\frac{3a}{2}+t=\frac{\theta-5}{6(4\theta^2+3\theta-3)}>0,$$

and its excesses over the final two terms are, respectively, $(a+t)/2>0$ and $-1/4+(a+t)/2>0$. Its negative exponent is

$$\frac{1}{2}-\frac{a}{2}-t=\frac{2\theta^2-5\theta-1}{3(4\theta^2+3\theta-3)}.$$

Therefore

$$\Delta \lesssim n^{-\frac{2\theta^2-5\theta-1}{3(4\theta^2+3\theta-3)}}(\log n)^{3+1/\zeta}.$$

Multiplying by $\sqrt{\tau_n}$ gives

$$\frac{2\theta^2-5\theta-1}{3(4\theta^2+3\theta-3)}-\frac{t}{2}=\frac{(\theta-5)(8\theta+1)}{12(4\theta^2+3\theta-3)},$$

which is positive for $\theta>5$. The active feasible-bootstrap term has logarithmic factor $(\log n)^{3+1/\zeta}$, while all remaining terms have strictly smaller powers of n.

Finally,

$$1-a-t=\frac{8\theta^2-\theta-3}{6(4\theta^2+3\theta-3)}>0,$$

so both sample-size restrictions hold for all sufficiently large n. Since the displayed coverage exponent is smaller than one, the n^{-1} term in Theorem 4.1 is absorbed. This proves the corollary. □

G Proofs of Technical Lemmas

This appendix collects auxiliary proofs and the derivation of the bootstrap matrix representation.

G.1 Additional lemmas

Lemma G.1. *Let $\mathcal{T}:\mathsf{H}\to\mathsf{H}$ be a bounded linear self-adjoint operator on H. Then $\mathcal{T}\mathcal{L}_Q:\mathsf{L}^2(\mathbb{Q})\to\mathsf{L}^2(\mathbb{Q})$ is a self-adjoint operator in $\mathsf{L}^2(\mathbb{Q})$.*

Lemma G.2. *For $0<\eta<1$ and $0<\lambda\le 1$, if*

$$\frac{(\tau_n\kappa^2+1)\log(e+\mathcal{N}_\lambda)}{n\lambda}<C_2$$

where C_2 is the constant in Lemma F.13, then, on the same event of probability at least $1-\eta$ as in Lemma F.12,

$$\left\|(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\right\|\le\sqrt{2}.$$

G.2 Proof of Lemma 2.2

Proof. Let $f = \sum_{\mu_j>0} a_j\phi_j \in \overline{\mathrm{Ran}(\mathcal{L}_Q)}$. Then

$$\mathcal{L}_Q^{1/2} f = \sum_{\mu_j>0} \mu_j^{1/2} a_j \phi_j.$$

By Lemma 2.1, this element belongs to H, and for $g = \sum_{\mu_j>0} b_j\phi_j \in \overline{\mathrm{Ran}(\mathcal{L}_Q)}$,

$$\left\langle \mathcal{L}_Q^{1/2} f, \mathcal{L}_Q^{1/2} g \right\rangle_{\mathsf{H}} = \sum_{\mu_j>0} a_j b_j = \langle f, g\rangle_{\mathsf{L}^2(\mathbb{Q})}.$$

Thus $\mathcal{L}_Q^{1/2}$ preserves inner products on $\overline{\mathrm{Ran}(\mathcal{L}_Q)}$. It remains to note that it is onto H: if $h = \sum_{\mu_j>0} h_j\phi_j \in \mathsf{H}$, then

$$f_h := \sum_{\mu_j>0} \frac{h_j}{\mu_j^{1/2}}\phi_j$$

belongs to $\overline{\mathrm{Ran}(\mathcal{L}_Q)}$, since $\|f_h\|^2_{\mathsf{L}^2(\mathbb{Q})} = \sum_{\mu_j>0} h_j^2/\mu_j = \|h\|^2_{\mathsf{H}} < \infty$, and $\mathcal{L}_Q^{1/2} f_h = h$. Hence $\mathcal{L}_Q^{1/2} : \overline{\mathrm{Ran}(\mathcal{L}_Q)} \to \mathsf{H}$ is a unitary isomorphism. □

G.3 Proof of Lemma F.1

Proof. Since the eigenvalues are ordered non-increasingly, the definition of s^* implies that $\mu_i > \lambda$ for $i < s^*$ and $\mu_i \le \lambda$ for $i \ge s^*$. For sufficiently small λ, $s^* \ge 2$. Hence, for $i < s^*$,

$$2^{-k} \le \left(\frac{\mu_i}{\mu_i + \lambda}\right)^k \le 1,$$

and therefore

$$\sum_{i=1}^{s^*-1} \left(\frac{\mu_i}{\mu_i + \lambda}\right)^k \asymp s^*.$$

On the other hand, for $i > s^*$ we have $\mu_i \le \lambda$, so

$$2^{-k}\frac{\mu_i^k}{\lambda^k} \le \left(\frac{\mu_i}{\mu_i + \lambda}\right)^k \le \frac{\mu_i^k}{\lambda^k}.$$

Consequently,

$$\sum_{i=s^*+1}^{\infty} \left(\frac{\mu_i}{\mu_i + \lambda}\right)^k \asymp \frac{1}{\lambda^k} \sum_{i=s^*+1}^{\infty} \mu_i^k.$$

The remaining term $i = s^*$ is bounded above by one and below by zero, and is therefore absorbed by the s^* term. Combining the preceding two displays gives the claimed equivalence. □

G.4 Proof of Lemma F.2

Proof. In the polynomial case, $s^* \asymp \lambda^{-1/\beta}$ and

$$\lambda^{-k} \sum_{i=s^*+1}^{\infty} \mu_i^k \asymp \lambda^{-k}(s^*)^{1-k\beta} \asymp \lambda^{-1/\beta}.$$

Lemma F.1 gives the first rate. In the exponential case, $s^* \asymp \{\log(1/\lambda)\}^{1/\gamma}$, while an integral comparison gives

$$\lambda^{-k} \sum_{i=s^*+1}^{\infty} \mu_i^k \lesssim s^*.$$

Lemma F.1 then gives the second rate, since the leading s^* term also provides the matching lower bound. □

G.5 Proof of Lemma F.3

Proof. Decompose the influence element U_i in (3.1) as $U_{i,1} + U_{i,2}$, where

$$U_{i,1} = \big(\mathcal{L}_{P^w} + \lambda\mathcal{I}\big)^{-1} w(X_i)\,\varepsilon_i k_{X_i}, \qquad U_{i,2} = \big(\mathcal{L}_{P^w} + \lambda\mathcal{I}\big)^{-1}\big(w(X_i)k_{X_i} \otimes k_{X_i} - \mathcal{L}_{P^w}\big)\big(f_* - f_{P^w}\big).$$

As $\mathbb{E}[\varepsilon_i \mid X_i] = 0$, we have

$$\begin{aligned}
\mathcal{V} &= \mathbb{E}\big[U_i \otimes_{\mathsf{L}^2(\mathbb{Q})} U_i\big] \\
&= \mathbb{E}\big[U_{i,1} \otimes_{\mathsf{L}^2(\mathbb{Q})} U_{i,1}\big] + \mathbb{E}\big[U_{i,2} \otimes_{\mathsf{L}^2(\mathbb{Q})} U_{i,2}\big] \\
&\succeq_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}\big[U_{i,1} \otimes_{\mathsf{L}^2(\mathbb{Q})} U_{i,1}\big] \\
&\succeq_{\mathsf{L}^2(\mathbb{Q})} \underline{\sigma}^2\, \mathbb{E}\Big\{\big[\big(\mathcal{L}_{P^w} + \lambda\mathcal{I}\big)^{-1} w(X_i)k_{X_i}\big] \otimes_{\mathsf{L}^2(\mathbb{Q})} \big[\big(\mathcal{L}_{P^w} + \lambda\mathcal{I}\big)^{-1} w(X_i)k_{X_i}\big]\Big\}.
\end{aligned}$$

□

G.6 Proof of Lemma F.4

Proof. For any $f \in \mathsf{L}^2(\mathbb{Q})$,

$$\begin{aligned}
&\mathbb{E}\big\{w(X_i)\big[(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i}\big] \otimes_{\mathsf{L}^2(\mathbb{Q})} \big[\big((\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i}\big)\big]\big\}\, f \\
=&\mathbb{E}\Big[w(X_i)\,\big\langle(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i}, f\big\rangle_{\mathsf{L}^2(\mathbb{Q})}\,(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i}\Big] \\
=&\mathbb{E}\big[w(X_i)\,\big\langle(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i}, \mathcal{L}_Q f\big\rangle\,(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i}\big] \\
=&\mathbb{E}\big[w(X_i)\,(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\big\langle k_{X_i}, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q f\big\rangle\,k_{X_i}\big] \\
=&(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q f.
\end{aligned}$$

□

G.7 Proof of Lemma F.5

Proof. Let

$$\mathsf{H}_Q := \overline{\mathrm{Ran}(\mathcal{L}_Q)} \quad \text{and} \quad U := \mathcal{L}_Q^{1/2} : \mathsf{H}_Q \to \mathsf{H}.$$

By Lemma 2.2, U is a unitary isomorphism. Moreover, $\mathsf{L}^2(\mathbb{Q}) = \mathsf{H}_Q \oplus \ker(\mathcal{L}_Q)$, and $\mathcal{T}\mathcal{L}_Q$ vanishes on $\ker(\mathcal{L}_Q)$. For $f, g \in \mathsf{H}_Q$,

$$\langle f, \mathcal{T}\mathcal{L}_Q g\rangle_{\mathsf{L}^2(\mathbb{Q})} = \Big\langle Uf,\, \mathcal{L}_Q^{1/2}\mathcal{T}\mathcal{L}_Q^{1/2} Ug\Big\rangle_{\mathsf{H}}.$$

Since $\mathcal{T}$ is positive self-adjoint, $\mathcal{L}_Q^{1/2}\mathcal{T}\mathcal{L}_Q^{1/2}$ is positive self-adjoint. The preceding identity therefore shows that the restriction of $\mathcal{T}\mathcal{L}_Q$ to H_Q is also positive self-adjoint. Its nonzero eigenvalues are precisely those of $\mathcal{T}\mathcal{L}_Q$ on $\mathsf{L}^2(\mathbb{Q})$.

We may now apply the min–max theorem and Lemma 2.2:

$$\begin{aligned}
\mu_k(\mathcal{T}\mathcal{L}_Q, \mathsf{L}^2(\mathbb{Q})) &= \max_{\substack{V\subset \mathsf{H}_Q \\ \dim(V)=k}} \min_{\substack{f\in V \\ \|f\|_{\mathsf{L}^2(\mathbb{Q})}=1}} \langle f, \mathcal{T}\mathcal{L}_Q f\rangle_{\mathsf{L}^2(\mathbb{Q})} \\
&= \max_{\substack{V\subset \mathsf{H}_Q \\ \dim(V)=k}} \min_{\substack{f\in V \\ \|\mathcal{L}_Q^{1/2} f\|_{\mathsf{H}}=1}} \langle \mathcal{L}_Q^{1/2} f, \mathcal{L}_Q^{1/2}\mathcal{T}\mathcal{L}_Q^{1/2}\mathcal{L}_Q^{1/2} f\rangle_{\mathsf{H}} \\
&= \max_{\substack{W\subset \mathsf{H} \\ \dim(W)=k}} \min_{\substack{g\in W \\ \|g\|_{\mathsf{H}}=1}} \langle g, \mathcal{L}_Q^{1/2}\mathcal{T}\mathcal{L}_Q^{1/2} g\rangle_{\mathsf{H}} \\
&= \mu_k(\mathcal{L}_Q^{1/2}\mathcal{T}\mathcal{L}_Q^{1/2}, \mathsf{H}).
\end{aligned}$$

□

G.8 Proof of Lemma F.10

Proof. If $0 \le s \le \frac{1}{2}$ in Assumption 4.2,

$$\begin{aligned}
\left\| \left(\mathcal{L}_{P^w} + \lambda\mathcal{I}\right)^{-\frac{1}{2}} f_* \right\| &= \left\| \left(\mathcal{L}_{P^w} + \lambda\mathcal{I}\right)^{-\frac{1}{2}} \mathcal{L}_{P^{w_0}}^{s} g_* \right\| \\
&\le \left\| \left(\mathcal{L}_{P^{w_0}} + \lambda\mathcal{I}\right)^{-\frac{1}{2}} \mathcal{L}_{P^{w_0}}^{s} g_* \right\| \\
&\le \lambda^{-\frac{1}{2}+s} \|g_*\|.
\end{aligned}$$

If $\frac{1}{2} < s \le 1$ in Assumption 4.2,

$$\begin{aligned}
\left\| (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} f_* \right\| &\le \left\| \mathcal{L}_{P^{w_0}}^{s-\frac{1}{2}} g_* \right\| \\
&\le \left\| \mathcal{L}_{Q}^{s-\frac{1}{2}} g_* \right\| \\
&\le \kappa^{2s-1} \|g_*\| .
\end{aligned}$$

In conclusion,

$$\left\| (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} f_* \right\| \le \max\{\kappa^{2s-1}, 1\}\, \lambda^{\min\{s-\frac{1}{2}, 0\}} \|g_*\|.$$

□

G.9 Proof of Lemma F.6

Proof. Since

$$f_{P^w} - f_* = -\lambda(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1} f_*,$$

we have

$$\begin{aligned}
\|f_{P^w} - f_*\| &= \lambda \left\| (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1} f_* \right\| \\
&\le \lambda^{1/2} \left\| (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2} f_* \right\| \\
&\le \max\{\kappa^{2s-1}, 1\} \lambda^{1/2+\min\{s-1/2, 0\}} \|g_*\| \\
&= \max\{\kappa^{2s-1}, 1\} \lambda^{\min\{s, 1/2\}} \|g_*\|.
\end{aligned}$$

The first inequality follows from

$$\left\|\lambda^{1/2}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1/2}\right\| \le 1,$$

and the second follows from Lemma F.10. □

G.10 Proof of Lemma F.7

Proof. With the same notation as in the proof of Lemma F.3,

$$\begin{aligned}\mathcal{V} &= \mathbb{E}[U_{i,1}\otimes_{\mathsf{L}^2(\mathbb{Q})} U_{i,1}] + \mathbb{E}[U_{i,2}\otimes_{\mathsf{L}^2(\mathbb{Q})} U_{i,2}]\\ &=: \mathcal{V}_1+\mathcal{V}_2\end{aligned}$$

$$\begin{aligned}\mathcal{V}_1 &= \mathbb{E}\left\{\left[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}w(X_i)\varepsilon_i k_{X_i}\right]\otimes_{\mathsf{L}^2(\mathbb{Q})}\left[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}w(X_i)\varepsilon_i k_{X_i}\right]\right\}\\ &\preceq_{\mathsf{L}^2(\mathbb{Q})} \tau_n\overline{\sigma}^2\mathbb{E}\left\{w(X_i)[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}]\otimes_{\mathsf{L}^2(\mathbb{Q})}[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}]\right\}\end{aligned}$$

By Lemma F.4,

$$\mathcal{V}_1 \preceq_{\mathsf{L}^2(\mathbb{Q})} \tau_n\overline{\sigma}^2\cdot(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q.$$

Denote $\mathcal{T}_1 = (\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q$, $\mathcal{T}_2 = \mathcal{L}_Q^{\frac{1}{2}}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q^{\frac{1}{2}}$.

With Lemma F.5,

$$\|\mathcal{T}_1\|_{\mathsf{L}^2(\mathbb{Q})} = \|\mathcal{T}_2\|_{\mathsf{H}} \le \left(\mathfrak{C}_1\right)^2. \tag{G.1}$$

Then,

$$\|\mathcal{V}_1\|_{\mathsf{L}^2(\mathbb{Q})} \le \tau_n\overline{\sigma}^2\|\mathcal{T}_1\|_{\mathsf{L}^2(\mathbb{Q})} \le \tau_n\overline{\sigma}^2\left(\mathfrak{C}_1\right)^2.$$

$\forall f\in\mathsf{L}^2(\mathbb{Q})$,

$$\langle f,\mathcal{V}_1 f\rangle_{\mathsf{L}^2(\mathbb{Q})} \le \left\langle f,\,\tau_n\overline{\sigma}^2\mathcal{T}_1 f\right\rangle_{\mathsf{L}^2(\mathbb{Q})} = \tau_n\overline{\sigma}^2\left\langle\mathcal{L}_Q^{1/2}f,\,\mathcal{T}_2\mathcal{L}_Q^{1/2}f\right\rangle.$$

Similarly with Lemma F.5,

$$\begin{aligned}\mathrm{tr}_{\mathsf{L}^2(\mathbb{Q})}(\mathcal{V}_1) &= \sum_{k=1}^{\infty}\mu_k(\mathcal{V}_1,\mathsf{L}^2(\mathbb{Q}))\\ &\le \tau_n\overline{\sigma}^2\sum_{k=1}^{\infty}\mu_k(\mathcal{T}_1,\mathsf{L}^2(\mathbb{Q}))\\ &= \tau_n\overline{\sigma}^2\sum_{k=1}^{\infty}\mu_k(\mathcal{T}_2,\mathsf{H})\\ &= \tau_n\overline{\sigma}^2\mathrm{tr}_{\mathsf{H}}(\mathcal{T}_2)\\ &\le \tau_n\overline{\sigma}^2\left(\mathfrak{C}_1\right)^2\mathrm{tr}_{\mathsf{H}}\left((\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}\right)\\ &\le \tau_n\overline{\sigma}^2\left(\mathfrak{C}_1\right)^2\mathcal{N}_\lambda.\end{aligned} \tag{G.2}$$

The last inequality follows from the domination of the truncated weighted-source operator by the target operator. Indeed, since $w\le\rho$, for every $h\in\mathsf{H}$,

$$\langle h,(\mathcal{L}_Q-\mathcal{L}_{P^w})h\rangle_{\mathsf{H}} = \int_{\mathfrak{X}}(\rho(x)-w(x))|h(x)|^2\,d\mathbb{P}(x)\ge 0,$$

and hence $\mathcal{L}_{P^w} \preceq_{\mathsf{H}} \mathcal{L}_Q$. The min–max principle and the monotonicity of $t \mapsto t/(t+\lambda)$ on $[0,\infty)$ therefore give

$$\mathrm{tr}_{\mathsf{H}}\big((\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}\big) \le \mathrm{tr}_{\mathsf{H}}\big((\mathcal{L}_{Q}+\lambda\mathcal{I})^{-1}\mathcal{L}_{Q}\big) = \mathcal{N}_\lambda.$$

$$\begin{aligned}
\mathcal{V}_2 &= \mathbb{E}\Big\{\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\big(w(X_i)\,k_{X_i}\otimes k_{X_i}-\mathcal{L}_{P^w}\big)\,(f_*-f_{P^w})\Big]\otimes_{\mathsf{L}^2(\mathbb{Q})}\\
&\qquad\qquad \Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\big(w(X_i)\,k_{X_i}\otimes k_{X_i}-\mathcal{L}_{P^w}\big)\,(f_*-f_{P^w})\Big]\Big\}\\
&\preceq_{\mathsf{L}^2(\mathbb{Q})} \mathbb{E}\Big\{\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\big(w(X_i)\,k_{X_i}\otimes k_{X_i}\big)\,(f_*-f_{P^w})\Big]\otimes_{\mathsf{L}^2(\mathbb{Q})}\\
&\qquad\qquad \Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\big(w(X_i)\,k_{X_i}\otimes k_{X_i}\big)\,(f_*-f_{P^w})\Big]\Big\}\\
&= \mathbb{E}\Big\{\Big[w(X_i)\big(f_*(X_i)-f_{P^w}(X_i)\big)\,(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}\Big]\otimes_{\mathsf{L}^2(\mathbb{Q})}\\
&\qquad\qquad \Big[w(X_i)\big(f_*(X_i)-f_{P^w}(X_i)\big)\,(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}\Big]\Big\}\\
&= \mathbb{E}\Big\{w(X_i)\big(f_*(X_i)-f_{P^w}(X_i)\big)^2 w(X_i)\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}\Big]\otimes_{\mathsf{L}^2(\mathbb{Q})}\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}\Big]\Big\}\\
&\preceq_{\mathsf{L}^2(\mathbb{Q})} \tau_n\kappa^2\max\{\kappa^{2s-1},1\}^2\lambda^{2\min\{s,1/2\}}\|g_*\|^2\\
&\qquad\times \mathbb{E}\Big\{w(X_i)\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}\Big]\otimes_{\mathsf{L}^2(\mathbb{Q})}\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}k_{X_i}\Big]\Big\}\\
&= \tau_n\kappa^2\max\{\kappa^{2s-1},1\}^2\lambda^{2\min\{s,1/2\}}\\
&\qquad\times \|g_*\|^2\,\mathcal{T}_1.
\end{aligned}$$

where we use Lemma F.6 to get the inequality

$$\big|f_*(X_i)-f_{P^w}(X_i)\big| = \big|\langle f_*-f_{P^w},\,k_{X_i}\rangle\big| \le \kappa\|f_*-f_{P^w}\| \le \kappa\max\{\kappa^{2s-1},1\}\lambda^{\min\{s,1/2\}}\|g_*\|.$$

Combining this bound with (G.1) and (G.2) proves the result. □

G.11 Proof of Lemma F.8

Proof. With Lemma F.3 and F.4,

$$\begin{aligned}
\mathcal{V} &\succeq_{\mathsf{L}^2(\mathbb{Q})} \underline{\sigma}^2\,\mathbb{E}\Big\{\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}w(X_i)k_{X_i}\Big]\otimes_{\mathsf{L}^2(\mathbb{Q})}\Big[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}w(X_i)k_{X_i}\Big]\Big\}\\
&\succeq_{\mathsf{L}^2(\mathbb{Q})} \underline{\rho\sigma}^2(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q.
\end{aligned}$$

For any $f \in \mathsf{L}^2(\mathbb{Q})$,

$$\begin{aligned}
\langle f,\mathcal{V}f\rangle_{\mathsf{L}^2(\mathbb{Q})} &\ge \underline{\rho\sigma}^2\langle f,(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q f\rangle_{\mathsf{L}^2(\mathbb{Q})}\\
&= \underline{\rho\sigma}^2\langle \mathcal{L}_Q f,(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q f\rangle_{\mathsf{H}}\\
&\ge \big(\mathfrak{C}_2\big)^{-2}\underline{\rho\sigma}^2\langle \mathcal{L}_Q f,(\mathcal{L}_{Q}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q(\mathcal{L}_{Q}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q f\rangle_{\mathsf{H}}\\
&= \big(\mathfrak{C}_2\big)^{-2}\underline{\rho\sigma}^2\langle f,(\mathcal{L}_{Q}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q(\mathcal{L}_{Q}+\lambda\mathcal{I})^{-1}\mathcal{L}_Q f\rangle_{\mathsf{L}^2(\mathbb{Q})}.
\end{aligned}$$

□

G.12 Proof of Lemma F.12

Proof. Let

$$\eta(x) = (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}}\, w(x)\, (k_x \otimes k_x)(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}}$$

$\forall h \in \mathsf{H}$,

$$\begin{aligned}
\eta(x)h &= w(x)\, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} \langle k_x, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} h\rangle\, k_x \\
[\eta(x)]^2\, h &= w^2(x)\, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} \langle k_x, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1} \langle k_x, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} h\rangle k_x\rangle\, k_x \\
&= w^2(x)\, \langle k_x, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1} k_x\rangle\, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} (k_x \otimes k_x)(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} h.
\end{aligned}$$

Thus,

$$[\eta(x)]^2 = w^2(x)\, \langle k_x, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1} k_x\rangle (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} (k_x \otimes k_x)(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}}.$$

As

$$\begin{aligned}
\big|\langle k_x, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1} k_x\rangle\big| &= \big|\mathrm{tr}\big((\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1} k_x \otimes k_x\big)\big| \\
&\le \lambda^{-1}\kappa^2
\end{aligned}$$

$$\begin{aligned}
\big\|\mathbb{E}[(\eta(x))^2]\big\| &\le \frac{\tau_n\, \kappa^2}{\lambda} \big\|\mathbb{E}\big[w(x)(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} (k_x \otimes k_x)(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}}\big]\big\| \\
&= \frac{\tau_n\, \kappa^2}{\lambda} \big\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}} \mathcal{L}_{P^w} (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-\frac{1}{2}}\big\| \\
&\le \frac{\tau_n\, \kappa^2}{\lambda}.
\end{aligned}$$

Since $w_n \le \rho$, we have $\mathcal{L}_{P^w} \preceq \mathcal{L}_Q$. The map $A \mapsto A(A + \lambda\mathcal{I})^{-1}$ is operator monotone on the cone of positive operators. Therefore,

$$\mathrm{Tr}\big\{(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_{P^w}\big\} \le \mathrm{Tr}\big\{(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\mathcal{L}_Q\big\} = \mathcal{N}_\lambda,$$

and hence

$$\mathrm{Tr}\Big(\mathbb{E}\big[(\eta(x))^2\big]\Big) \le \tau_n\, \kappa^2 \lambda^{-1}\, \mathcal{N}_\lambda.$$

Let

$$\xi(x) = \eta(x) - \mathbb{E}[\eta(x)].$$

$$\|\xi(x)\| \le \|\eta(x)\| + \|\mathbb{E}[\eta(x)]\| \le \frac{\tau_n \kappa^2}{\lambda} + 1.$$

$$\begin{aligned}
\big\|\mathbb{E}\Big[\Big(\sum_{i=1}^n \xi(X_i)\Big)^2\Big]\big\| &= \Big\|\sum_{i=1}^n \mathbb{E}[\xi(X_i)^2]\Big\| = n\big\|\mathbb{E}\eta^2 - (\mathbb{E}\eta)^2\big\| \\
&\le n\big\|\mathbb{E}\eta^2\big\| + n\big\|(\mathbb{E}\eta)^2\big\| \le n\Big(\frac{\tau_n\kappa^2}{\lambda} + 1\Big).
\end{aligned}$$

$$\begin{aligned}
\mathrm{Tr}\Big(\mathbb{E}\Big[\Big(\sum_{i=1}^n \xi(X_i)\Big)^2\Big]\Big) &= n\ \mathrm{Tr}\Big(\mathbb{E}\big[(\eta)^2\big] - (\mathbb{E}\eta)^2\Big) \\
&\le n\, \mathrm{Tr}\big(\mathbb{E}[(\eta)^2]\big) \\
&\le n\tau_n\, \kappa^2 \lambda^{-1} \mathcal{N}_\lambda \\
&\le n\Big(\frac{\tau_n\kappa^2}{\lambda} + 1\Big)\mathcal{N}_\lambda.
\end{aligned}$$

Consider $0 < \lambda \le 1$.

Applying Lemma F.11 with

$$R = \frac{\tau_n\kappa^2+1}{\lambda}, \qquad V = n\frac{\tau_n\kappa^2+1}{\lambda}, \qquad W = e + \mathcal{N}_\lambda$$

$$\forall\, t \ge \sqrt{\frac{\tau_n\kappa^2+1}{n\lambda}} + \frac{\tau_n\kappa^2+1}{3n\lambda}$$

$$\begin{aligned}
&\mathbb{P}\Big(\big\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P_n^w}-\mathcal{L}_{P^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\big\| \ge t\Big)\\
=&\mathbb{P}\Big(\Big\|\frac{1}{n}\sum_{i=1}^{n}\xi(X_i)\Big\| \ge t\Big)\\
\le& 4(e+\mathcal{N}_\lambda)\exp\Big(-\frac{\lambda\, n\, t^2}{2\,(\tau_n\kappa^2+1)+\dfrac{2t(\tau_n\kappa^2+1)}{3}}\Big).
\end{aligned}$$

Let $\epsilon > 0$ be the positive solution of

$$\frac{n\lambda\,\epsilon^2}{2(\tau_n\kappa^2+1)\big(1+\frac{\epsilon}{3}\big)} = \log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big).$$

Solving this quadratic equation and using $\sqrt{x^2+y} \le x + \sqrt{y}$ for $x, y \ge 0$ give

$$\begin{aligned}
\epsilon =& \frac{\tau_n\kappa^2+1}{3n\lambda}\log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big)\\
&+\sqrt{\Big\{\frac{\tau_n\kappa^2+1}{3n\lambda}\log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big)\Big\}^2+\frac{2(\tau_n\kappa^2+1)}{n\lambda}\log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big)}\\
\le& \frac{2(\tau_n\kappa^2+1)}{3n\lambda}\log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big)+\sqrt{\frac{2(\tau_n\kappa^2+1)}{n\lambda}\log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big)}.
\end{aligned}$$

Since $\log\{4(e+\mathcal{N}_\lambda)/\eta\} > 1$, this choice also satisfies the applicability threshold of Lemma F.11; indeed,

$$\epsilon \ge \sqrt{\frac{\tau_n\kappa^2+1}{n\lambda}} + \frac{\tau_n\kappa^2+1}{3n\lambda}.$$

Consequently,

$$\begin{aligned}
&\mathbb{P}\Big(\big\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P_n^w}-\mathcal{L}_{P^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\big\| \ge \epsilon\Big)\\
&\qquad \le 4(e+\mathcal{N}_\lambda)\exp\Big\{-\log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big)\Big\} = \eta.
\end{aligned}$$

Moreover, because $\log(e+\mathcal{N}_\lambda) \ge 1$ and $\log(2/\eta) \ge \log 2$,

$$\log\Big(\frac{4(e+\mathcal{N}_\lambda)}{\eta}\Big) = \log(e+\mathcal{N}_\lambda) + \log\Big(\frac{4}{\eta}\Big) \lesssim \log(e+\mathcal{N}_\lambda)\log\Big(\frac{2}{\eta}\Big).$$

Together with $\{\log(2/\eta)\}^{1/2} \lesssim \log(2/\eta)$, substituting this comparison into the preceding upper bound for ϵ proves the first assertion after increasing C_1 by a universal constant. Under the additional condition in the second assertion, the nonnegative ratio satisfies

$$\frac{\big(\tau_n\kappa^2+1\big)\log(e+\mathcal{N}_\lambda)}{n\lambda} \le \sqrt{\frac{\big(\tau_n\kappa^2+1\big)\log(e+\mathcal{N}_\lambda)}{n\lambda}}.$$

Thus, the second assertion follows from the first after absorbing the factor $2C_1$ into the implicit constant. □

G.13 Proof of Lemma F.13

Proof. With Lemma F.12, when

$$\frac{(\tau_n\kappa^2+1)\log(e+\mathcal{N}_\lambda)}{n\lambda} < \min\left\{1, \frac{1}{\left(4C_1\log\frac{2}{\eta}\right)^2}\right\},$$

$$\left\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w}-\mathcal{L}_{P_n^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\right\| \le \frac{1}{2},$$

then

$$\begin{aligned}
&(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\\
=&(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\left[(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}-(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1}\right](\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}+\mathcal{I}\\
=&I+(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w}-\mathcal{L}_{P_n^w})(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}.
\end{aligned}$$

Thus,

$$\begin{aligned}
&\left\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\right\|\\
\le&1+\frac{1}{2}\left\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\right\|,
\end{aligned}$$

and hence

$$\left\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w}+\lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{\frac{1}{2}}\right\| \le 2.$$

□

G.14 Proof of Lemma F.14

Proof.

$$(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}\left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top\mathbf{W}\boldsymbol{\varepsilon}\right) = \frac{1}{n}\sum_{i=1}^n(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)\varepsilon_i k_{X_i}$$

$$\mathbb{E}[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)\varepsilon_i k_{X_i}] = 0.$$

$$\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)\varepsilon_i k_{X_i}\| \le \tau_n\frac{\kappa\cdot\overline{\sigma}}{\sqrt{\lambda}}$$

$$\begin{aligned}
&\mathbb{E}\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)\varepsilon_i k_{X_i}\|^2\\
&\le\overline{\sigma}^2\cdot\mathbb{E}\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)k_{X_i}\|^2\\
&=\overline{\sigma}^2\cdot\mathbb{E}\mathrm{tr}([(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)k_{X_i}]\otimes[(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)k_{X_i}])\\
&\le\tau_n\cdot\overline{\sigma}^2\cdot\mathcal{N}_\lambda
\end{aligned}$$

For

$$Z_i := (\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-\frac{1}{2}}w(X_i)\varepsilon_i k_{X_i},$$

the preceding bounds imply

$$\sum_{i=1}^{n} \mathbb{E}\|Z_i\|^2 \le n\tau_n \overline{\sigma}^2 \mathcal{N}_\lambda, \qquad \|Z_i\| \le \frac{\tau_n \kappa \overline{\sigma}}{\sqrt{\lambda}} \quad \text{a.s.}$$

Therefore, the Hilbert-space Bernstein inequality [Pinelis and Sakhanenko, 1986, Corollary 1] yields, with probability at least $1-\eta$,

$$\left\| \frac{1}{n} \sum_{i=1}^{n} Z_i \right\| \lesssim \overline{\sigma} \left[\sqrt{\frac{\tau_n \mathcal{N}_\lambda}{n} \log\left(\frac{2}{\eta}\right)} + \frac{\tau_n \kappa}{n\sqrt{\lambda}} \log\left(\frac{2}{\eta}\right) \right].$$

Since $\sqrt{\log(2/\eta)} \lesssim \log(2/\eta)$ and $a+b \le 2\max\{a,b\}$, the asserted bound follows. □

G.15 Proof of Lemma F.15

Proof.

$$\begin{aligned} B - B_{\mathsf{ora}} &= \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij} \frac{\widehat{V}_i - \widehat{V}_j}{\sqrt{2}} - \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij} \frac{V_i - V_j}{\sqrt{2}} \\ &= \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\, \mathcal{L}^{-1}_{P_n^w,\lambda} \hat{\alpha}_{ij} - \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\, \mathcal{L}^{-1}_{P^w,\lambda} \alpha_{ij} \\ &= \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij} \left(\mathcal{L}^{-1}_{P_n^w,\lambda} - \mathcal{L}^{-1}_{P^w,\lambda}\right)\hat{\alpha}_{ij} + \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\, \mathcal{L}^{-1}_{P^w,\lambda}\left(\hat{\alpha}_{ij} - \alpha_{ij}\right). \end{aligned}$$

□

G.16 Proof of Lemma F.16

Lemma G.3. *Under Assumption 4.2, when $\tau_n \ge \tau_0$, if*

$$\left\| (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-\frac{1}{2}} (\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w}) (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-\frac{1}{2}} \right\| < \delta \le \frac{1}{2}$$

and

$$\left\| (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-\frac{1}{2}} \left(\frac{1}{n} \mathbf{k}_{\mathbf{x}}^\top \mathbf{W} \boldsymbol{\varepsilon} \right) \right\| < \gamma,$$

then

$$\begin{aligned} \|\hat{f} - f_{P^w}\| &\le \frac{2}{\sqrt{\lambda}} \left(\gamma + \delta \cdot \max\{\kappa^{2s-1}, 1\} \cdot \lambda^{\min\{s+\frac{1}{2}, 1\}} \cdot \|g_*\| \right), \\ \|\hat{f} - f_*\| &\le \frac{2}{\sqrt{\lambda}} \left(\gamma + \max\{\kappa^{2s-1}, 1\} \cdot \lambda^{\min\{s+\frac{1}{2}, 1\}} \cdot \|g_*\| \right). \end{aligned}$$

Proof of Lemma G.3. The Bahadur representation gives

$$\begin{aligned} \hat{f} - f_{P^w} &= (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \left[\left(\frac{1}{n} \mathbf{k}_{\mathbf{x}}^\top \mathbf{W} \boldsymbol{\varepsilon} \right) + (\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w})(f_* - f_{P^w}) \right] + u \\ &= \mathcal{L}^{-1}_{P^w,\lambda} \left\{ \mathbb{E}_n\left[w(X_i) k_{X_i} \varepsilon_i \right] + (\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w})(f_* - f_{P^w}) \right\} + u. \end{aligned}$$

Here u satisfies

$$\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}} u\| \le 2\delta \left(\gamma + \delta \cdot \max\{\kappa^{2s-1}, 1\} \cdot \lambda^{\min\{s+\frac{1}{2},1\}} \|g_*\|\right).$$

Then,

$$\begin{aligned}\left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}(\hat{f} - f_{P^w})\right\| &\le \gamma + \delta \cdot \left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}(f_* - f_{P^w})\right\| \\ &\quad + 2\delta \left(\gamma + \delta \cdot \left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}(f_* - f_{P^w})\right\|\right) \\ &\le 2\left(\gamma + \delta \cdot \max\{\kappa^{2s-1}, 1\} \cdot \lambda^{\min\{s+\frac{1}{2},1\}} \cdot \|g_*\|\right).\end{aligned}$$

$$\begin{aligned}\left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}(\hat{f} - f_*)\right\| &\le 2\left(\gamma + \delta \cdot \max\{\kappa^{2s-1}, 1\} \cdot \lambda^{\min\{s+\frac{1}{2},1\}} \cdot \|g_*\|\right) \\ &\quad + \max\{\kappa^{2s-1}, 1\} \cdot \lambda^{\min\{s+\frac{1}{2},1\}} \cdot \|g_*\| \\ &\le 2\left(\gamma + \max\{\kappa^{2s-1}, 1\} \cdot \lambda^{\min\{s+\frac{1}{2},1\}} \cdot \|g_*\|\right).\end{aligned}$$

□

Proof of Lemma F.16. Apply Lemmas F.12, F.13, and F.14 with suitably split failure probabilities. The second sample-size condition implies

$$\max\left\{\sqrt{\frac{\tau_n \mathcal{N}_\lambda}{n}}, \frac{2\tau_n \kappa}{n\sqrt{\lambda}}\right\} \lesssim \kappa\sqrt{\frac{\tau_n \mathcal{N}_\lambda}{n}}.$$

The sample-size condition is meaningful only if $\mathcal{N}_\lambda > 0$, and hence $\mathcal{L}_Q \neq 0$. Let $\mu_1 > 0$ be a positive eigenvalue of $\mathcal{L}_Q$. Since $\lambda \to 0$, for all sufficiently large n,

$$\mathcal{N}_\lambda \ge \frac{\mu_1}{\mu_1 + \lambda} \ge \frac{1}{2}.$$

It follows that $\log(e + \mathcal{N}_\lambda) \lesssim \mathcal{N}_\lambda$. Hence Lemma F.12 and the first sample-size condition give

$$\left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2}(\mathcal{L}_{P_n^w} - \mathcal{L}_{P^w})(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2}\right\| \lesssim \kappa\sqrt{\frac{\tau_n \mathcal{N}_\lambda}{n\lambda}} \log\left(\frac{2}{\eta}\right),$$

whereas Lemma F.14 gives

$$\frac{1}{\sqrt{\lambda}} \left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2} \left(\frac{1}{n}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\boldsymbol{\varepsilon}\right)\right\| \lesssim \overline{\sigma}\kappa\sqrt{\frac{\tau_n \mathcal{N}_\lambda}{n\lambda}} \log\left(\frac{2}{\eta}\right).$$

Substituting these two bounds into Lemma G.3, and using $\max\{\kappa^{2s-1}, 1\} \le \kappa^{2s}$, $0 < \lambda \le 1$, and $\|g_*\|_{\mathsf{H}} \le 1 + \|g_*\|_{\mathsf{H}}$, proves both assertions. □

G.17 Proof of Lemma F.17

Denote $\Delta_{\text{inv}} = (\mathcal{L}^{-1}_{P^w_n,\lambda} - \mathcal{L}^{-1}_{P^w,\lambda})u$, where

$$u = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\,\hat{\alpha}_{ij}.$$

Lemma G.4. *Under Assumption 4.4, writing $\lambda = \lambda_n$, we have*

$$\|\Delta_{\text{inv}}\|_{\mathsf{L}^2(\mathbb{Q})} \le \mathfrak{C}_1 \|\mathcal{L}^{1/2}_{P^w,\lambda}\mathcal{L}^{-1}_{P^w_n,\lambda}\mathcal{L}^{1/2}_{P^w,\lambda}\| \|\mathcal{L}^{-1/2}_{P^w,\lambda}(\mathcal{L}_{P^w} - \mathcal{L}_{P^w_n})\mathcal{L}^{-1/2}_{P^w,\lambda}\| \|\mathcal{L}^{-1/2}_{P^w,\lambda}u\|.$$

Proof of Lemma G.4.

$$\begin{aligned}
\|\Delta_{\text{inv}}\|_{\mathsf{L}^2(\mathbb{Q})} &= \big\|\mathcal{L}_Q^{1/2}\Delta_{\text{inv}}\big\| \\
&= \big\|\mathcal{L}_Q^{1/2}(\mathcal{L}^{-1}_{P^w_n,\lambda} - \mathcal{L}^{-1}_{P^w,\lambda})u\big\| \\
&= \big\|\mathcal{L}_Q^{1/2}\mathcal{L}^{-1}_{P^w_n,\lambda}(\mathcal{L}_{P^w} - \mathcal{L}_{P^w_n})\mathcal{L}^{-1}_{P^w,\lambda}u\big\| \\
&= \Big\|\Big(\mathcal{L}_Q^{1/2}\mathcal{L}^{-1/2}_{P^w,\lambda}\Big)\Big(\mathcal{L}^{1/2}_{P^w,\lambda}\mathcal{L}^{-1}_{P^w_n,\lambda}\mathcal{L}^{1/2}_{P^w,\lambda}\Big)\Big[\mathcal{L}^{-1/2}_{P^w,\lambda}(\mathcal{L}_{P^w} - \mathcal{L}_{P^w_n})\mathcal{L}^{-1/2}_{P^w,\lambda}\Big]\mathcal{L}^{-1/2}_{P^w,\lambda}u\Big\| \\
&\le \mathfrak{C}_1 \|\mathcal{L}^{1/2}_{P^w,\lambda}\mathcal{L}^{-1}_{P^w_n,\lambda}\mathcal{L}^{1/2}_{P^w,\lambda}\| \|\mathcal{L}^{-1/2}_{P^w,\lambda}(\mathcal{L}_{P^w} - \mathcal{L}_{P^w_n})\mathcal{L}^{-1/2}_{P^w,\lambda}\| \|\mathcal{L}^{-1/2}_{P^w,\lambda}u\|.
\end{aligned}$$

□

Lemma G.5. *For $0 < \eta < 1$ and $\lambda > 0$, conditionally on $\mathcal{D}$, with probability at least $1-\eta$,*

$$\|\mathcal{L}^{-1/2}_{P^w,\lambda}u\| \le \Big\{1 + \sqrt{2\log\Big(\frac{1}{\eta}\Big)}\Big\}\sqrt{\frac{4}{n}\sum_{i=1}^{n}\big\|w(X_i)\,\varepsilon_i\,\mathcal{L}^{-1/2}_{P^w,\lambda}k_{X_i}\big\|^2 + \frac{4}{n}\sum_{i=1}^{n}\big\|\mathcal{L}^{-1/2}_{P^w,\lambda}\mathcal{L}_{P^w_i}(f_* - \hat{f})\big\|^2}.$$

Proof of Lemma G.5. $\mathcal{L}^{-1/2}_{P^w,\lambda}u \mid \mathcal{D}$ is Gaussian. By Borell's inequality under $\mathbb{P}(\cdot \mid \mathcal{D})$, with probability $1-\eta$,

$$\|\mathcal{L}^{-1/2}_{P^w,\lambda}u\| \le \Big\{1 + \sqrt{2\log\Big(\frac{1}{\eta}\Big)}\Big\}\sqrt{\mathbb{E}_h\big\|\mathcal{L}^{-1/2}_{P^w,\lambda}u\big\|^2}.$$

$$\mathbb{E}_h\big\|\mathcal{L}^{-1/2}_{P^w,\lambda}u\big\|^2 = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\big\|\mathcal{L}^{-1/2}_{P^w,\lambda}\hat{\alpha}_{ij}\big\|^2.$$

$$\begin{aligned}
\big\|\mathcal{L}^{-1/2}_{P^w,\lambda}\hat{\alpha}_{ij}\big\|^2 &= \frac{1}{2}\big\|w(X_i)\widehat{\varepsilon}_i\,\mathcal{L}^{-1/2}_{P^w,\lambda}k_{X_i} - w(X_j)\widehat{\varepsilon}_j\,\mathcal{L}^{-1/2}_{P^w,\lambda}k_{X_j}\big\|^2 \\
&\le \big\|w(X_i)\widehat{\varepsilon}_i\,\mathcal{L}^{-1/2}_{P^w,\lambda}k_{X_i}\big\|^2 + \big\|w(X_j)\widehat{\varepsilon}_j\,\mathcal{L}^{-1/2}_{P^w,\lambda}k_{X_j}\big\|^2.
\end{aligned}$$

$$\begin{aligned}
w(X_i)\widehat{\varepsilon}_i\,k_{X_i} &= w(X_i)\big[Y_i - \hat{f}(X_i)\big]\,k_{X_i} \\
&= w(X_i)\Big[(Y_i - f_*(X_i)) + (f_*(X_i) - \hat{f}(X_i))\Big]\,k_{X_i} \\
&= w(X_i)\varepsilon_i\,k_{X_i} + w(X_i)\big(f_*(X_i) - \hat{f}(X_i)\big)\,k_{X_i} \\
&= w(X_i)\varepsilon_i\,k_{X_i} + \mathcal{L}_{P^w_i}(f_* - \hat{f}).
\end{aligned}$$

Then,

$$\begin{aligned}\mathbb{E}_h\big\|\mathcal{L}_{P^w,\lambda}^{-1/2}u\big\|^2 &\le \frac{2}{n}\sum_{i=1}^n\big\|w(X_i)\widehat{\varepsilon}_i\,\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\big\|^2\\ &\le \frac{4}{n}\sum_{i=1}^n\big\|w(X_i)\varepsilon_i\,\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\big\|^2+\frac{4}{n}\sum_{i=1}^n\big\|\mathcal{L}_{P^w,\lambda}^{-1/2}\mathcal{L}_{P_i^w}(f_*-\hat{f})\big\|^2.\end{aligned}$$

□

Lemma G.6. *Under Assumption 4.1, for $0<\eta<1$ and $\lambda>0$, with probability at least $1-\eta$,*

$$\frac{1}{n}\sum_{i=1}^n\big\|w(X_i)\varepsilon_i\,\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\big\|^2\le\tau_n\overline{\sigma}^2\mathcal{N}_\lambda+2\log\left(\frac{2}{\eta}\right)\left\{\frac{\overline{\sigma}^2\kappa^2\tau_n^2}{n\lambda}+\sqrt{\frac{\overline{\sigma}^4\kappa^2\tau_n^3\mathcal{N}_\lambda}{n\lambda}}\right\}.$$

Proof of Lemma G.6. Let

$$\xi_i:=\big\|w(X_i)\varepsilon_i\,\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\big\|^2.$$

Then

$$\xi_i\le\frac{\tau_n^2\,\overline{\sigma}^2\,\kappa^2}{\lambda}.$$

Moreover,

$$\begin{aligned}\mathbb{E}\xi_i&=\mathbb{E}\big\|w(X_i)\varepsilon_i\,\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\big\|^2\\ &\le\tau_n\overline{\sigma}^2\,\mathbb{E}\Big[w(X_i)\,\big\|\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\big\|^2\Big]\\ &=\tau_n\overline{\sigma}^2\,\mathbb{E}\Big[w(X_i)\,\big\langle\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i},\,\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\big\rangle\Big]\\ &=\tau_n\overline{\sigma}^2\,\mathrm{tr}\Big(\mathcal{L}_{P^w,\lambda}^{-1}\mathcal{L}_{P^w}\Big)\\ &\le\tau_n\overline{\sigma}^2\,\mathcal{N}_\lambda.\end{aligned}$$

The last inequality follows from $w\le\rho$, which implies $\mathcal{L}_{P^w}\preceq_{\mathsf{H}}\mathcal{L}_Q$, and the operator monotonicity of $x\mapsto x/(x+\lambda)$.

Then,

$$\mathbb{E}\xi_i^2\le\frac{\tau_n^3\overline{\sigma}^4\kappa^2}{\lambda}\,\mathcal{N}_\lambda.$$

Therefore, by Bernstein's inequality, with probability $1-\eta$,

$$\mathbb{E}_n\xi_i\le\mathbb{E}\xi_i+2\log\left(\frac{2}{\eta}\right)\left\{\frac{\tau_n^2\overline{\sigma}^2\kappa^2}{n\lambda}+\sqrt{\frac{\tau_n^3\overline{\sigma}^4\kappa^2\mathcal{N}_\lambda}{n\lambda}}\right\}.$$

□

Lemma G.7. *For $0<\eta<1$ and $\lambda>0$, with probability at least $1-\eta$,*

$$\frac{1}{n}\sum_{i=1}^n\big\|\mathcal{L}_{P^w,\lambda}^{-1/2}\mathcal{L}_{P_i^w}\big\|_{HS}^2\le\tau_n\kappa^2\mathcal{N}_\lambda+2\log\left(\frac{2}{\eta}\right)\left\{\frac{2\kappa^4\tau_n^2}{n\lambda}+\sqrt{\frac{\kappa^6\tau_n^3\mathcal{N}_\lambda}{n\lambda}}\right\}.$$

Proof of Lemma G.7. Let

$$\begin{aligned}\xi_i&:=\big\|\mathcal{L}_{P^w,\lambda}^{-1/2}\mathcal{L}_{P_i^w}\big\|_{HS}^2\\ &=\big\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1/2}\,w(X_i)\,k_{X_i}\otimes k_{X_i}\big\|_{HS}^2.\end{aligned}$$

Then

$$0 \le \xi_i \le \frac{1}{\lambda}\,\tau_n^2\kappa^4.$$

Moreover,

$$\begin{aligned}
\mathbb{E}\xi_i &\le \tau_n\,\mathbb{E}\Big[w(X_i)\,\big\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1/2}(k_{X_i}\otimes k_{X_i})\big\|_{HS}^2\Big] \\
&= \tau_n\,\mathbb{E}\left[w(X_i)\sum_{j=1}^{\infty}\big\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1/2}(k_{X_i}\otimes k_{X_i})e_j\big\|^2\right] \\
&= \tau_n\,\mathbb{E}\left[w(X_i)\sum_{j=1}^{\infty}\big\|\langle k_{X_i},e_j\rangle\,(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1/2}k_{X_i}\big\|^2\right] \\
&= \tau_n\,\mathbb{E}\left[w(X_i)\,\big\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1/2}k_{X_i}\big\|^2\sum_{j=1}^{\infty}\langle k_{X_i},e_j\rangle^2\right] \\
&= \tau_n\,\mathbb{E}\Big[w(X_i)\,\big\|(\mathcal{L}_{P^w}+\lambda\mathcal{I})^{-1/2}k_{X_i}\big\|^2\,\|k_{X_i}\|^2\Big] \\
&\le \tau_n\kappa^2\,\mathcal{N}_\lambda.
\end{aligned}$$

Thus,

$$\mathbb{E}\,\xi_i^2 \le \frac{1}{\lambda}\,\tau_n^3\,\kappa^6\,\mathcal{N}_\lambda.$$

By Bernstein's inequality, with probability $1-\eta$,

$$\mathbb{E}_n\xi_i \le \mathbb{E}\xi_i + 2\log\left(\frac{2}{\eta}\right)\left\{\tau_n^2\frac{2\kappa^4}{n\lambda} + \sqrt{\tau_n^3\frac{\kappa^6\mathcal{N}_\lambda}{n\lambda}}\right\}.$$

□

Lemma G.8. *For $0<\eta<1$, under Assumptions 4.1 and 4.2, suppose $\tau_n \ge \tau_0$ for all sufficiently large n, $\tau_n\to\infty$, and $\lambda=\lambda_n\to 0$. If, for all sufficiently large n,*

$$n > \max\left\{\frac{(\tau_n\kappa^2+1)\log(e+\mathcal{N}_\lambda)}{C_2\lambda}, \frac{\tau_n}{\lambda\mathcal{N}_\lambda}\right\}$$

where C_2 is the constant in Lemma F.13 after replacing its failure probability by the fixed fraction of η used in the union bound, then, for all sufficiently large n, with probability at least $1-\eta$ over the data, conditionally on the data with probability at least $1-\eta$,

$$\|\mathcal{L}_{P^w,\lambda}^{-1/2}u\| \lesssim \overline{\sigma}\,\kappa^{2s+3}\big(1+\|g_*\|_{\mathsf{H}}\big)\sqrt{\tau_n\mathcal{N}_\lambda}\left(1+\sqrt{\frac{\tau_n\mathcal{N}_\lambda}{n\lambda}}\right)\left(\log\left(\frac{2}{\eta}\right)\right)^2.$$

Proof of Lemma G.8. Apply Lemmas G.6, G.7, and F.16 with suitably split failure probabilities. The second sample-size condition implies

$$\begin{aligned}
\frac{1}{n}\sum_{i=1}^{n}\|w(X_i)\varepsilon_i\mathcal{L}_{P^w,\lambda}^{-1/2}k_{X_i}\|^2 &\lesssim \overline{\sigma}^2\kappa^2\tau_n\mathcal{N}_\lambda\log\left(\frac{2}{\eta}\right), \\
\frac{1}{n}\sum_{i=1}^{n}\|\mathcal{L}_{P^w,\lambda}^{-1/2}\mathcal{L}_{P_i^w}\|_{HS}^2 &\lesssim \kappa^4\tau_n\mathcal{N}_\lambda\log\left(\frac{2}{\eta}\right).
\end{aligned}$$

Furthermore,

$$\frac{1}{n}\sum_{i=1}^{n}\|\mathcal{L}_{P^w,\lambda}^{-1/2}\mathcal{L}_{P_i^w}(f_*-\hat{f})\|^2 \le \|f_*-\hat{f}\|^2\,\frac{1}{n}\sum_{i=1}^{n}\|\mathcal{L}_{P^w,\lambda}^{-1/2}\mathcal{L}_{P_i^w}\|_{HS}^2.$$

Substituting these bounds into Lemma G.5, followed by the second bound in Lemma F.16, yields

$$\begin{aligned}\|\mathcal{L}_{P^w,\lambda}^{-1/2}u\| \lesssim \overline{\sigma}\,\kappa^{2s+3}\big(1+\|g_*\|_{\mathsf{H}}\big)\sqrt{\tau_n\mathcal{N}_\lambda}\\ \times\left[1+\sqrt{\frac{\tau_n\mathcal{N}_\lambda}{n\lambda}}+\lambda^{\min\{s,1/2\}}\right]\left(\log\left(\frac{2}{\eta}\right)\right)^2.\end{aligned}$$

Since $0<\lambda\le 1$ for all sufficiently large n, the last λ-term is absorbed by 1, proving the stated bound. Replacing η by fixed fractions of η in the data and conditional multiplier bounds changes only universal constants inside the logarithm. □

Proof of Lemma F.17. Apply Lemmas F.12, F.13, G.4, and G.8, with suitably split failure probabilities. This gives

$$\begin{aligned}\|\Delta_{\mathrm{inv}}\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \mathfrak{C}_1\overline{\sigma}\,\kappa^{2s+4}\big(1+\|g_*\|_{\mathsf{H}}\big)\tau_n\sqrt{\frac{\mathcal{N}_\lambda\log(e+\mathcal{N}_\lambda)}{n\lambda}}\\ \times\left(1+\sqrt{\frac{\tau_n\mathcal{N}_\lambda}{n\lambda}}\right)\left(\log\left(\frac{2}{\eta}\right)\right)^3.\end{aligned}$$

By the argument in the proof of Lemma F.16, $\log(e+\mathcal{N}_\lambda)\lesssim\mathcal{N}_\lambda$ for all sufficiently large n. Hence

$$\sqrt{\frac{\log(e+\mathcal{N}_\lambda)}{\mathcal{N}_\lambda}}\lesssim 1.$$

The first sample-size condition also gives

$$\sqrt{\frac{\log(e+\mathcal{N}_\lambda)}{\mathcal{N}_\lambda}}\sqrt{\frac{\tau_n\mathcal{N}_\lambda}{n\lambda}}=\sqrt{\frac{\tau_n\log(e+\mathcal{N}_\lambda)}{n\lambda}}\le\frac{\sqrt{C_2}}{\kappa}\le 1.$$

Both terms are therefore bounded by $\tau_n\mathcal{N}_\lambda/\sqrt{n\lambda}$, which proves the claim. □

G.18 Proof of Lemma F.18

Lemma G.9. *Under Assumption 4.4, writing $\lambda=\lambda_n$, for $0<\eta<1$, conditionally on $\mathcal{D}$, with probability at least $1-\eta$,*

$$\|\Delta_{\mathrm{res}}\|_{\mathsf{L}^2(\mathbb{Q})}\le\mathfrak{C}_1\,\|f_{P^w}-\hat{f}\|\,\Big(1+\sqrt{2\log\left(\frac{1}{\eta}\right)}\Big)\sqrt{\frac{2}{n}\sum_{i=1}^{n}\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}}\mathcal{L}_{P_i^w}\|_{HS}^2}.$$

Proof of Lemma G.9.

$$\begin{aligned}\hat{\alpha}_{ij}-\alpha_{ij}&=\frac{1}{\sqrt{2}}\Big(w(X_i)\widehat{\varepsilon}_i k_{X_i}-w(X_j)\widehat{\varepsilon}_j k_{X_j}\Big)-\frac{1}{\sqrt{2}}\Big(w(X_i)\epsilon_i^{P^w}k_{X_i}-w(X_j)\epsilon_j^{P^w}k_{X_j}\Big)\\&=\frac{1}{\sqrt{2}}\Big[w(X_i)\big(f_{P^w}(X_i)-\hat{f}(X_i)\big)k_{X_i}-w(X_j)\big(f_{P^w}(X_j)-\hat{f}(X_j)\big)k_{X_j}\Big]\\&=\frac{1}{\sqrt{2}}\Big[\mathcal{L}_{P_i^w}(f_{P^w}-\hat{f})-\mathcal{L}_{P_j^w}(f_{P^w}-\hat{f})\Big]\\&=\frac{1}{\sqrt{2}}\big(\mathcal{L}_{P_i^w}-\mathcal{L}_{P_j^w}\big)(f_{P^w}-\hat{f}).\end{aligned}$$

Hence,

$$\Delta_{\mathrm{res}} = \mathcal{L}_{P^w,\lambda}^{-1}\left[\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\frac{1}{\sqrt{2}}(\mathcal{L}_{P_i^w} - \mathcal{L}_{P_j^w})(f_{P^w} - \hat{f})\right].$$

$$\|\Delta_{\mathrm{res}}\|_{\mathsf{L}^2(\mathbb{Q})} \le \mathfrak{C}_1 \|f_{P^w} - \hat{f}\| \left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}} u\right\|_{HS},$$

where

$$u = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} h_{ij}\frac{1}{\sqrt{2}}(\mathcal{L}_{P_i^w} - \mathcal{L}_{P_j^w}).$$

By Borell's inequality under $\mathbb{P}(\cdot \mid \mathcal{D})$, with probability at least $1-\eta$,

$$\left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}} u\right\|_{HS} \le \left(1 + \sqrt{2\log\left(\frac{1}{\eta}\right)}\right)\sqrt{\mathbb{E}_h\left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}} u\right\|_{HS}^2}.$$

And

$$\begin{aligned}
\mathbb{E}_h\left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}} u\right\|_{HS}^2 &= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{1}{2}\left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}}\mathcal{L}_{P_i^w} - \mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}}\mathcal{L}_{P_j^w}\right\|_{HS}^2 \\
&\le \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left[\left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}}\mathcal{L}_{P_i^w}\right\|_{HS}^2 + \left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}}\mathcal{L}_{P_j^w}\right\|_{HS}^2\right] \\
&= \frac{2}{n}\sum_{i=1}^{n}\left\|\mathcal{L}_{P^w,\lambda}^{-\frac{1}{2}}\mathcal{L}_{P_i^w}\right\|_{HS}^2.
\end{aligned}$$

□

Proof of Lemma F.18. On the intersection of the data events in Lemmas F.16 and G.7, the second sample-size condition gives

$$\sqrt{\frac{2}{n}\sum_{i=1}^{n}\|\mathcal{L}_{P^w,\lambda}^{-1/2}\mathcal{L}_{P_i^w}\|_{HS}^2} \lesssim \kappa^2\sqrt{\tau_n\mathcal{N}_\lambda}\sqrt{\log\left(\frac{2}{\eta}\right)}.$$

Conditionally on the data, Lemma G.9 and the first bound in Lemma F.16 therefore imply

$$\|\Delta_{\mathrm{res}}\|_{\mathsf{L}^2(\mathbb{Q})} \lesssim \mathfrak{C}_1\overline{\sigma}\,\kappa^{2s+3}(1+\|g_*\|_{\mathsf{H}})\frac{\tau_n\mathcal{N}_\lambda}{\sqrt{n\lambda}}\left(\log\left(\frac{2}{\eta}\right)\right)^2.$$

Using fixed fractions of η for the two data events and the conditional multiplier event gives the asserted two-layer probability statement without changing the displayed logarithmic order. □

G.19 Proof of Lemma F.19

Proof. Let $H(\cdot \mid \mathcal{A})$ be a regular conditional distribution of the pair (X', Y') given $\mathcal{A}$. Define its two marginal kernels by

$$F(C \mid \mathcal{A}) = H(C \times \mathsf{S} \mid \mathcal{A}), \qquad G(C \mid \mathcal{A}) = H(\mathsf{S} \times C \mid \mathcal{A}),$$

for Borel $C \subset \mathsf{S}$. Then $G(\cdot \mid \mathcal{A})$ is a regular conditional distribution of Y' given $\mathcal{A}$, and, because $X \mid \mathcal{A} \overset{d}{=} X' \mid \mathcal{A}$, $F(\cdot \mid \mathcal{A})$ is also a version of the regular conditional distribution of X given $\mathcal{A}$.

The conditional Strassen–Dudley theorem of Monrad and Philipp [1991, Theorem 4] gives a random element Y, with conditional law $G(\cdot \mid \mathcal{A})$ given $\mathcal{A}$, provided that

$$\mathbb{E} \sup_C [F(C \mid \mathcal{A}) - G(C^r \mid \mathcal{A})] \le \varepsilon,$$

where the supremum is over Borel sets $C \subset \mathsf{S}$ and

$$C^r = \{s \in \mathsf{S} : \inf_{t \in C} d(s,t) \le r\}.$$

The extra uniform random variable in the assumptions is the randomization needed in that theorem to realize the resulting conditional transition kernel on the original probability space.

It remains to check this displayed condition. Let

$$D_r = \{(s,t) \in \mathsf{S} \times \mathsf{S} : d(s,t) \le r\}.$$

For any Borel set $C \subset \mathsf{S}$, the set inclusion

$$(C \times \mathsf{S}) \cap D_r \subset \mathsf{S} \times C^r$$

holds by the definition of C^r. Hence

$$C \times \mathsf{S} \subset (\mathsf{S} \times C^r) \cup D_r^c.$$

Applying the conditional probability kernel $H(\cdot \mid \mathcal{A})$ to this inclusion gives, outside a fixed null set in the underlying probability space and simultaneously for all Borel C,

$$\begin{aligned} F(C \mid \mathcal{A}) - G(C^r \mid \mathcal{A}) &\le H(D_r^c \mid \mathcal{A}) \\ &= P\{d(X', Y') > r \mid \mathcal{A}\}. \end{aligned}$$

The right-hand side does not depend on C, and therefore the same bound holds after taking the supremum over Borel C. Hence

$$\begin{aligned} \mathbb{E} \sup_C [F(C \mid \mathcal{A}) - G(C^r \mid \mathcal{A})] &\le \mathbb{E} P\{d(X', Y') > r \mid \mathcal{A}\} \\ &= P\{d(X', Y') > r\} \le \varepsilon. \end{aligned}$$

The conditional Strassen–Dudley theorem applies and yields an S-valued random element Y satisfying $Y \mid \mathcal{A} \overset{d}{=} Y' \mid \mathcal{A}$ and $P\{d(X,Y) > r\} \le \varepsilon$. □

G.20 Proof of Lemma F.27

Proof. Because $w(X) = \min\{\rho(X), \tau_n\}$,

$$1 - \frac{w(X)}{\rho(X)} = \left(1 - \frac{\tau_n}{\rho(X)}\right) \mathbf{1}\{\rho(X) \ge \tau_n\}.$$

In particular,

$$\left(1 - \frac{w(X)}{\rho(X)}\right)^2 \le \mathbf{1}\{\rho(X) \ge \tau_n\}.$$

Let $p_m = (m-1)/\gamma$. Since $\gamma \geq 1$, Lyapunov's inequality and Definition 4.1 give

$$\mathbb{E}_{\mathbb{Q}}\rho(X)^{p_m} \leq \left\{\mathbb{E}_{\mathbb{Q}}\rho(X)^{\gamma(m-1)}\right\}^{1/\gamma^2} \leq \left(\frac{1}{2}m!W^{m-2}\sigma^2\right)^{1/\gamma}.$$

Consequently, Markov's inequality yields

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}\left(1-\frac{w(X)}{\rho(X)}\right)^2 &\leq \mathbb{Q}\{\rho(X) \geq \tau_n\} \\
&\leq \tau_n^{-p_m}\mathbb{E}_{\mathbb{Q}}\rho(X)^{p_m} \\
&\leq \tau_n^{-\frac{m-1}{\gamma}}\left(\frac{1}{2}m!W^{m-2}\sigma^2\right)^{\frac{1}{\gamma}} \\
&= \left(\frac{1}{2}\tau_n^{-(m-1)}m!W^{m-2}\sigma^2\right)^{\frac{1}{\gamma}}.
\end{aligned}$$

□

G.21 Proof of Lemma F.28

Proof. Since $q = \rho p$ and $w(x) = \min\{\rho(x), \tau_n\}$,

$$\mathcal{L}_{P^w} f = \int w(x)k_x f(x)\, p(x)\, dx = \int \frac{w(x)}{\rho(x)}k_x f(x)\, q(x)\, dx.$$

Hence, with $a(x) := 1 - w(x)/\rho(x)$,

$$\mathcal{L}_Q - \mathcal{L}_{P^w} = \int a(x)\, k_x \otimes_{\mathsf{H}} k_x\, \mathbb{Q}(dx).$$

For $A_\lambda = (\mathcal{L}_Q + \lambda\mathcal{I})^{-1}$, self-adjointness gives

$$(k_x \otimes_{\mathsf{H}} k_x)A_\lambda = k_x \otimes_{\mathsf{H}} A_\lambda k_x.$$

Thus

$$(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1} = \int a(x)\, k_x \otimes_{\mathsf{H}} A_\lambda k_x\, \mathbb{Q}(dx).$$

Using $\|T\| \leq \|T\|_{HS}$, Jensen's inequality in Hilbert–Schmidt norm, and Cauchy–Schwarz,

$$\begin{aligned}
\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\| &\leq \int |a(x)|\, \|k_x \otimes_{\mathsf{H}} A_\lambda k_x\|_{HS}\, \mathbb{Q}(dx) \\
&\leq \left(\mathbb{E}_{\mathbb{Q}}a(X)^2\right)^{1/2}\left(\mathbb{E}_{\mathbb{Q}}\|k_X \otimes_{\mathsf{H}} A_\lambda k_X\|_{HS}^2\right)^{1/2}.
\end{aligned}$$

For a rank-one operator, $\|u \otimes v\|_{HS} = \|u\|\,\|v\|$, and $\|k_X\| \leq \kappa$. Therefore,

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}\|k_X \otimes_{\mathsf{H}} A_\lambda k_X\|_{HS}^2 &\leq \kappa^2\, \mathbb{E}_{\mathbb{Q}}\|A_\lambda k_X\|^2 \\
&= \kappa^2\, \mathrm{tr}_{\mathsf{H}}\left(A_\lambda^2 \mathcal{L}_Q\right) \\
&\leq \kappa^2\lambda^{-1}\mathrm{tr}_{\mathsf{H}}(A_\lambda \mathcal{L}_Q) = \kappa^2\lambda^{-1}\mathcal{N}_\lambda,
\end{aligned}$$

because $A_\lambda \preceq \lambda^{-1}\mathcal{I}$. Combining this with Lemma F.27 yields

$$\left\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\right\| \leq \kappa\lambda^{-1/2}\mathcal{N}_\lambda^{1/2}\left(\frac{1}{2}\tau_n^{-(m-1)}m!W^{m-2}\sigma^2\right)^{\frac{1}{2\gamma}}.$$

□

G.22 Proof of Lemma F.29

Proof. Let

$$\Delta_\lambda := (\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}.$$

By Lemma F.28 with $m = m_0$,

$$\begin{aligned}\|\Delta_\lambda\| &\le \kappa\lambda^{-1/2}\mathcal{N}_\lambda^{1/2}\left(\frac{1}{2}\tau_n^{-(m_0-1)}m_0!W^{m_0-2}\sigma^2\right)^{\frac{1}{2\gamma}} \\ &\le \kappa E_d\lambda^{-\frac{1+d}{2}}\left(\frac{1}{2}\tau_n^{-(m_0-1)}m_0!W^{m_0-2}\sigma^2\right)^{\frac{1}{2\gamma}},\end{aligned}$$

where the second inequality follows from $\mathcal{N}_\lambda \le E_d^2\lambda^{-d}$. The lower bound imposed on τ_n is exactly the condition that the last display is at most $1/2$. This proves the first claim.

For the second claim, use the factorization

$$\begin{aligned}\mathcal{L}_{P^w} + \lambda\mathcal{I} &= \mathcal{L}_Q + \lambda\mathcal{I} - (\mathcal{L}_Q - \mathcal{L}_{P^w}) \\ &= (\mathcal{I} - \Delta_\lambda)(\mathcal{L}_Q + \lambda\mathcal{I}).\end{aligned}$$

Since $\|\Delta_\lambda\| \le 1/2$, the operator $\mathcal{I} - \Delta_\lambda$ is invertible and

$$\|(\mathcal{I} - \Delta_\lambda)^{-1}\| \le \frac{1}{1 - \|\Delta_\lambda\|} \le 2.$$

Consequently,

$$\begin{aligned}\|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\| &= \left\|\mathcal{L}_Q(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}(\mathcal{I} - \Delta_\lambda)^{-1}\right\| \\ &\le \|\mathcal{L}_Q(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\|\|(\mathcal{I} - \Delta_\lambda)^{-1}\| \le 2,\end{aligned}$$

because $0 \preceq \mathcal{L}_Q(\mathcal{L}_Q + \lambda\mathcal{I})^{-1} \preceq \mathcal{I}$.

It remains to prove the third claim. By the Cordes inequality in Lemma F.26, applied with $A = \mathcal{L}_Q$, $B = (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}$, and $s = 1/2$,

$$\|\mathcal{L}_Q^{1/2}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2}\| \le \|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\|^{1/2} \le \sqrt{2}.$$

□

G.23 Proof of Lemma G.10

Lemma G.10. *Under Definition 4.2 with $\theta = 2$, for $\tau_n > 0$,*

$$\mathbb{E}_{\mathbb{Q}}\left(1 - \frac{w(X)}{\rho(X)}\right)^2 \le \tau_n^{-1}\Omega.$$

Proof.

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}}\left(1 - \frac{w(X)}{\rho(X)}\right)^2 &= \int\left(1 - \frac{w(x)}{\rho(x)}\right)^2 \mathbb{Q}(dx) \\ &= \int_{\rho\ge\tau_n}\left(1 - \frac{\tau_n}{\rho(x)}\right)^2 \mathbb{Q}(dx)\end{aligned}$$

$$
\begin{aligned}
&\leq \int_{\rho \geq \tau_n} 1 \, \mathbb{Q}(dx) \\
&= \mathbb{Q}(\rho(X) \geq \tau_n) \\
&\leq \tau_n^{-1} \mathbb{E}_{\mathbb{Q}} \rho(X) \leq \tau_n^{-1} \Omega.
\end{aligned}
$$

□

G.24 Proof of Lemma G.11

Lemma G.11. *Under Definition 4.2 with $\theta = 2$, for $\tau_n > 0$ and $\lambda > 0$,*

$$
\left\| (\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda \mathcal{I})^{-1} \right\| \leq \kappa \lambda^{-1/2} \mathcal{N}_\lambda^{1/2} \tau_n^{-1/2} \Omega^{1/2}.
$$

Proof. The proof is identical to the Hilbert–Schmidt argument in Lemma F.28; the only input that changes is the moment bound for the truncated density ratio. Indeed, the same calculation gives

$$
\left\| (\mathcal{L}_Q - \mathcal{L}_{P^w}) (\mathcal{L}_Q + \lambda \mathcal{I})^{-1} \right\| \leq \kappa \lambda^{-1/2} \mathcal{N}_\lambda^{1/2} \left[\mathbb{E}_{\mathbb{Q}} \left(1 - \frac{w(X)}{\rho(X)} \right)^2 \right]^{1/2}.
$$

Applying Lemma G.10 yields

$$
\left\| (\mathcal{L}_Q - \mathcal{L}_{P^w}) (\mathcal{L}_Q + \lambda \mathcal{I})^{-1} \right\| \leq \kappa \lambda^{-1/2} \mathcal{N}_\lambda^{1/2} \tau_n^{-1/2} \Omega^{1/2}.
$$

□

G.25 Proof of Lemma G.12

Lemma G.12. *Under Assumption 4.3 and Definition 4.2 with $\theta = 2$, for $0 < \lambda \leq 1$, if*

$$
\tau_n \geq 4 \kappa^2 \Omega E_d^2 \lambda^{-(1+d)}, \tag{G.3}
$$

then, as operators on H,

1. $\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda \mathcal{I})^{-1}\| \leq 1/2$;
2. $\|\mathcal{L}_Q (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1}\| \leq 2$;
3. $\|\mathcal{L}_Q^{1/2} (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1/2}\| \leq \sqrt{2}$.

Proof. Let again

$$
\Delta_\lambda := (\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda \mathcal{I})^{-1}.
$$

By Lemma G.11 and $\mathcal{N}_\lambda \leq E_d^2 \lambda^{-d}$,

$$
\begin{aligned}
\|\Delta_\lambda\| &\leq \kappa \lambda^{-1/2} \mathcal{N}_\lambda^{1/2} \tau_n^{-1/2} \Omega^{1/2} \\
&\leq \kappa E_d \Omega^{1/2} \lambda^{-\frac{1+d}{2}} \tau_n^{-1/2}.
\end{aligned}
$$

Condition (G.3) implies that the right-hand side is at most $1/2$, proving the first claim.

The remaining two claims follow from the same deterministic perturbation argument as in Lemma F.29. Namely,

$$\mathcal{L}_{P^w} + \lambda\mathcal{I} = (\mathcal{I} - \Delta_\lambda)(\mathcal{L}_Q + \lambda\mathcal{I}), \qquad \|(\mathcal{I} - \Delta_\lambda)^{-1}\| \le 2.$$

Therefore,

$$\|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\| \le \|\mathcal{L}_Q(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\|\|(\mathcal{I} - \Delta_\lambda)^{-1}\| \le 2,$$

and Lemma F.26 gives

$$\|\mathcal{L}_Q^{1/2}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2}\| \le \|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\|^{1/2} \le \sqrt{2}.$$

□

G.26 Proof of Lemma G.13

Lemma G.13. *Under Definition F.1, for $\tau_n > 0$,*

$$\mathbb{E}_{\mathbb{Q}}\left(1 - \frac{w(X)}{\rho(X)}\right)^2 \le \tau_n^{-(\vartheta-1)}\Omega_\vartheta.$$

Proof. Since $w(X) = \min\{\rho(X), \tau_n\}$,

$$\begin{aligned}\mathbb{E}_{\mathbb{Q}}\left(1 - \frac{w(X)}{\rho(X)}\right)^2 &\le \mathbb{Q}\{\rho(X) \ge \tau_n\} \\ &\le \tau_n^{-(\vartheta-1)}\mathbb{E}_{\mathbb{Q}}\left[\rho(X)^{\vartheta-1}\right] \\ &\le \tau_n^{-(\vartheta-1)}\Omega_\vartheta,\end{aligned}$$

where the second inequality follows from Markov's inequality. □

G.27 Proof of Lemma F.30

Proof. The Hilbert–Schmidt argument used in Lemma F.28 gives

$$\left\|(\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}\right\| \le \kappa\lambda^{-1/2}\mathcal{N}_\lambda^{1/2}\left[\mathbb{E}_{\mathbb{Q}}\left(1 - \frac{w(X)}{\rho(X)}\right)^2\right]^{1/2}.$$

The claim follows by applying Lemma G.13. □

G.28 Proof of Lemma F.31

Proof. Let

$$\Delta_\lambda := (\mathcal{L}_Q - \mathcal{L}_{P^w})(\mathcal{L}_Q + \lambda\mathcal{I})^{-1}.$$

By Lemma F.30 and $\mathcal{N}_\lambda \le E_d^2\lambda^{-d}$,

$$\begin{aligned}\|\Delta_\lambda\| &\le \kappa\lambda^{-1/2}\mathcal{N}_\lambda^{1/2}\tau_n^{-\frac{\vartheta-1}{2}}\Omega_\vartheta^{1/2} \\ &\le \kappa E_d\Omega_\vartheta^{1/2}\lambda^{-\frac{1+d}{2}}\tau_n^{-\frac{\vartheta-1}{2}}.\end{aligned}$$

Condition (F.4) makes the last display at most $1/2$, proving the first claim.

The remaining claims follow from the deterministic perturbation argument in Lemma F.29. In particular,

$$\mathcal{L}_{P^w} + \lambda\mathcal{I} = (\mathcal{I} - \Delta_\lambda)(\mathcal{L}_Q + \lambda\mathcal{I}), \qquad \|(\mathcal{I} - \Delta_\lambda)^{-1}\| \le 2.$$

Consequently,

$$\|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\| \le 2,$$

and Lemma F.26 yields

$$\|\mathcal{L}_Q^{1/2}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1/2}\| \le \|\mathcal{L}_Q(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\|^{1/2} \le \sqrt{2}.$$

□

G.29 Proof of Lemma F.20

Proof.

$$\begin{aligned}
\mathcal{N}_\lambda &= \sum_{i=1}^{\infty} \frac{\mu_i}{\mu_i + \lambda} \\
&\le \sum_{i=1}^{\infty} \frac{\alpha i^{-\beta}}{\alpha i^{-\beta} + \lambda} \\
&= \sum_{i=1}^{\infty} \frac{\alpha}{\alpha + \lambda i^{\beta}} \\
&\le \int_0^{\infty} \frac{\alpha}{\alpha + \lambda x^{\beta}}\, dx
\end{aligned}$$

Let $t = (\lambda/\alpha)^{1/\beta} x$. Then

$$\begin{aligned}
\mathcal{N}_\lambda &\le \left(\frac{\alpha}{\lambda}\right)^{1/\beta} \left[\int_0^1 \frac{1}{1+t^{\beta}}\, dt + \int_1^{\infty} \frac{1}{1+t^{\beta}}\, dt\right] \\
&\le \left(\frac{\alpha}{\lambda}\right)^{1/\beta} \left[1 + \frac{1}{\beta - 1}\right] \\
&= \alpha^{1/\beta} \frac{\beta}{\beta - 1} \lambda^{-1/\beta}.
\end{aligned}$$

□

G.30 Proof of Lemma F.21

Proof.

$$\begin{aligned}
\sigma^2(m, \mathcal{L}_Q) &= \sum_{i=m+1}^{\infty} \mu_i \\
&\le \int_m^{\infty} \alpha \cdot t^{-\beta}\, dt \\
&= \alpha \left[\frac{1}{1-\beta} t^{1-\beta}\right]_m^{\infty} \\
&= \frac{\alpha}{\beta - 1} m^{1-\beta}.
\end{aligned}$$

Taking square roots proves the claim. □

G.31 Proof of Lemma F.22

Proof. Let $u = \beta x^\gamma$. Then

$$\int_a^\infty e^{-\beta x^\gamma}\,dx = \frac{1}{\gamma\beta^{1/\gamma}}\int_{\beta a^\gamma}^\infty u^{1/\gamma-1}e^{-u}\,du.$$

The standard incomplete-gamma tail estimate

$$\int_t^\infty u^{q-1}e^{-u}\,du \sim t^{q-1}e^{-t}, \qquad t\to\infty,$$

with $q = 1/\gamma$, yields

$$\int_a^\infty e^{-\beta x^\gamma}\,dx \sim \frac{1}{\gamma\beta^{1/\gamma}}(\beta a^\gamma)^{1/\gamma-1}e^{-\beta a^\gamma} = \frac{e^{-\beta a^\gamma}}{\beta\gamma a^{\gamma-1}}.$$

□

G.32 Proof of Lemma F.23

Proof.

$$\begin{aligned}
\mathcal{N}_\lambda &= \sum_{i=1}^\infty \frac{\mu_i}{\mu_i+\lambda}\\
&\le \sum_{i=1}^\infty \frac{\alpha\exp(-\beta i^\gamma)}{\alpha\exp(-\beta i^\gamma)+\lambda}\\
&= \sum_{i=1}^\infty \frac{\alpha}{\alpha+\lambda\exp(\beta i^\gamma)}\\
&\le \int_0^\infty \frac{\alpha}{\alpha+\lambda\exp(\beta x^\gamma)}\,dx\\
&= \left[\int_0^{\left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{1}{\gamma}}} + \int_{\left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{1}{\gamma}}}^\infty\right]\frac{\alpha}{\alpha+\lambda\exp(\beta x^\gamma)}\,dx\\
&\le \left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{1}{\gamma}} + \frac{\alpha}{\lambda}\int_{\left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{1}{\gamma}}}^\infty \exp(-\beta x^\gamma)\,dx\\
&\lesssim \left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{1}{\gamma}} + \frac{\alpha}{\lambda}\cdot\frac{\exp\left(-\beta\left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]\right)}{\beta\gamma\left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{\gamma-1}{\gamma}}}\\
&= \left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{1}{\gamma}} + \frac{1}{\beta\gamma}\cdot\left[\frac{1}{\beta}\ln\left(\frac{\alpha}{\lambda}\right)\right]^{\frac{1}{\gamma}-1}\\
&\lesssim \left(\log\frac{1}{\lambda}\right)^{\frac{1}{\gamma}}.
\end{aligned}$$

The last comparison holds for all sufficiently small $\lambda > 0$, which gives constants $\lambda_0 \in (0,1)$ and $C_{\alpha,\beta,\gamma} > 0$ as stated. □

G.33 Proof of Lemma F.24

Proof.

$$\begin{aligned}\sigma^2(m, \mathcal{L}_Q) &= \sum_{i=m+1}^{\infty} \mu_i \\ &\le \int_m^{\infty} \alpha \cdot \exp(-\beta t^\gamma)\, dt \\ &\le C_{\alpha,\beta,\gamma}^2 m^{1-\gamma} \exp(-\beta m^\gamma).\end{aligned}$$

Here Lemma F.22 gives the last inequality for all sufficiently large m; increasing $C_{\alpha,\beta,\gamma}$ covers the finitely many remaining positive integers. Taking square roots proves the claim. □

G.34 Proof of Lemma F.25

Proof. As in the proof of Lemma F.7,

$$\begin{aligned}\mathcal{V} &= \mathcal{V}_1 + \mathcal{V}_2, \\ \mathcal{V}_1 &\preceq_{\mathsf{L}^2(\mathbb{Q})} \tau_n \overline{\sigma}^2 \cdot \mathcal{T}_1, \\ \mathcal{V}_2 &\preceq_{\mathsf{L}^2(\mathbb{Q})} \tau_n \kappa^2 \max\{\kappa^{2s-1}, 1\}^2 \lambda^{2\min\{s,1/2\}} \|g_*\|^2 \mathcal{T}_1, \\ \mathcal{V}_1 + \mathcal{V}_2 &\preceq_{\mathsf{L}^2(\mathbb{Q})} \tau_n \Big(\overline{\sigma}^2 + \kappa^2 \max\{\kappa^{2s-1}, 1\}^2 \lambda^{2\min\{s,1/2\}} \|g_*\|^2\Big) \mathcal{T}_1.\end{aligned}$$

where

$$\mathcal{T}_1 = (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_{P^w} (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_Q.$$

$\forall\, f \in \mathsf{L}^2(\mathbb{Q})$,

$$\begin{aligned}\langle f, \mathcal{T}_1 f \rangle_{\mathsf{L}^2(\mathbb{Q})} &= \big\langle f,\, (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_{P^w} (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_Q f \big\rangle_{\mathsf{L}^2(\mathbb{Q})} \\ &= \big\langle \mathcal{L}_Q f,\, (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_{P^w} (\mathcal{L}_{P^w} + \lambda \mathcal{I})^{-1} \mathcal{L}_Q f \big\rangle \\ &\le \left\langle \mathcal{L}_Q f, \frac{1}{\lambda} \mathcal{L}_Q f \right\rangle \\ &= \frac{1}{\lambda} \langle f, \mathcal{L}_Q f \rangle_{\mathsf{L}^2(\mathbb{Q})}.\end{aligned}$$

Thus,

$$\mathcal{T}_1 \preceq_{\mathsf{L}^2(\mathbb{Q})} \frac{1}{\lambda} \mathcal{L}_Q.$$

By the min–max eigenvalue monotonicity for compact positive operators and the identity

$$\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{L}_Q) = \sigma^2(m, \mathcal{L}_Q),$$

we have

$$\sigma^2_{\mathsf{L}^2(\mathbb{Q})}(m, \mathcal{V}) \le \frac{\tau_n}{\lambda} \Big(\overline{\sigma}^2 + \kappa^2 \max\{\kappa^{2s-1}, 1\}^2 \lambda^{2\min\{s,1/2\}} \|g_*\|^2\Big) \sigma^2(m, \mathcal{L}_Q).$$

□

G.35 Proof of Lemma A.1

Proof.

$$\begin{aligned}\widetilde{U}_i &= (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\Big[w(X_i)\,\varepsilon_i\,k_{X_i} + w(X_i)\big(k_{X_i}\otimes k_{X_i}\big)\,(f_* - f_{P^w})\Big]\\ &= w(X_i)\big(\varepsilon_i + f_*(X_i) - f_{P^w}(X_i)\big)\,(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i}.\end{aligned}$$

Fix $t \in \mathsf{L}^2(\mathbb{Q})$. Since $(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q t \in \mathsf{H}$ is deterministic, the weighted-feature part of Assumption A.1 and [Vershynin, 2018, Proposition 2.5.2] imply

$$\begin{aligned}&\big\|\langle w(X_i)k_{X_i}, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q t\rangle - \mathbb{E}\big[\langle w(X_i)k_{X_i}, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q t\rangle\big]\big\|_{\psi_2}\\ &\le C_{sg}b_0\left(\mathbb{E}\left[\langle w(X_i)k_{X_i}, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q t\rangle^2\right]\right)^{1/2}.\end{aligned}$$

Here the condition applies with $w = w_n = w_{\tau_n}$, since $\tau_n \ge \tau_0$. The same bound holds for the uncentered scalar projection after changing the universal constant, because $b_0 \ge 1$ and the absolute mean is bounded by the square root of the second moment.

$$\begin{aligned}\big|\langle\widetilde{U}_i, t\rangle_{\mathsf{L}^2(\mathbb{Q})}\big| &= \big|\langle\widetilde{U}_i, \mathcal{L}_Q t\rangle\big|\\ &\le (\overline{\sigma} + \kappa\|f_*\|)\,\big|\langle w(X_i)\,(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}k_{X_i},\, \mathcal{L}_Q t\rangle\big|\\ &= (\overline{\sigma} + \kappa\|f_*\|)\,\big|\langle w(X_i)k_{X_i},\, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q t\rangle\big|.\\ \big\|\langle\widetilde{U}_i, t\rangle_{\mathsf{L}^2(\mathbb{Q})}\big\|_{\psi_2} &\le (\overline{\sigma} + \kappa\|f_*\|)\,\big\|\langle w(X_i)k_{X_i},\, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q t\rangle\big\|_{\psi_2}\\ &\le C_{sg}\cdot b_0\,(\overline{\sigma} + \kappa\|f_*\|)\,\left(\mathbb{E}\Big[\big\langle w(X_i)k_{X_i},\, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{-1}\mathcal{L}_Q t\big\rangle^2\Big]\right)^{1/2}\\ &\le C_{sg}\cdot b_0\,\frac{\overline{\sigma} + \kappa\|f_*\|}{\underline{\sigma}}\,\left\{\mathbb{E}\Big[\langle U_i, t\rangle^2_{\mathsf{L}^2(\mathbb{Q})}\Big]\right\}^{1/2}.\end{aligned}$$

where we used Lemma F.3.

Since $U_i = \widetilde{U}_i - \mathbb{E}\widetilde{U}_i$, centering changes only the universal constant. Hence

$$\big\|\langle U_i, t\rangle_{\mathsf{L}^2(\mathbb{Q})}\big\|_{\psi_2} \le C_{sg}^2\cdot b_0\,\frac{\overline{\sigma} + \kappa\|f_*\|}{\underline{\sigma}}\,\left(\mathbb{E}\Big[\langle U_i, t\rangle^2_{\mathsf{L}^2(\mathbb{Q})}\Big]\right)^{1/2}.$$

Again by [Vershynin, 2018, Proposition 2.5.2], U_i is $\mathsf{L}^2(\mathbb{Q})$-$C_{sg}^3 b_0\dfrac{\overline{\sigma} + \kappa\|f_*\|}{\underline{\sigma}}$-sub-Gaussian. □

G.36 Proof of Lemma G.1

Proof. By the spectral decomposition of $\mathcal{L}_Q$,

$$\mathsf{L}^2(\mathbb{Q}) = \overline{\mathrm{Ran}(\mathcal{L}_Q)} \oplus^{\perp}_{\mathsf{L}^2(\mathbb{Q})} \ker(\mathcal{L}_Q), \qquad \overline{\mathrm{Ran}(\mathcal{L}_Q)} = \overline{\mathrm{span}\{\phi_j : \mu_j > 0\}}.$$

Moreover, when H is viewed as a subspace of $\mathsf{L}^2(\mathbb{Q})$, Lemma 2.1 gives $\mathsf{H} \subset \overline{\mathrm{Ran}(\mathcal{L}_Q)}$. We first record the identity

$$\langle u, h\rangle_{\mathsf{L}^2(\mathbb{Q})} = \langle\mathcal{L}_Q u, h\rangle_{\mathsf{H}}, \qquad u \in \mathsf{L}^2(\mathbb{Q}),\ h \in \mathsf{H}. \tag{G.4}$$

Indeed, writing $u = \sum_j u_j\phi_j$ in $\mathsf{L}^2(\mathbb{Q})$ and $h = \sum_{\mu_j>0} h_j\phi_j \in \mathsf{H}$, the kernel component of u is orthogonal to h, and

$$\langle\mathcal{L}_Q u, h\rangle_{\mathsf{H}} = \sum_{\mu_j>0}\frac{\mu_j u_j h_j}{\mu_j} = \sum_{\mu_j>0} u_j h_j = \langle u, h\rangle_{\mathsf{L}^2(\mathbb{Q})}.$$

Now take $f, g \in \mathsf{L}^2(\mathbb{Q})$ and decompose $f = f_r + f_0$, $g = g_r + g_0$, with $f_r, g_r \in \overline{\mathrm{Ran}(\mathcal{L}_Q)}$ and $f_0, g_0 \in \ker(\mathcal{L}_Q)$. Since $\mathcal{L}_Q f_0 = \mathcal{L}_Q g_0 = 0$ and $\mathcal{T}\mathcal{L}_Q f_r, \mathcal{T}\mathcal{L}_Q g_r \in \mathsf{H} \subset \overline{\mathrm{Ran}(\mathcal{L}_Q)}$, the kernel components do not contribute to the $\mathsf{L}^2(\mathbb{Q})$ inner products. Hence, by (G.4) and the self-adjointness of $\mathcal{T}$ on H,

$$\begin{aligned}
\langle f, \mathcal{T}\mathcal{L}_Q g\rangle_{\mathsf{L}^2(\mathbb{Q})} &= \langle f_r, \mathcal{T}\mathcal{L}_Q g_r\rangle_{\mathsf{L}^2(\mathbb{Q})} \\
&= \langle \mathcal{L}_Q f_r, \mathcal{T}\mathcal{L}_Q g_r\rangle_{\mathsf{H}} \\
&= \langle \mathcal{T}\mathcal{L}_Q f_r, \mathcal{L}_Q g_r\rangle_{\mathsf{H}} \\
&= \langle \mathcal{T}\mathcal{L}_Q f_r, g_r\rangle_{\mathsf{L}^2(\mathbb{Q})} \\
&= \langle \mathcal{T}\mathcal{L}_Q f, g\rangle_{\mathsf{L}^2(\mathbb{Q})}.
\end{aligned}$$

Thus $\mathcal{T}\mathcal{L}_Q$ is self-adjoint in $\mathsf{L}^2(\mathbb{Q})$. □

G.37 Proof of Lemma G.2

Proof. As $\mathcal{L}_{P^w}$ and $\mathcal{L}_{P_n^w}$ are self-adjoint, we have

$$\begin{aligned}
&\left\|(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}\right\|^2 \\
=& \sup_{\|f\|=1} \left\langle (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}} f, (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-\frac{1}{2}}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}} f \right\rangle \\
=& \sup_{\|f\|=1} \left\langle f, (\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}} f \right\rangle \\
=& \left\|(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}(\mathcal{L}_{P^w} + \lambda\mathcal{I})^{\frac{1}{2}}\right\|.
\end{aligned}$$

Combining this with Lemma F.13, we have the desired result. □

G.38 Derivation of Equation (3.2)

Starting from the first representation of B in (3.2), we have

$$\begin{aligned}
B &= \frac{1}{n}\sum_{i=1}^n\sum_{j=1}^n h_{ij}\frac{\widehat{V}_i - \widehat{V}_j}{\sqrt{2}} \\
&= \frac{1}{\sqrt{2}}\cdot\frac{1}{n}\left[\sum_{i=1}^n \widehat{V}_i \sum_{j=1}^n h_{ij} - \sum_{j=1}^n \widehat{V}_j \sum_{i=1}^n h_{ij}\right] \\
&= \frac{1}{\sqrt{2}}\cdot\frac{1}{n}\left[\sum_{i=1}^n \widehat{V}_i \sum_{j=1}^n h_{ij} - \sum_{i=1}^n \widehat{V}_i \sum_{j=1}^n h_{ji}\right] \\
&= \frac{1}{\sqrt{2}}\cdot\frac{1}{n}\sum_{i=1}^n \widehat{V}_i \sum_{j=1}^n (h_{ij} - h_{ji}) \\
&= \frac{1}{\sqrt{2}}\mathbf{v}_{\mathbf{x}}^\top \mathrm{diag}(\widehat{\varepsilon})(\mathbf{h} - \mathbf{h}^\top)\mathbf{1}.
\end{aligned}$$

where

$$\begin{aligned}
\mathbf{v}_{\mathbf{x}}^\top &= \left[\frac{1}{n}w(X_1)(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}k_{X_1}, \ldots, \frac{1}{n}w(X_n)(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}k_{X_n}\right] \\
&= \frac{1}{n}(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}.
\end{aligned}$$

As

$$
\begin{aligned}
(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})\mathbf{k}_{\mathbf{x}}^\top &= [(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})k_{X_1}, \ldots, (\mathcal{L}_{P_n^w} + \lambda\mathcal{I})k_{X_n}] \\
\mathcal{L}_{P_n^w} k_{X_j} &= \frac{1}{n}\sum_{i=1}^{n} w(X_i)k(X_i, X_j)k_{X_i} \\
&= \frac{1}{n}(\mathbf{k}_{\mathbf{x}}^\top \mathbf{W}\mathbf{K})_j,
\end{aligned}
$$

we have

$$
\begin{aligned}
(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})\mathbf{k}_{\mathbf{x}}^\top &= \mathbf{k}_{\mathbf{x}}^\top \left(\frac{1}{n}\mathbf{W}\mathbf{K} + \lambda\mathbf{I}\right) \\
(\mathcal{L}_{P_n^w} + \lambda\mathcal{I})^{-1}\mathbf{k}_{\mathbf{x}}^\top &= \mathbf{k}_{\mathbf{x}}^\top \left(\frac{1}{n}\mathbf{W}\mathbf{K} + \lambda\mathbf{I}\right)^{-1}.
\end{aligned}
$$

Then,

$$
\mathbf{v}_{\mathbf{x}}^\top = \mathbf{k}_{\mathbf{x}}^\top(\mathbf{W}\mathbf{K} + n\lambda\mathbf{I})^{-1}\mathbf{W} = \mathbf{k}_{\mathbf{x}}^\top(\mathbf{K} + n\lambda\mathbf{W}^{-1})^{-1}.
$$